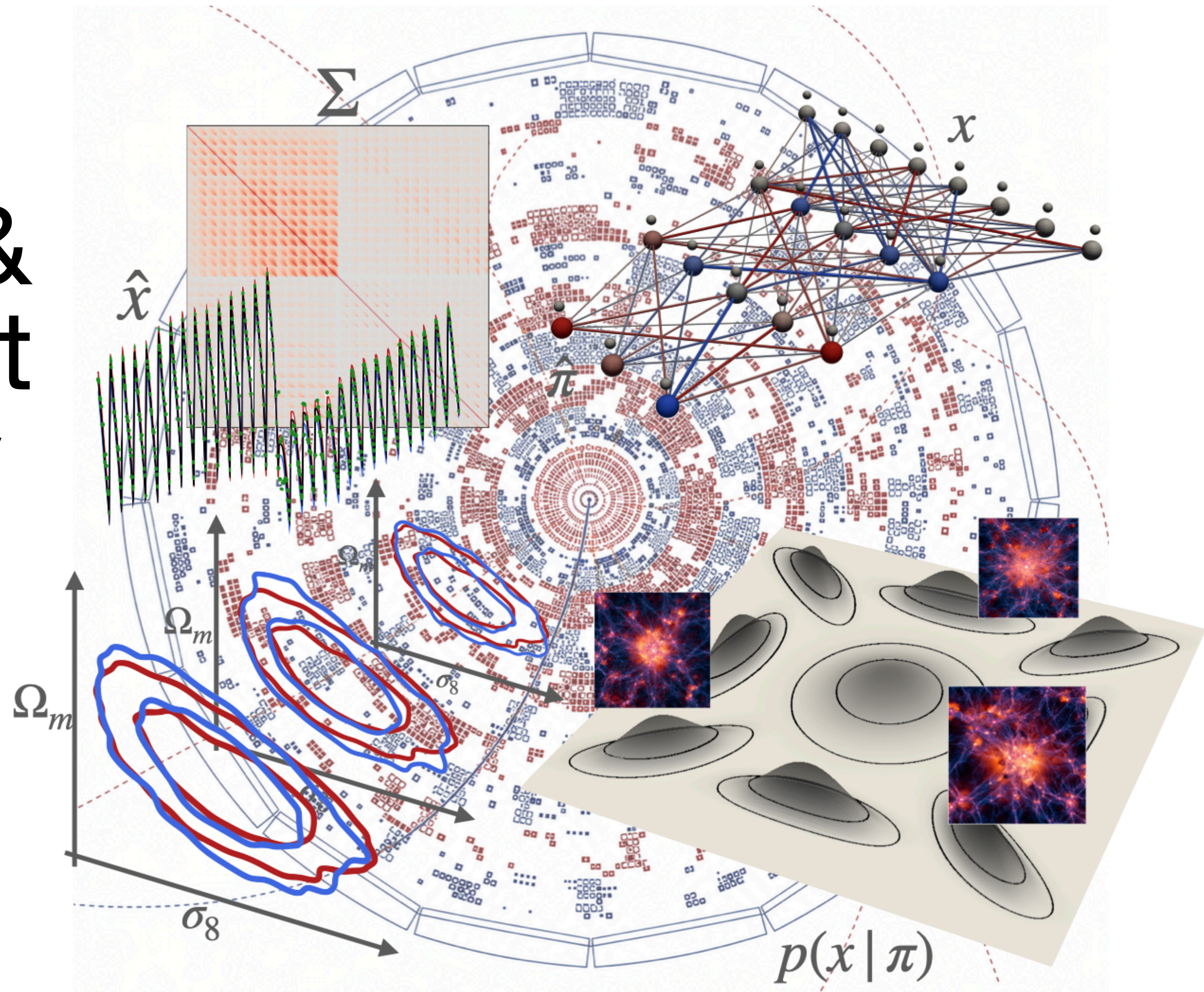


Bayesian & Frequentist Probability

ODSL Block Course



*Deadline for exam signup

		Monday	Tuesday	Wednesday	Thursday	Friday
Week 1	AM	What is probability?	Estimators & Fisher information	Bayesian and Frequentist probability	Frequentist Analysis	Bayesian Analysis
	PM	<i>JAX & statistics basics</i>	<i>Monte-Carlo Estimation in JAX</i>	<i>Confidence, credibility & the interpretation of probability</i>	<i>Frequentist Analysis</i>	<i>Bayesian Analysis</i>
Week 2	AM	Covariance Matrix Estimation	MCMC	Hypothesis testing and model selection	<i>Frontiers: SBI</i>	?
	PM	Analysis with estimated covariance			Normalising flows (affine + continuous)	

Exam

- 6 ECTS (TUM can still take the exam)
- Written exam (1hr30min)
- Guidance questions by Friday
- Sign-up 1st week only via LSF
- Example exam
- One sheet A4 (both sides), hand-written (creation via iPad/tablet allowed)

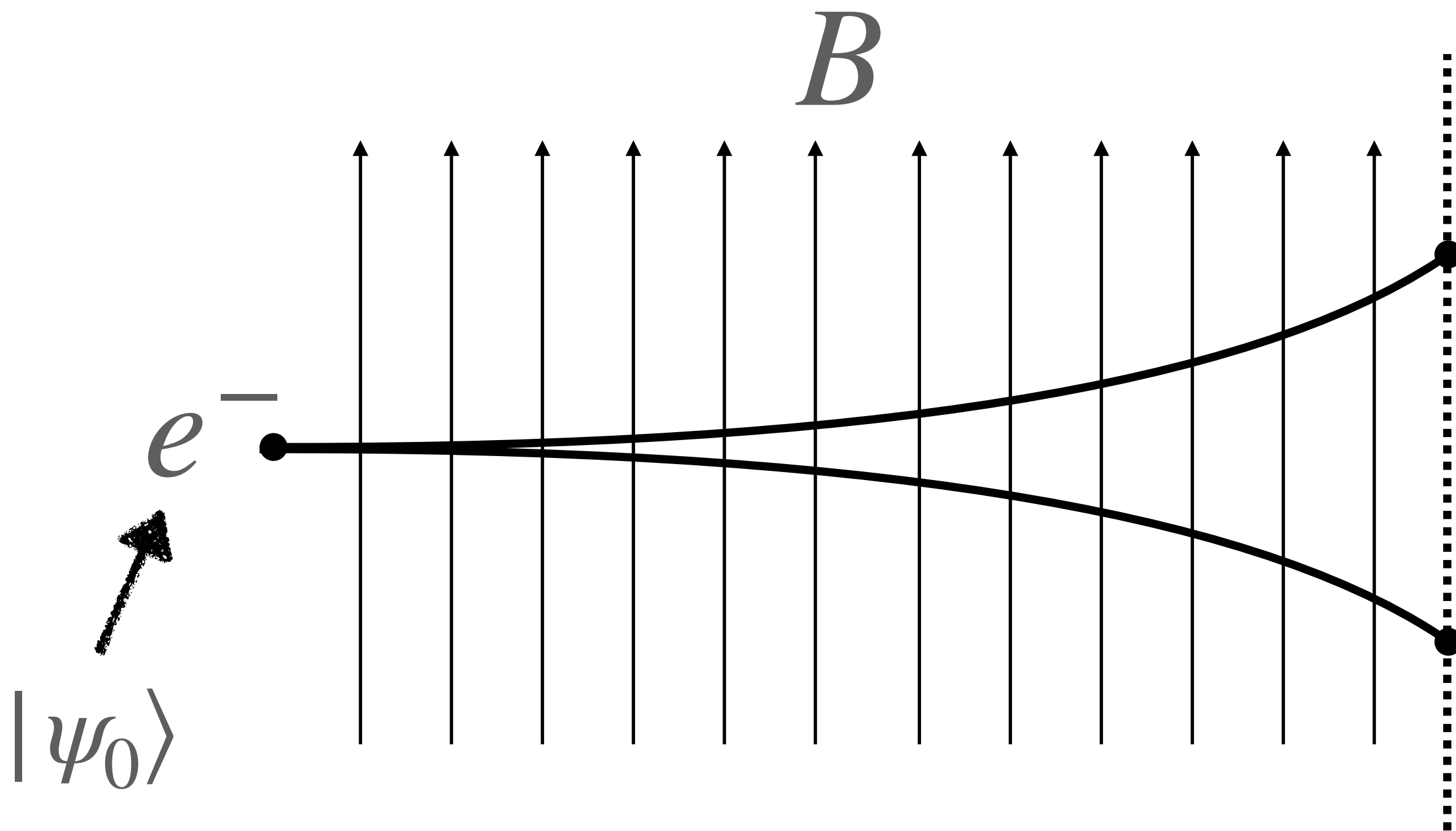
Why?

Probability is the most important concept in modern science, especially as nobody has the slightest notion what it means.

—Bertrand Russell

Why?

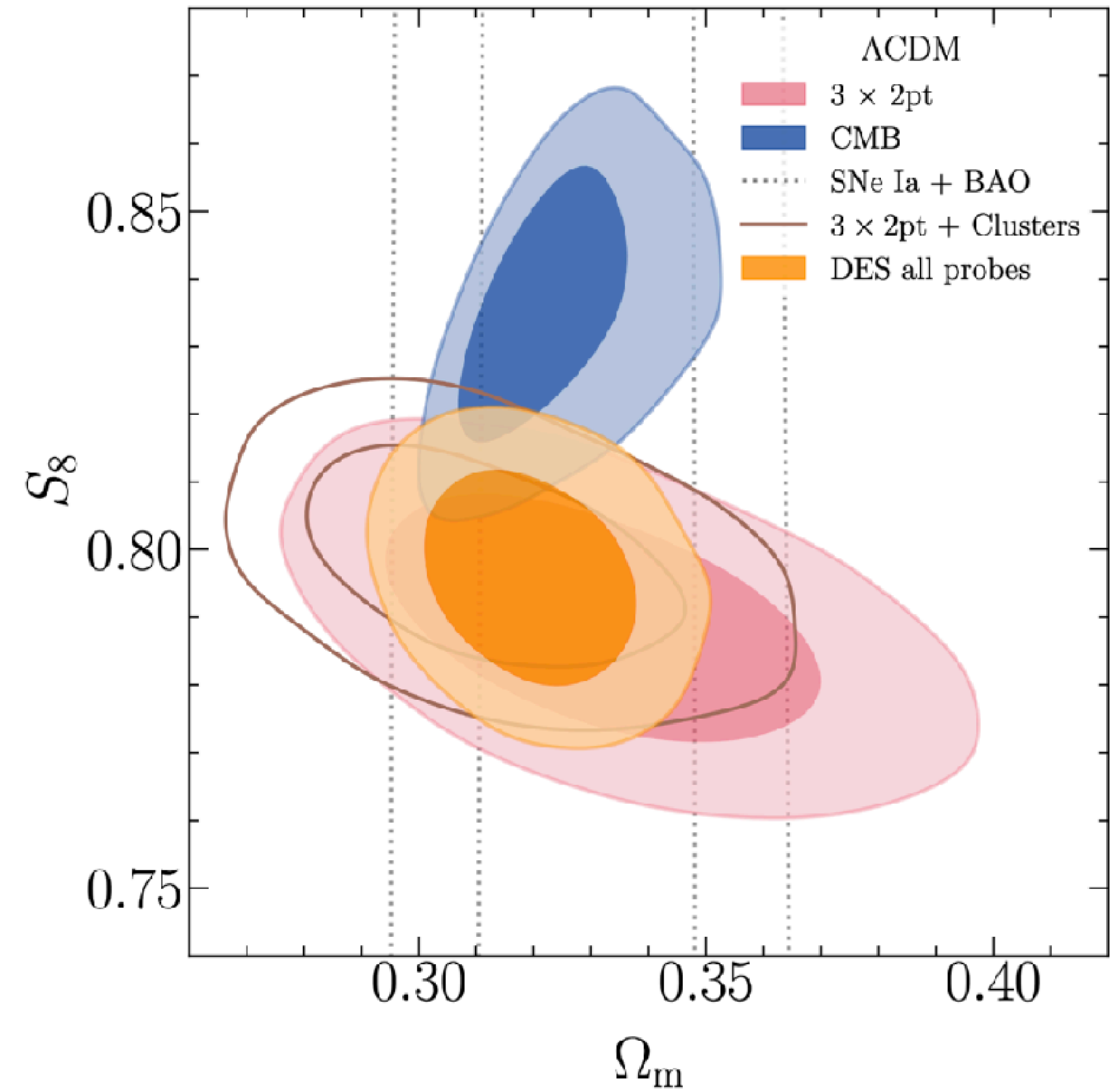
- Probability is at the core of physics.
- QM does not explain why the coefficients of $|\psi_0\rangle$ act as probabilities.



$$|\psi_0\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle + |\downarrow\rangle)$$

Why?

- Comparing theory to data is what provides insights upon fundamental physics.
- Modern physics concerns vast many-body systems that incorporate physics on many scales.

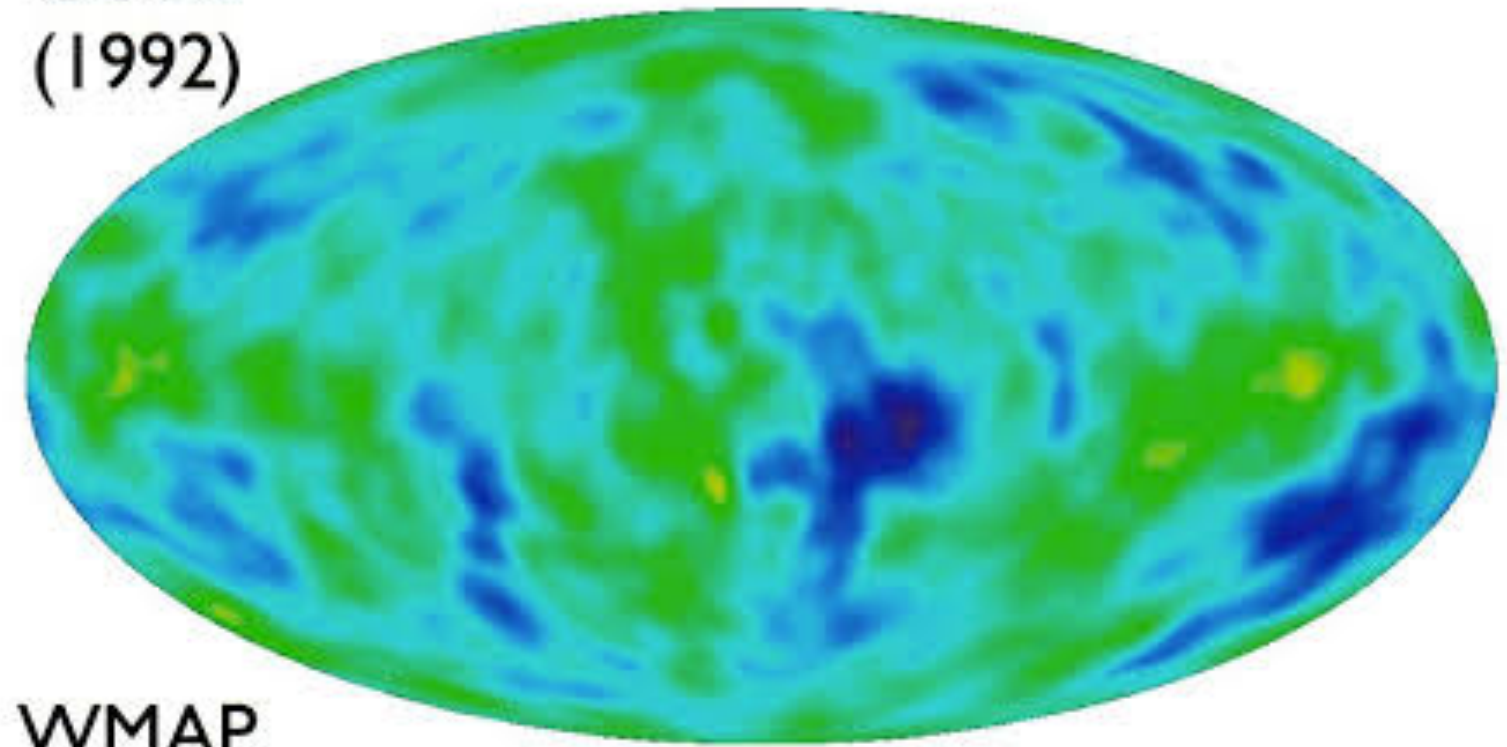


Dark Energy Survey 2026

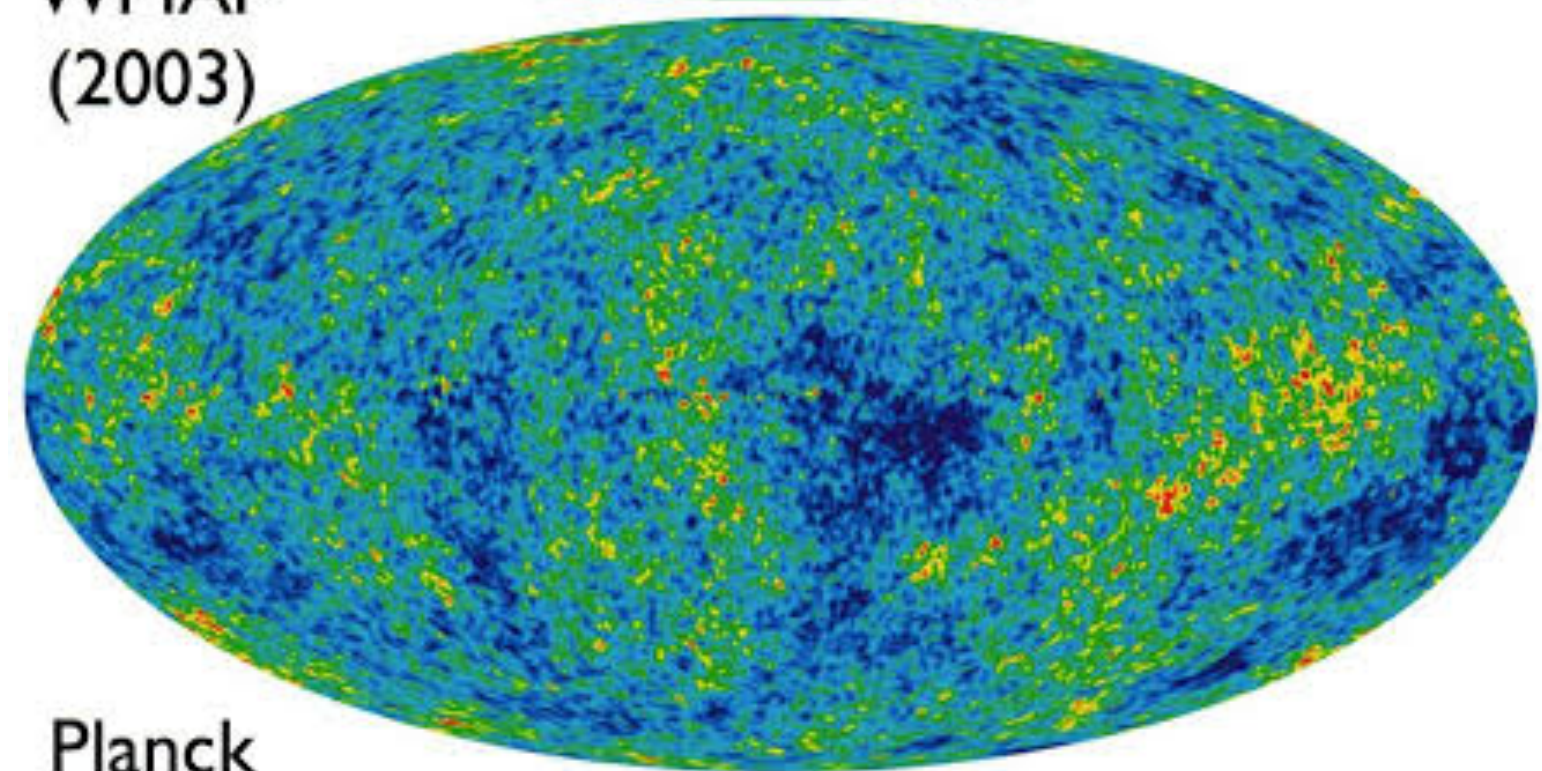
Why?

- Theory and data will always increase in complexity and abundance.
- Erroneously analysing this data wastes resources, erodes trust in science and leads the field astray.

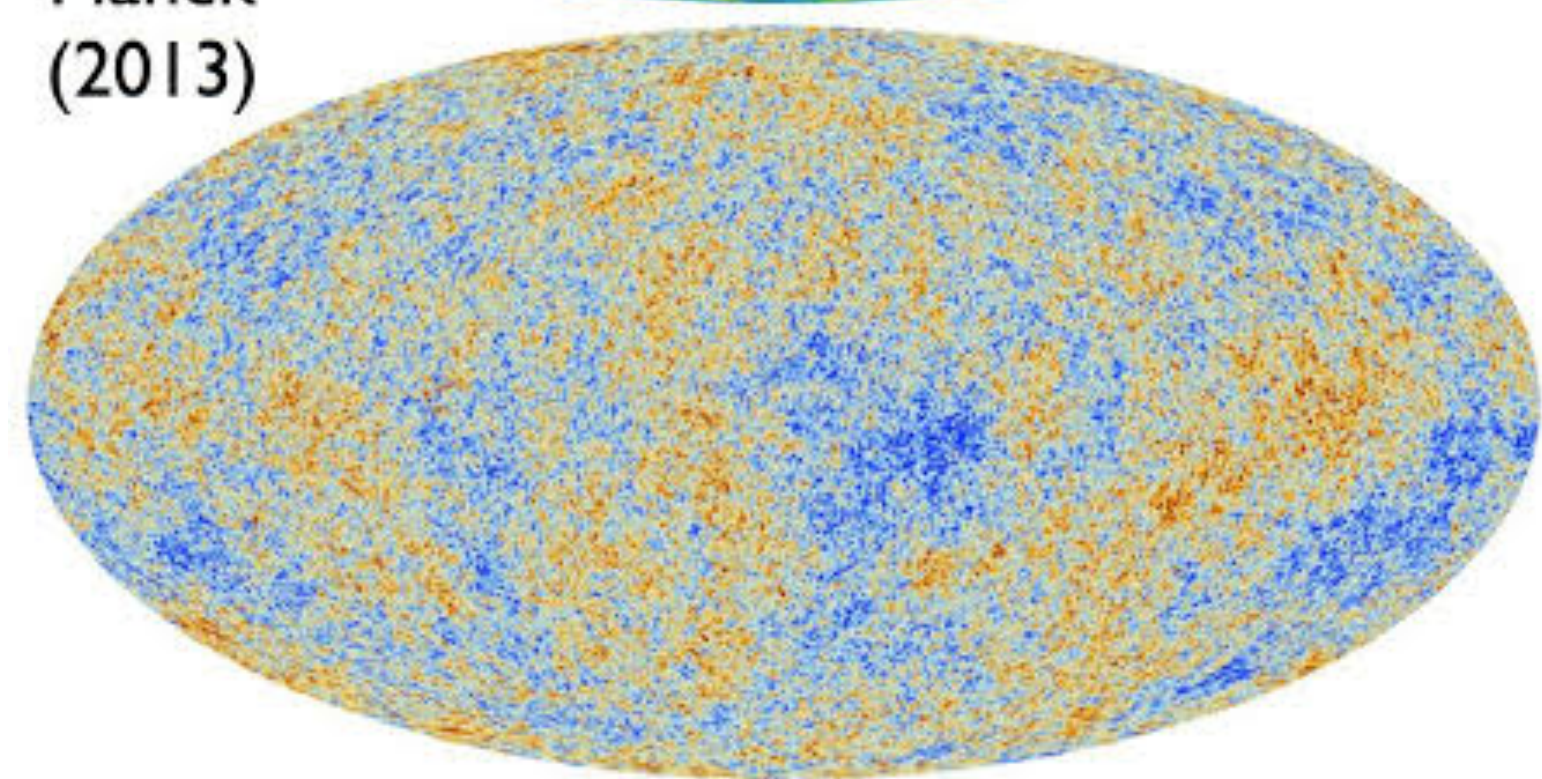
COBE
(1992)



WMAP
(2003)



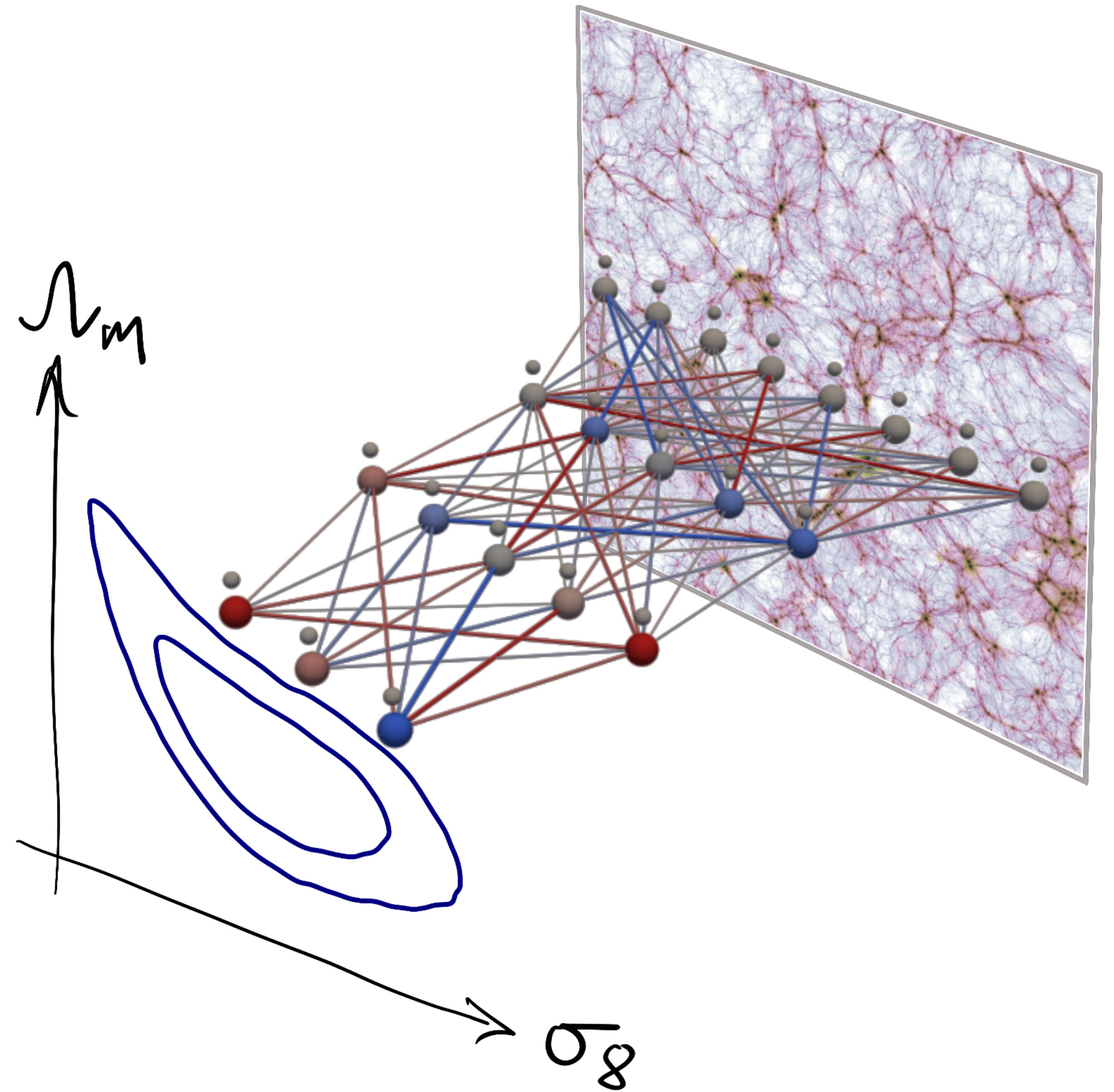
Planck
(2013)



The Cosmic Microwave Background as
probed by COBE, WMAP and Planck

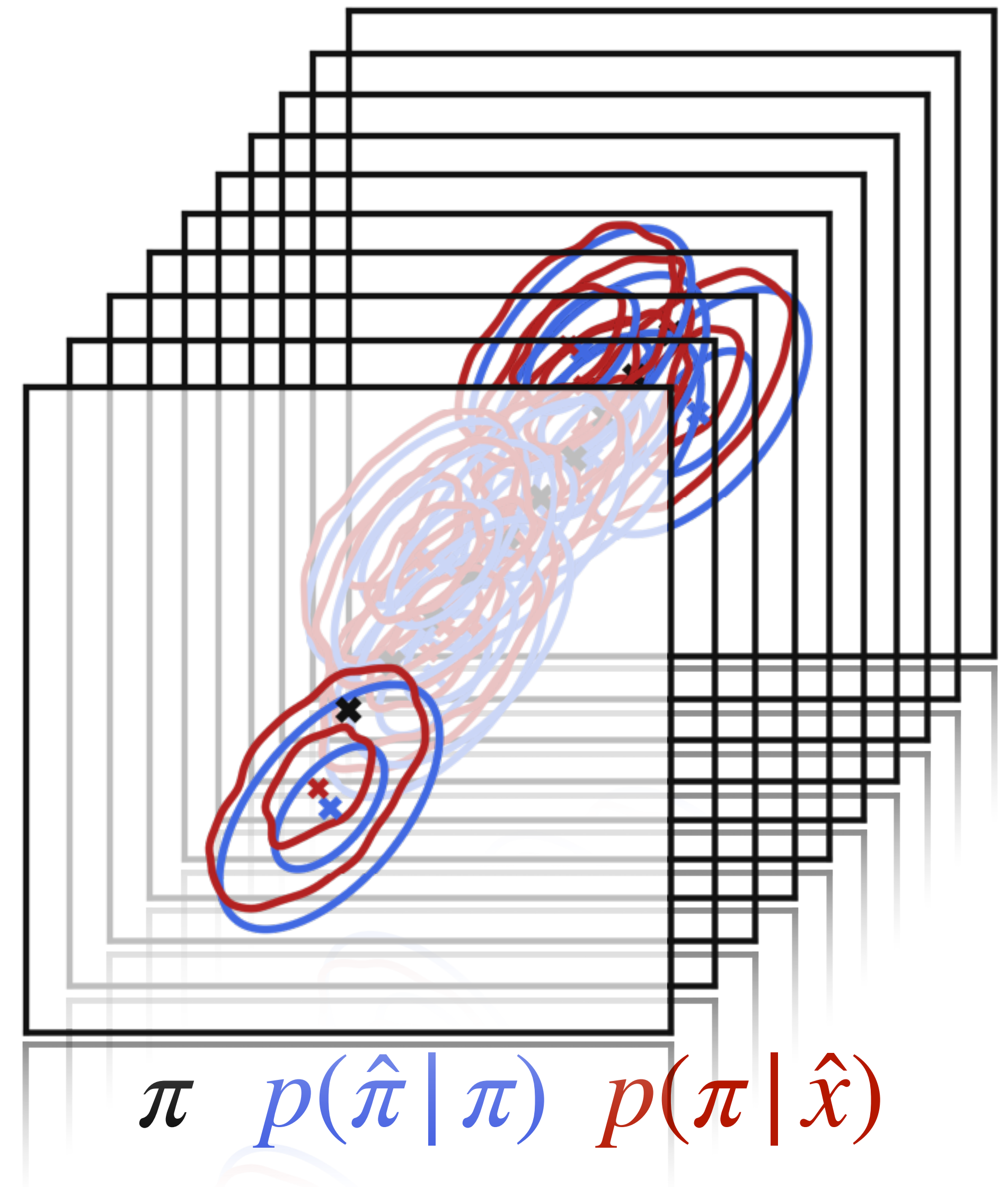
Why?

- Machine learning extends standard statistical techniques beyond theory-based analysis of complex data.
- Machine learning analyses can only be diagnosed *statistically*, by construction.



What is probability?

Jed Homer LMU/ODSL



What?

- Understand what random variables and their distributions are
- Understand basic properties we require of probability distributions
- PMFs, PDFs & CDFs
- Understand how to manipulate probability towards inference

Random variables

- $X : \Omega \rightarrow \mathbb{R}$ (Ω is a set of possible outcomes)
- *Convention:* X is a random variable, x is an observed value.

Random variables

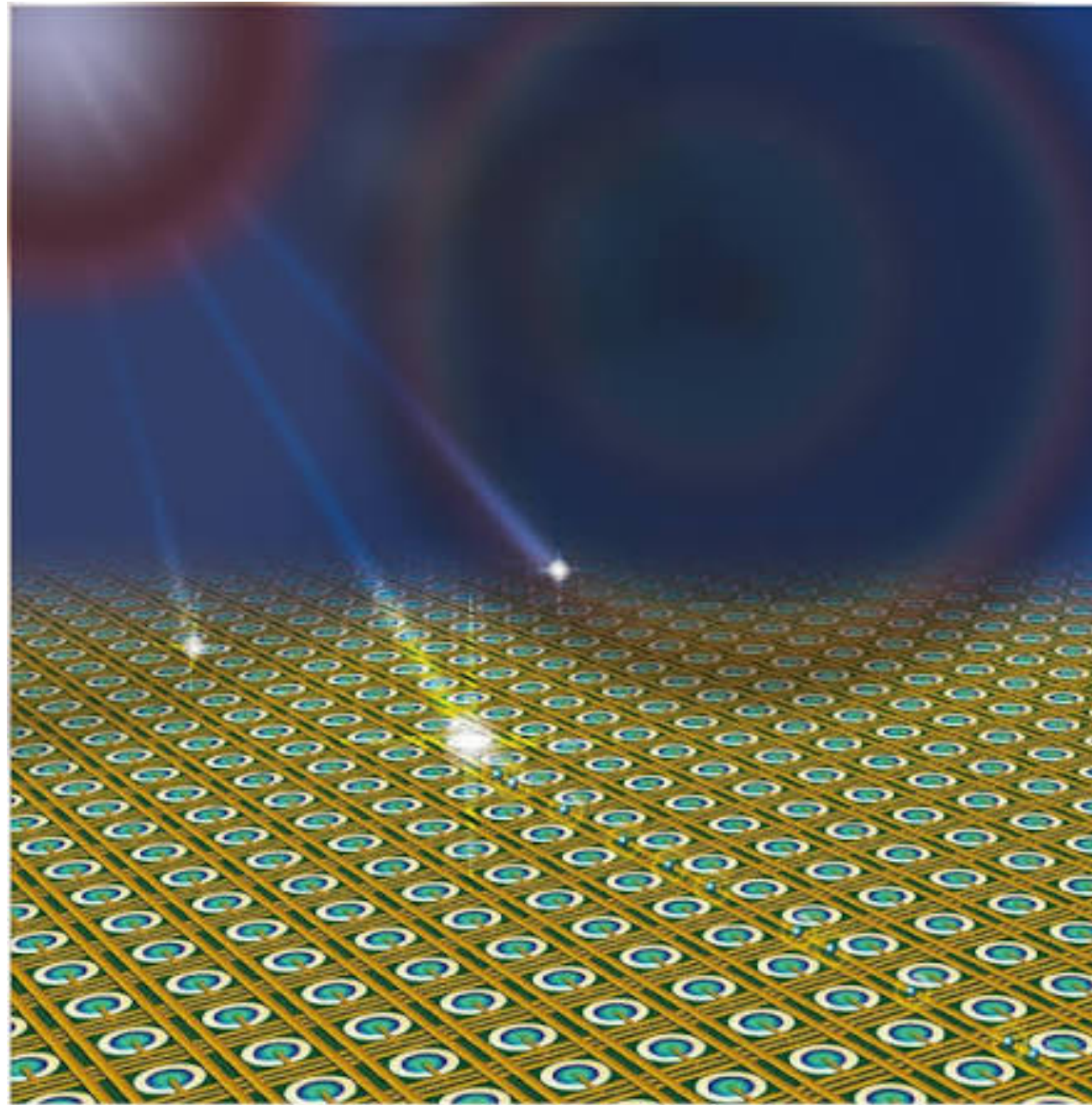
- $X : \Omega \rightarrow \mathbb{R}$ (Ω is a set of possible outcomes)
- E.g. if X is the number of heads observed in two coin flips

$$X(HH) = 2, \quad X(HT) = X(TH) = 1, \quad X(TT) = 0.$$

Random variables

Discrete random variables

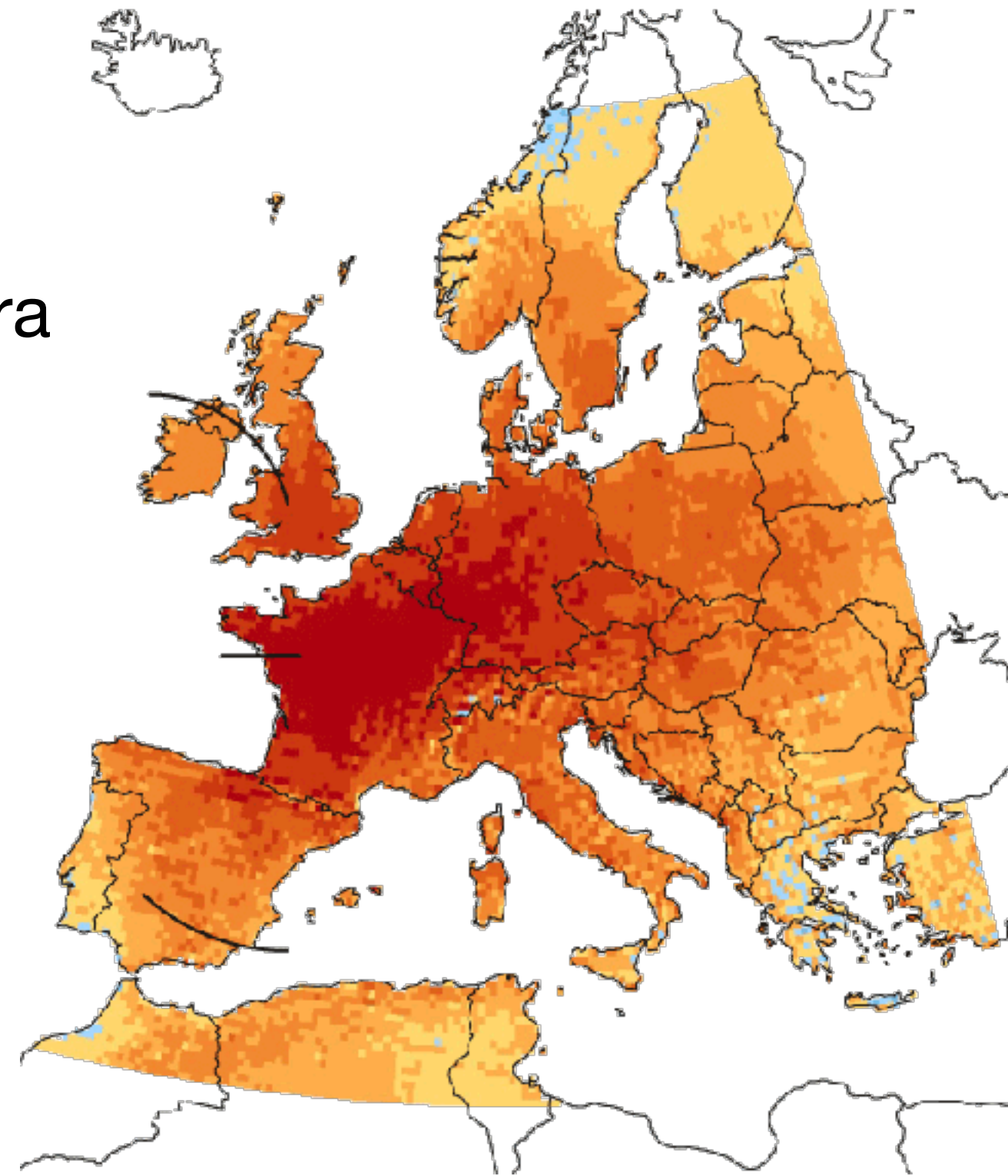
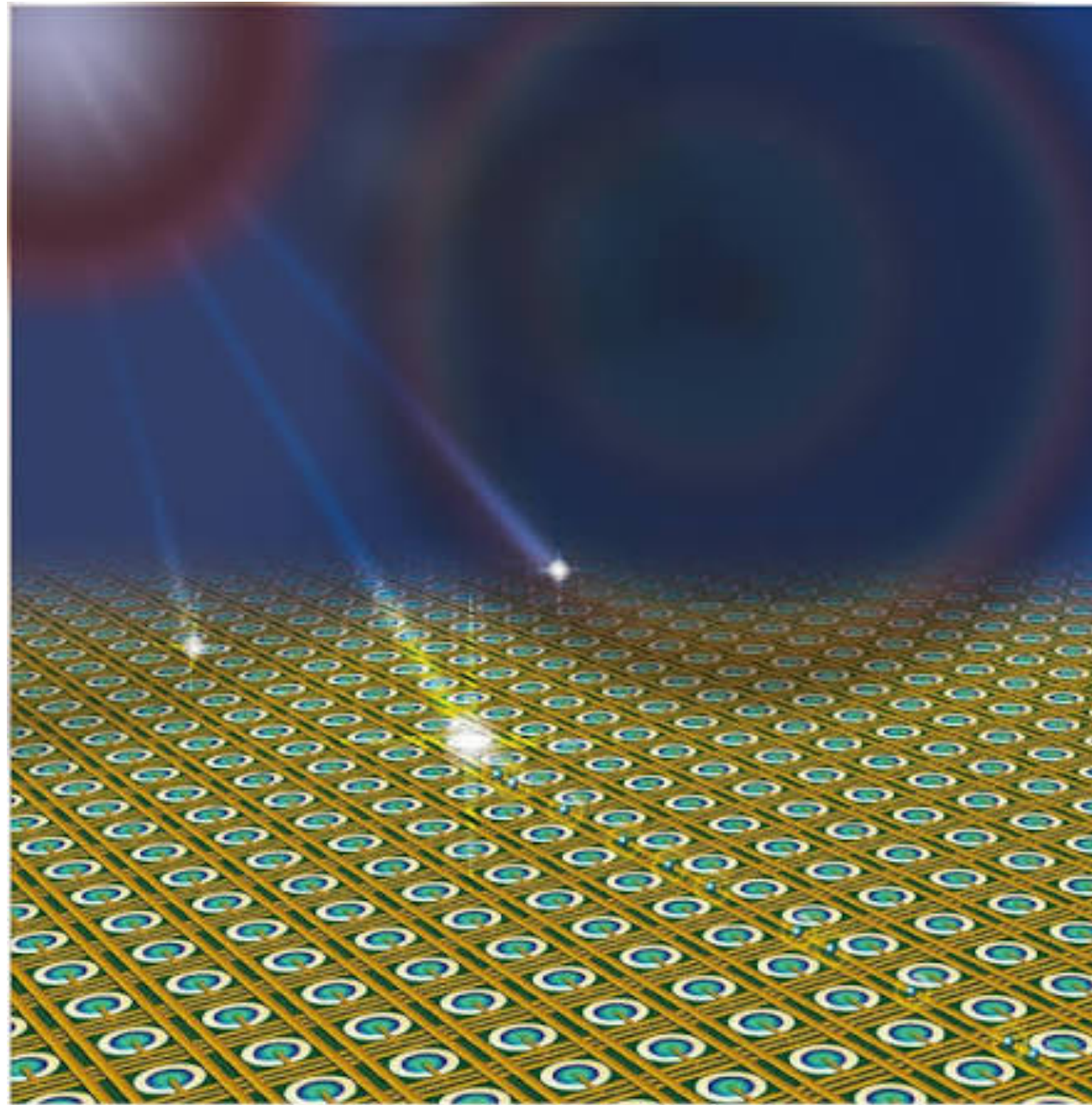
- e.g. Dice rolls, photon counts on a camera



Random variables

Discrete random variables

- e.g. Dice rolls, photon counts on a camera



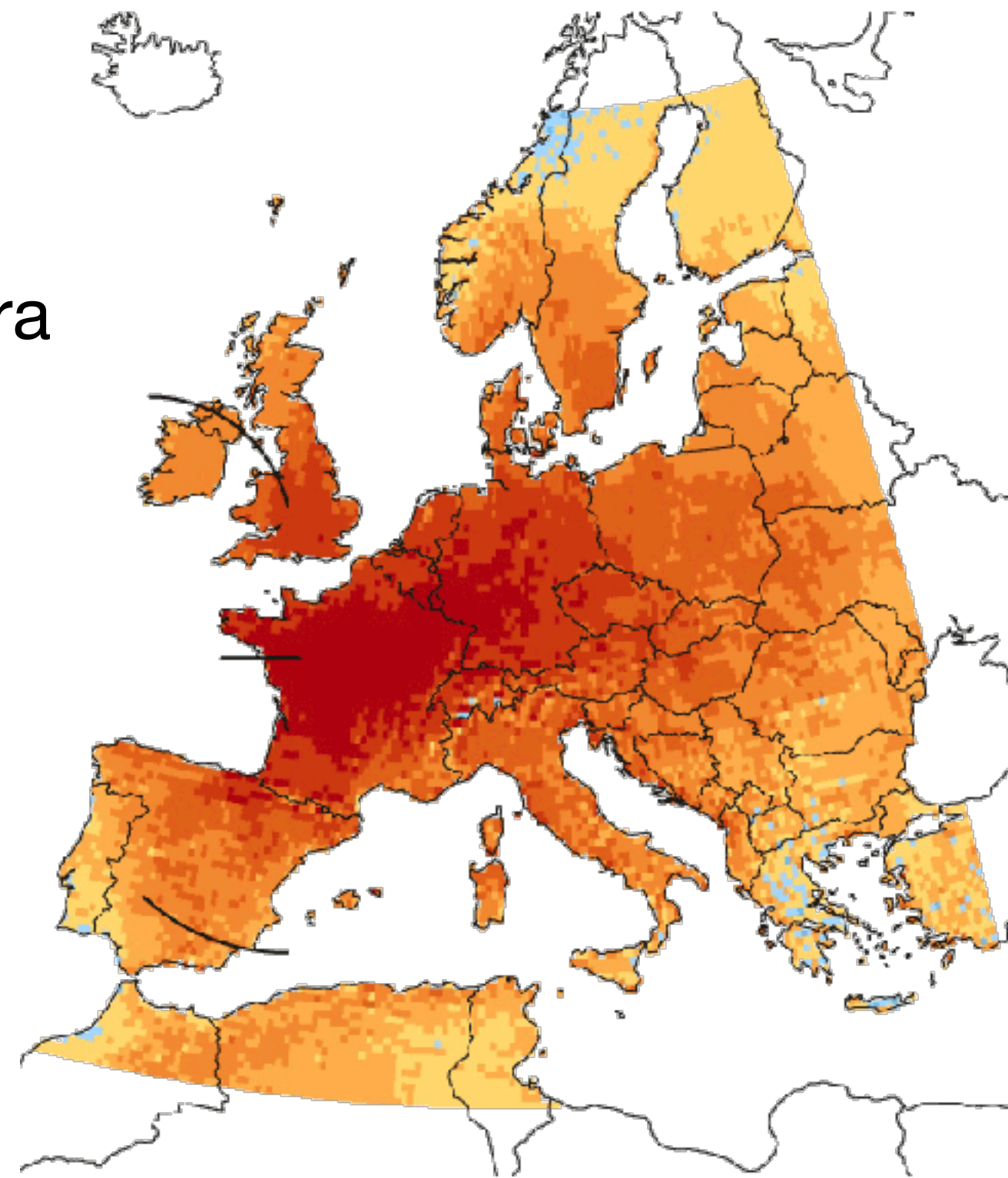
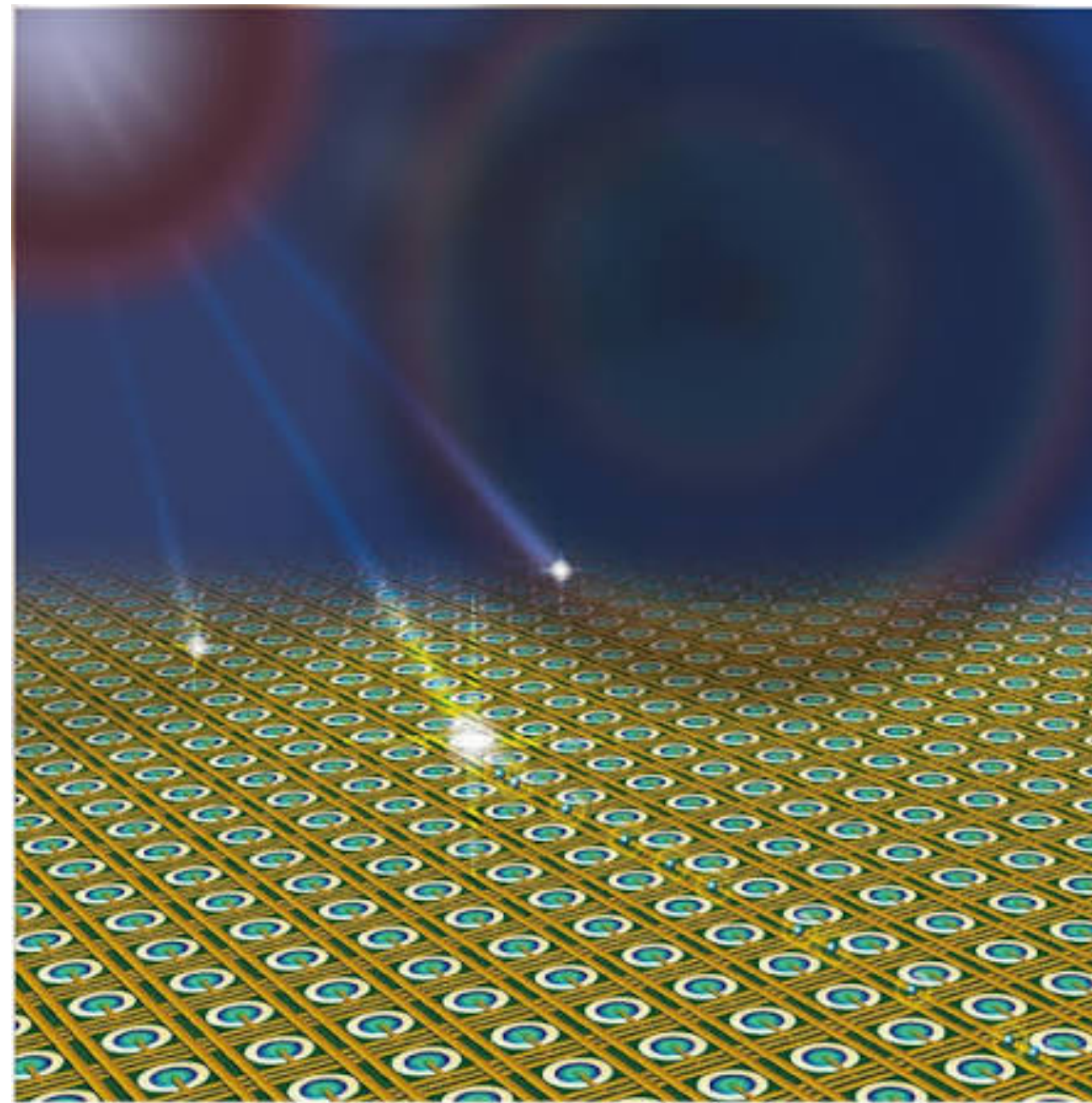
Continuous random variables

- e.g. the hottest temperature in Europe

Random variables

Discrete random variables

- e.g. Dice rolls, photon counts on a camera



Continuous random variables

- e.g. the hottest temperature in Europe

Multivariate random variables

- e.g. the temperature on Earth at N points



How do we define probability?

- A deeper question is "how do we define probability?"

How do we define probability?

- Let's say we flip a coin N times, where we have $\Omega = \{H, T\}$.



How do we define probability?

- Let's say we flip a coin N times, where we have $\Omega = \{H, T\}$.
- It is normal to say something like...

"As $N \rightarrow \infty$, $\frac{H}{N}$ will fall within $0.5 \pm \epsilon$ with probability $P \geq 1 - \delta$."

ϵ, δ are small numbers



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Frequentist interpretation of probability



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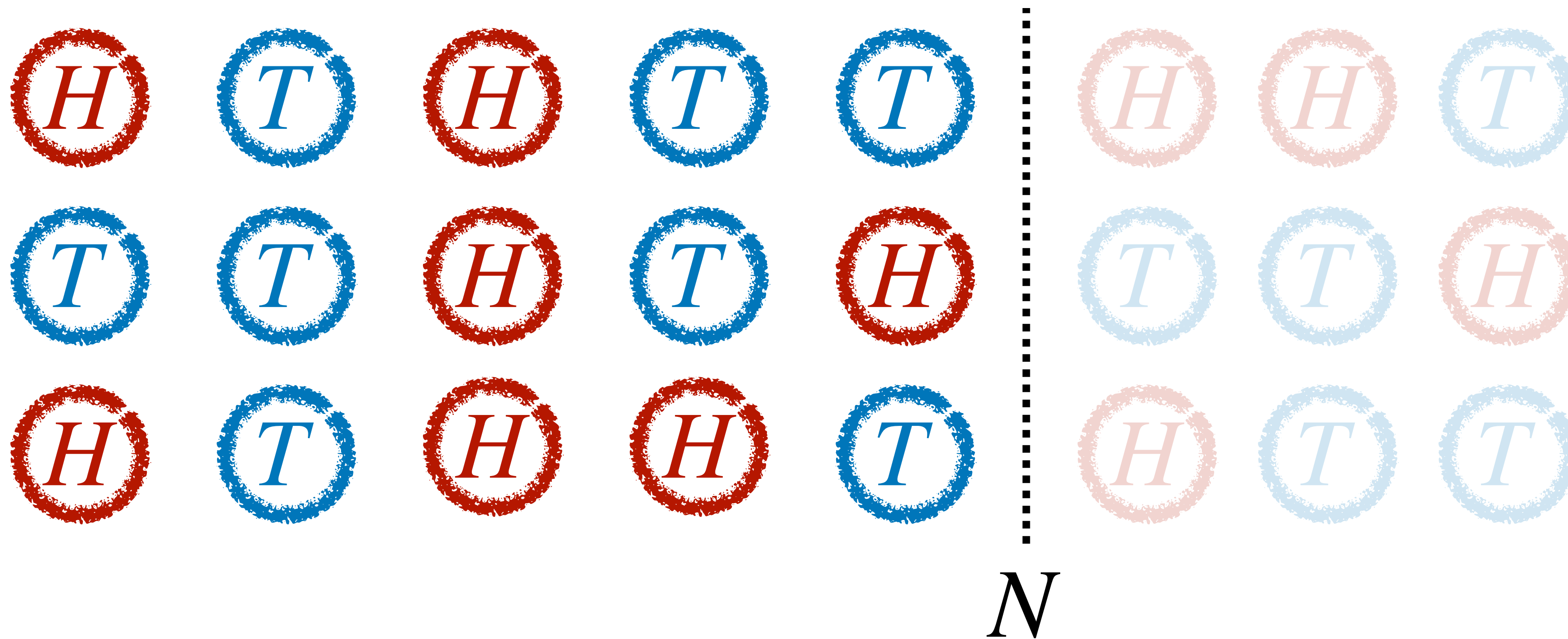
ϵ, δ are small numbers

Frequentist interpretation of probability



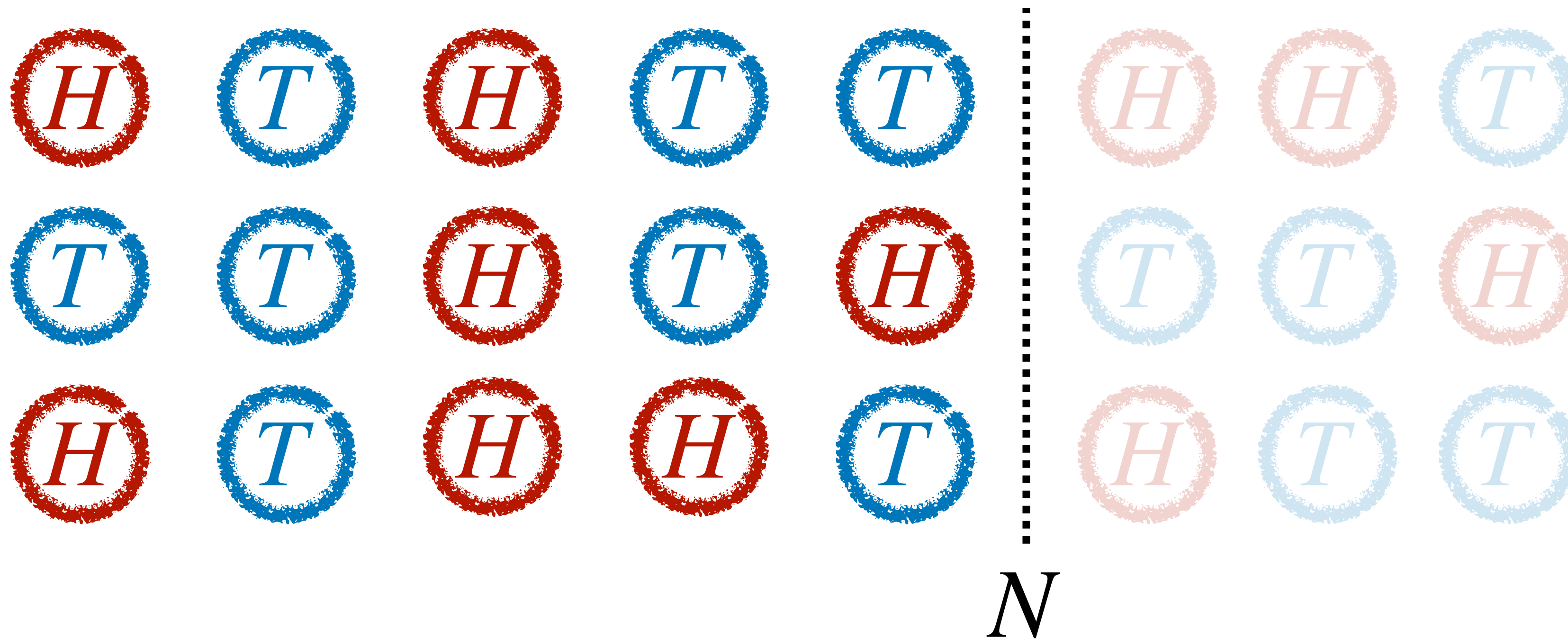
How do we define probability?

- What is wrong with this statement?



How do we define probability?

- What is wrong with this statement?



- Why does each *finite* sequence give a different probability?

- Are these experiments *really identical*?

Kolmogorov Axioms

- Assumptions for manipulating probability between random variables.
- **Require:**
 - Space of outcomes Ω ,
 - Event $E \in \Omega$,
 - Probability measure of E , denoted $P(E)$.

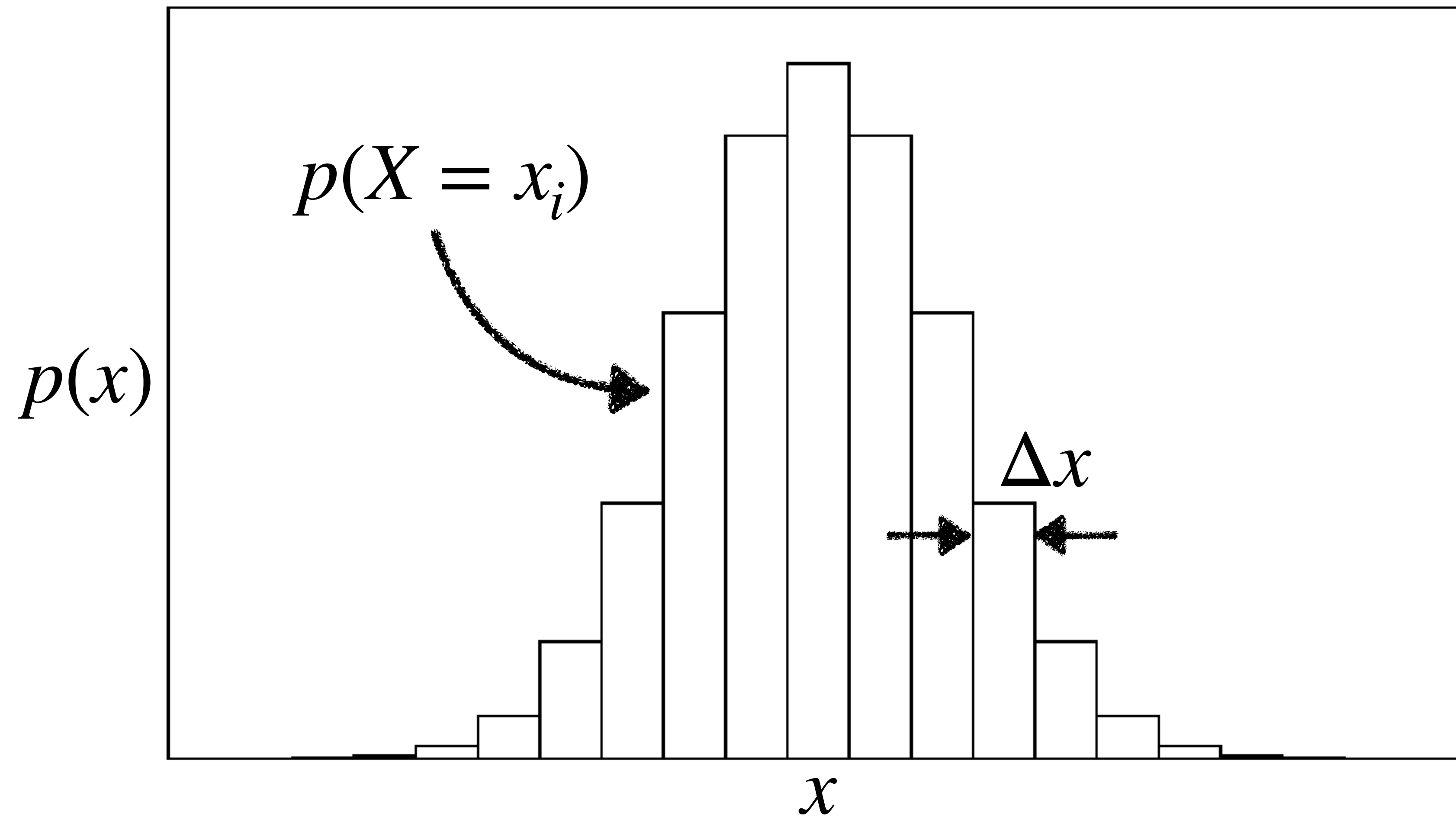
Kolmogorov Axioms

1. **Positivity:** $P(E) \geq 0$
2. **Normalisation:** $P(\Omega) = 1$
3. **Additivity** under exclusive A & B : $P(A \cup B) = P(A) + P(B)$

Probability Mass Functions

- Let X take values in a finite set, so $p(x) = P(X = x)$.
- Then $P(X \in A) = \sum_{x \in A} p(x)$.

Probability Mass Functions



Probability Density Functions

- **Continuous random variable** X has $P(X = x) = 0$ for all x .
- Probabilities are non-zero only for **intervals** e.g. $P(a \leq X \leq B) \geq 0$.

$$P(x \leq X \leq x + \Delta x) \approx p(x)\Delta x$$

Probability Density Functions

- **Continuous random variable** X has $P(X = x) = 0$ for all x

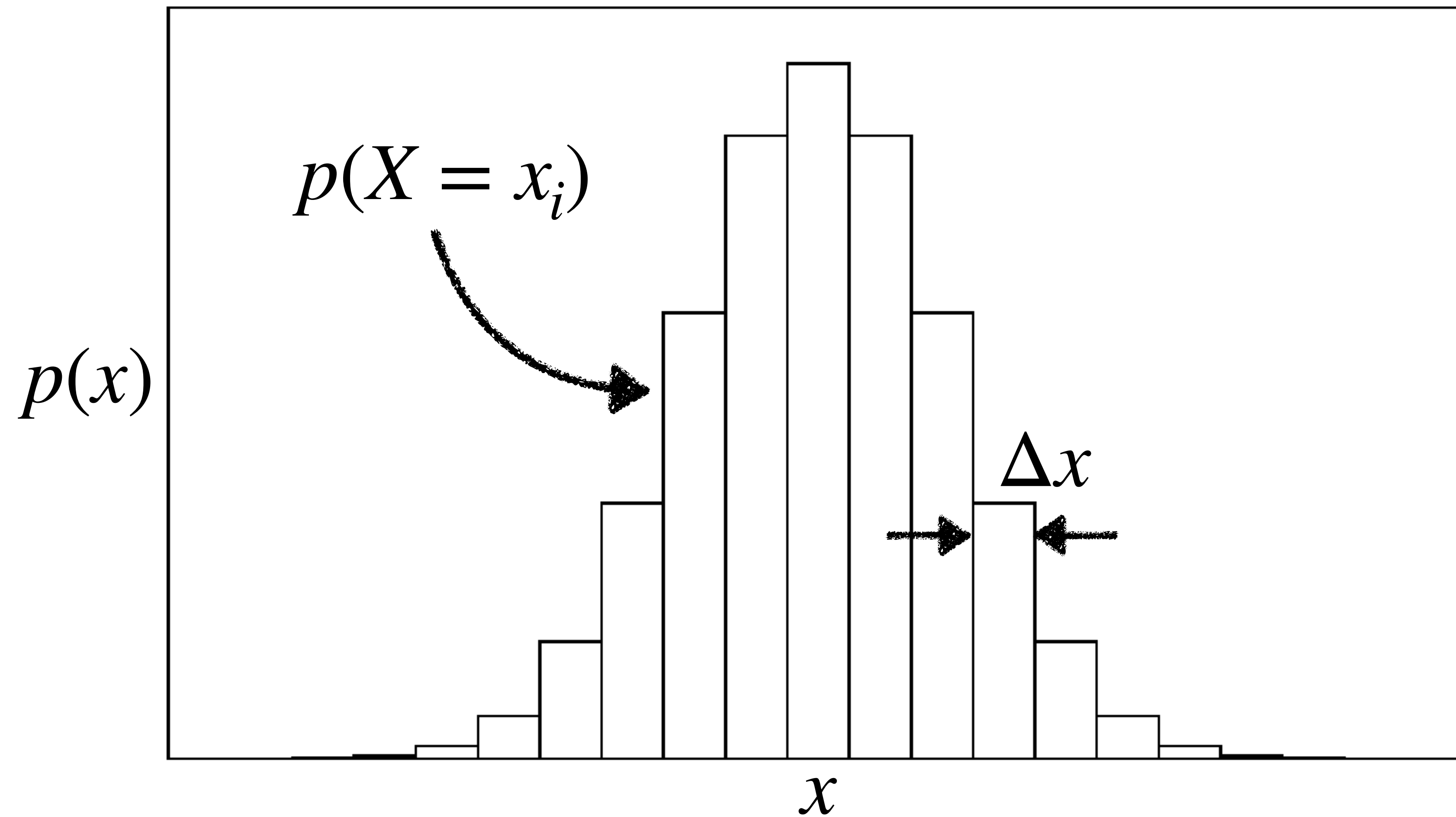
$$p(x) = \lim_{\Delta x \rightarrow 0} \frac{P(x \leq X \leq x + \Delta x)}{\Delta x}$$

Probability Density Functions

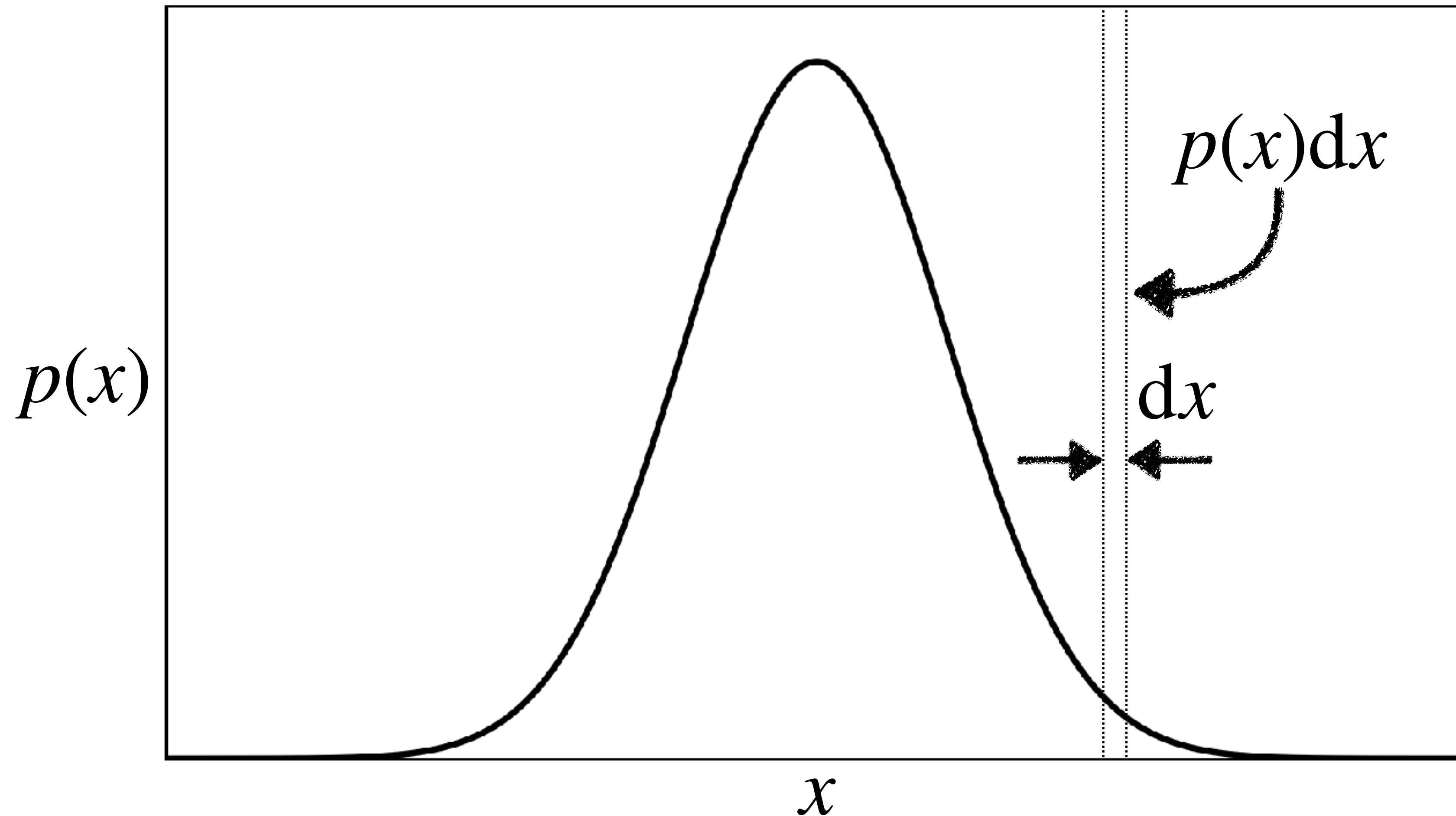
- **Continuous random variable** X has $P(X = x) = 0$ for all x

$$P(a \leq X \leq b) = \int_a^b dx p(x)$$

Probability Mass Functions

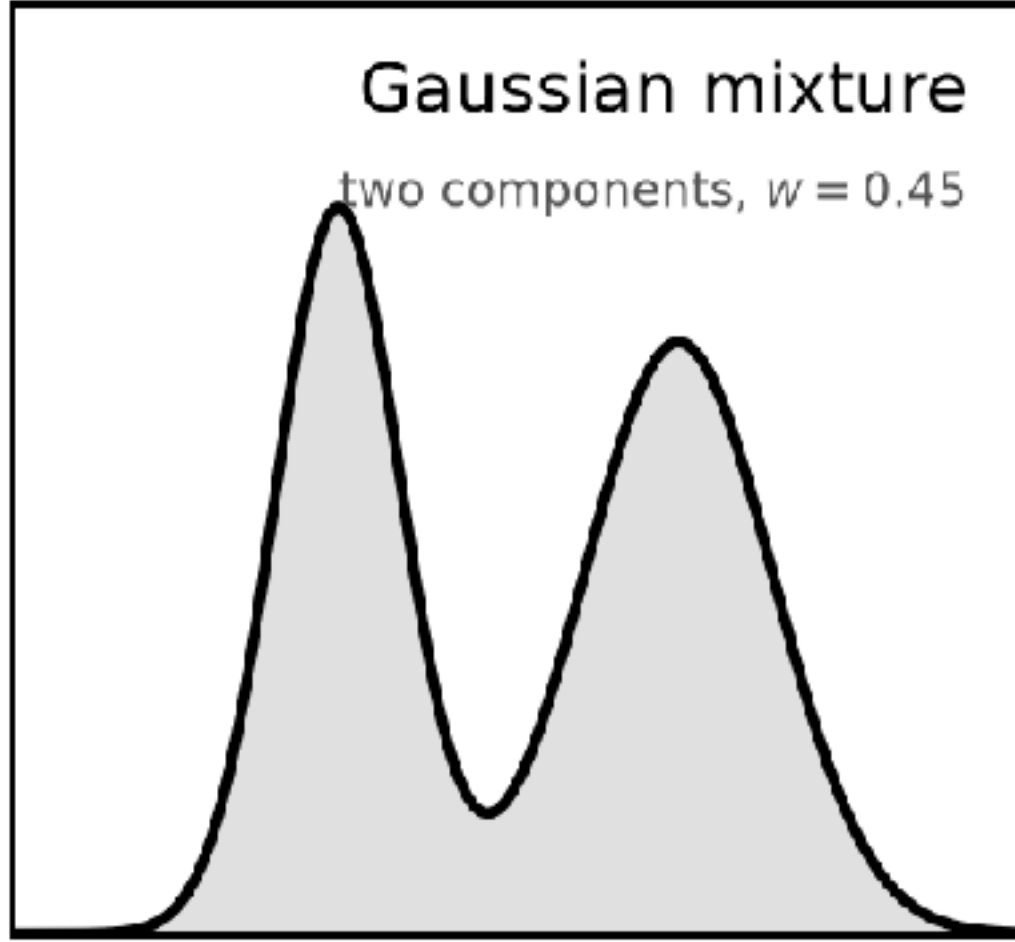
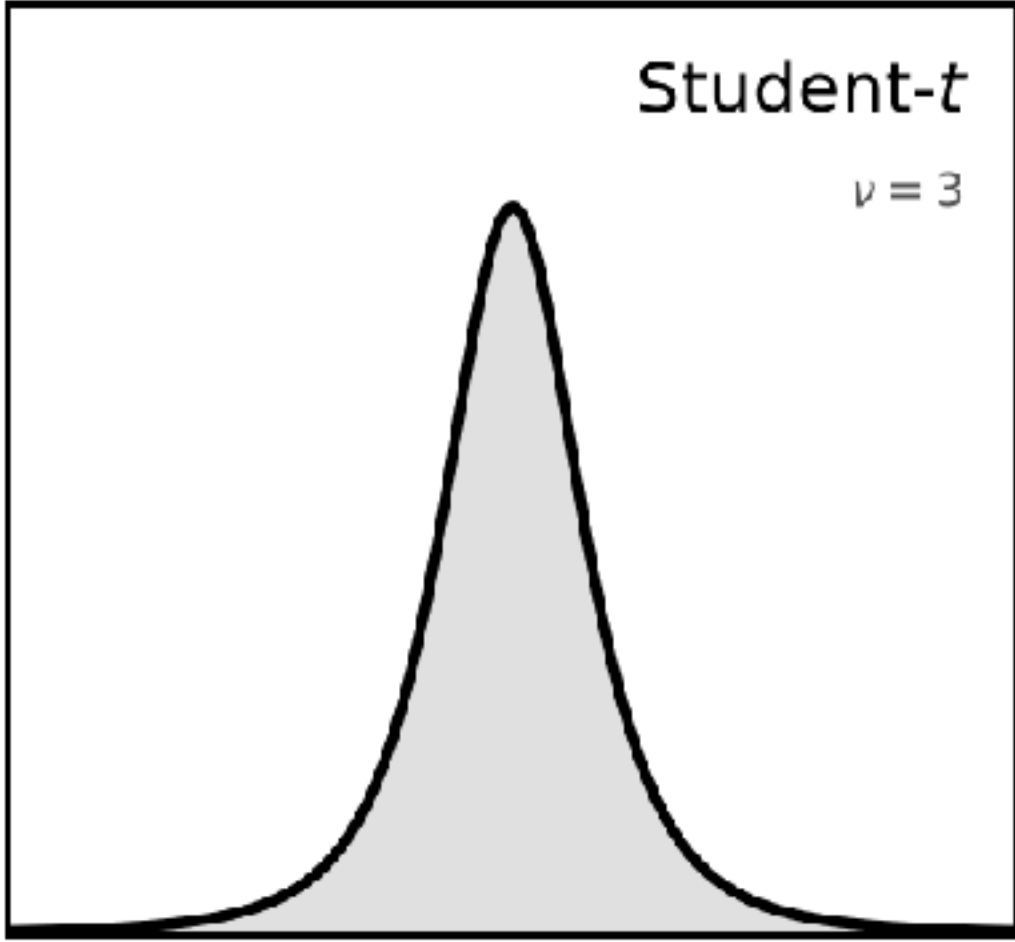
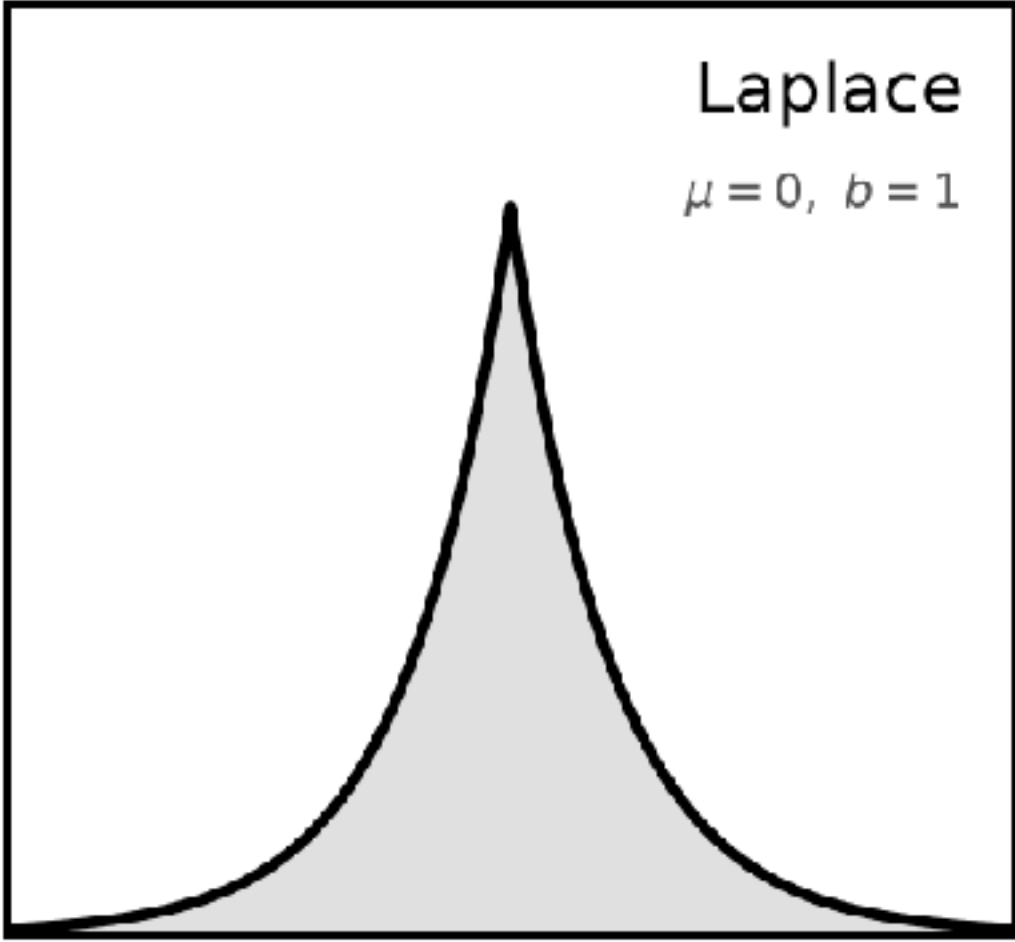
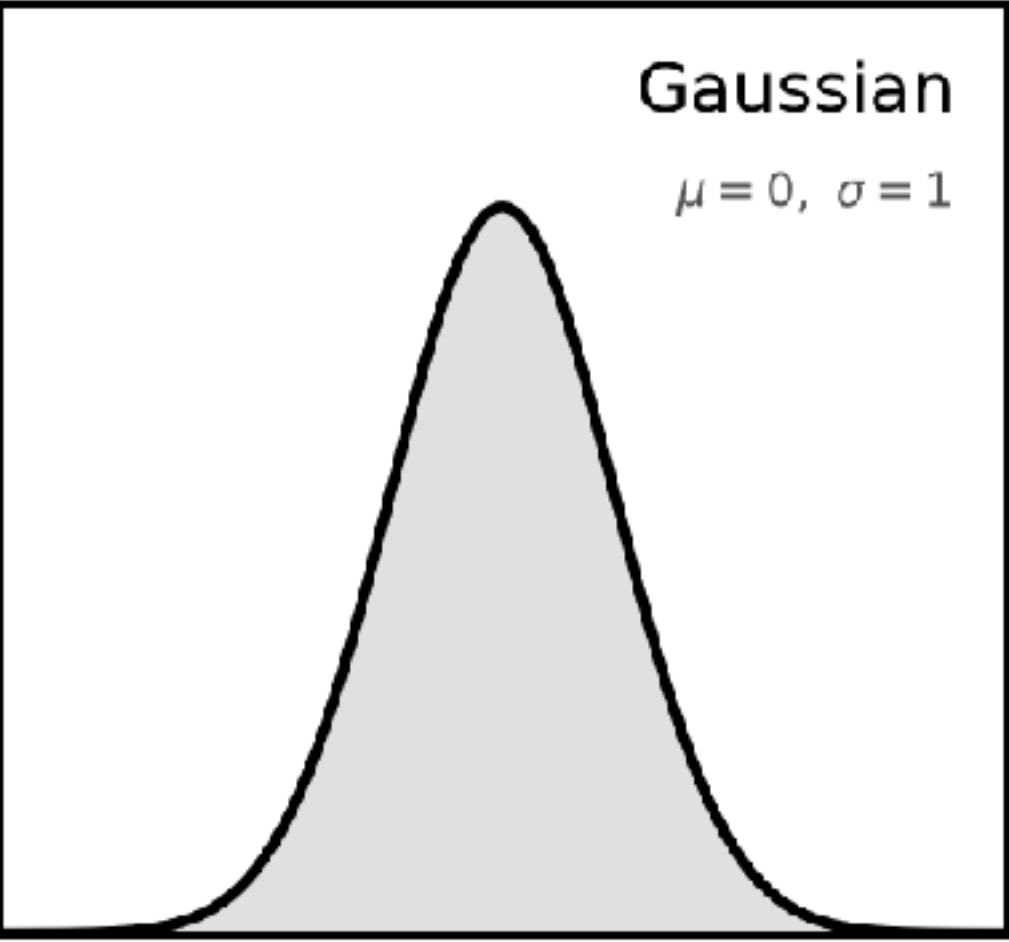


Probability Density Functions

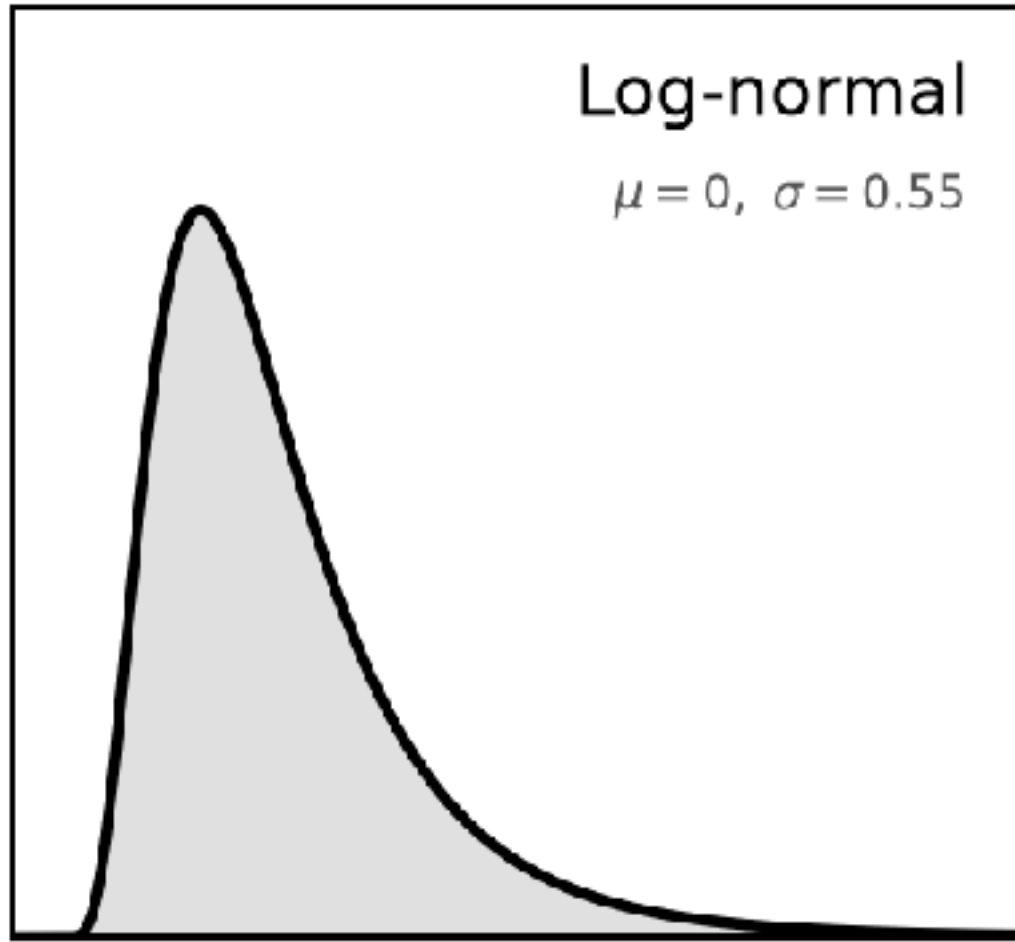
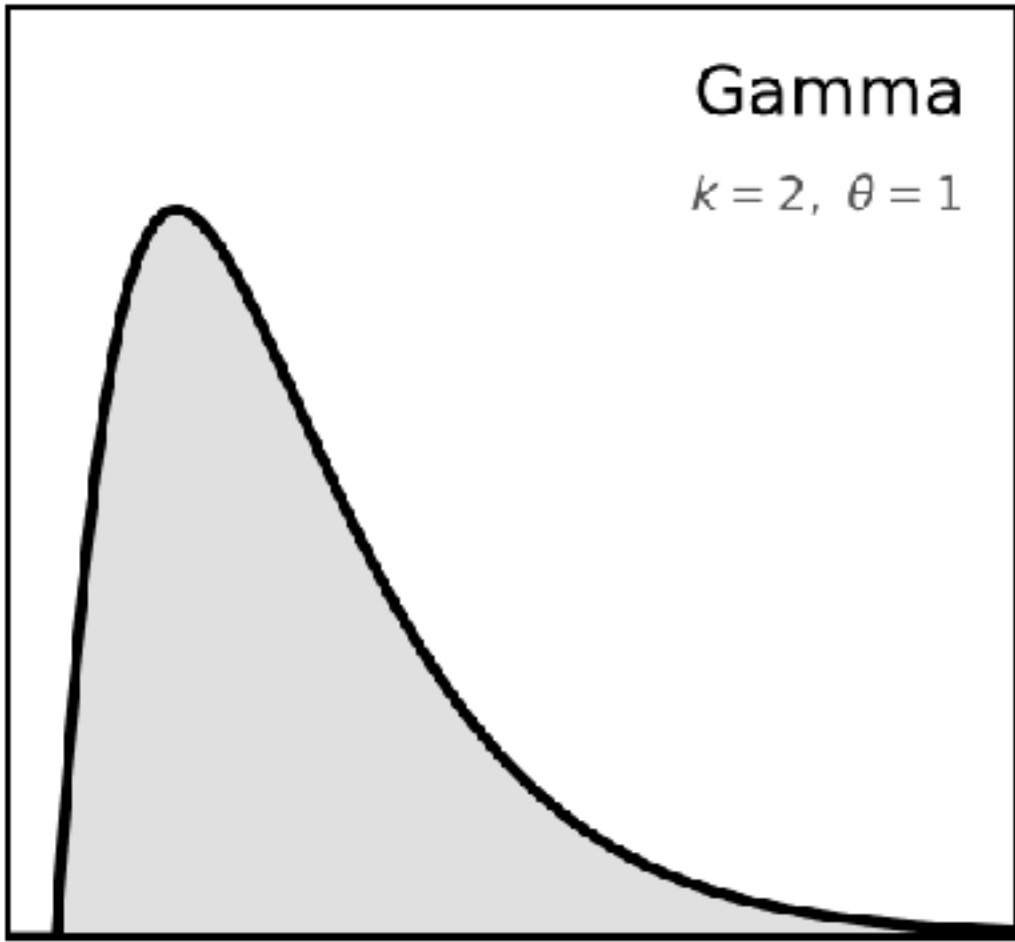
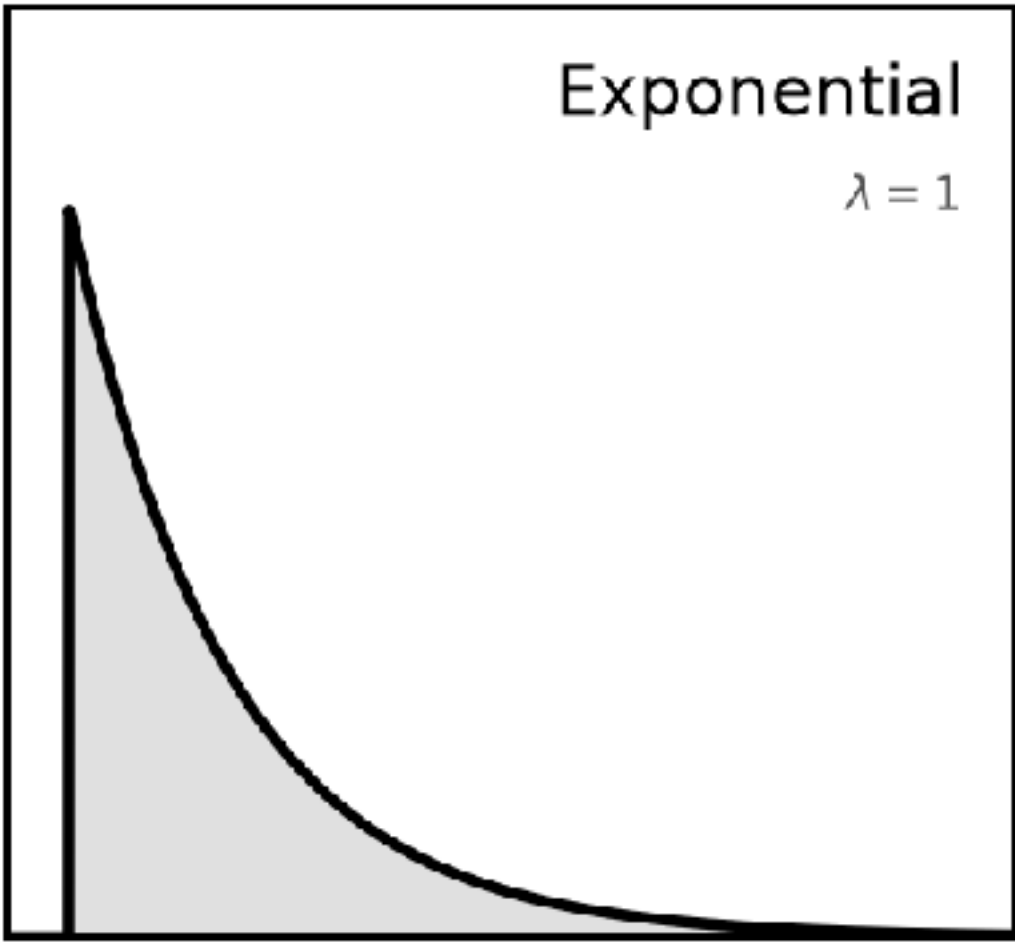
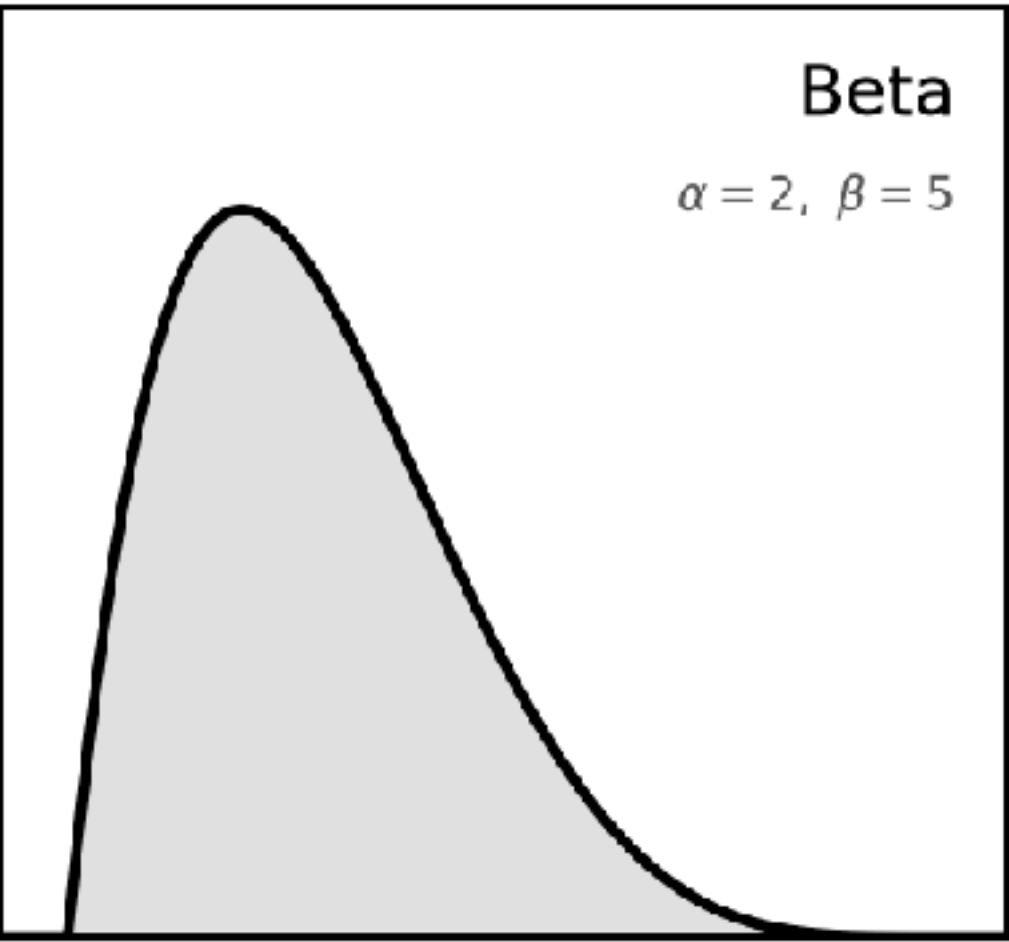


Probability Density Functions

$p(x)$



$p(x)$



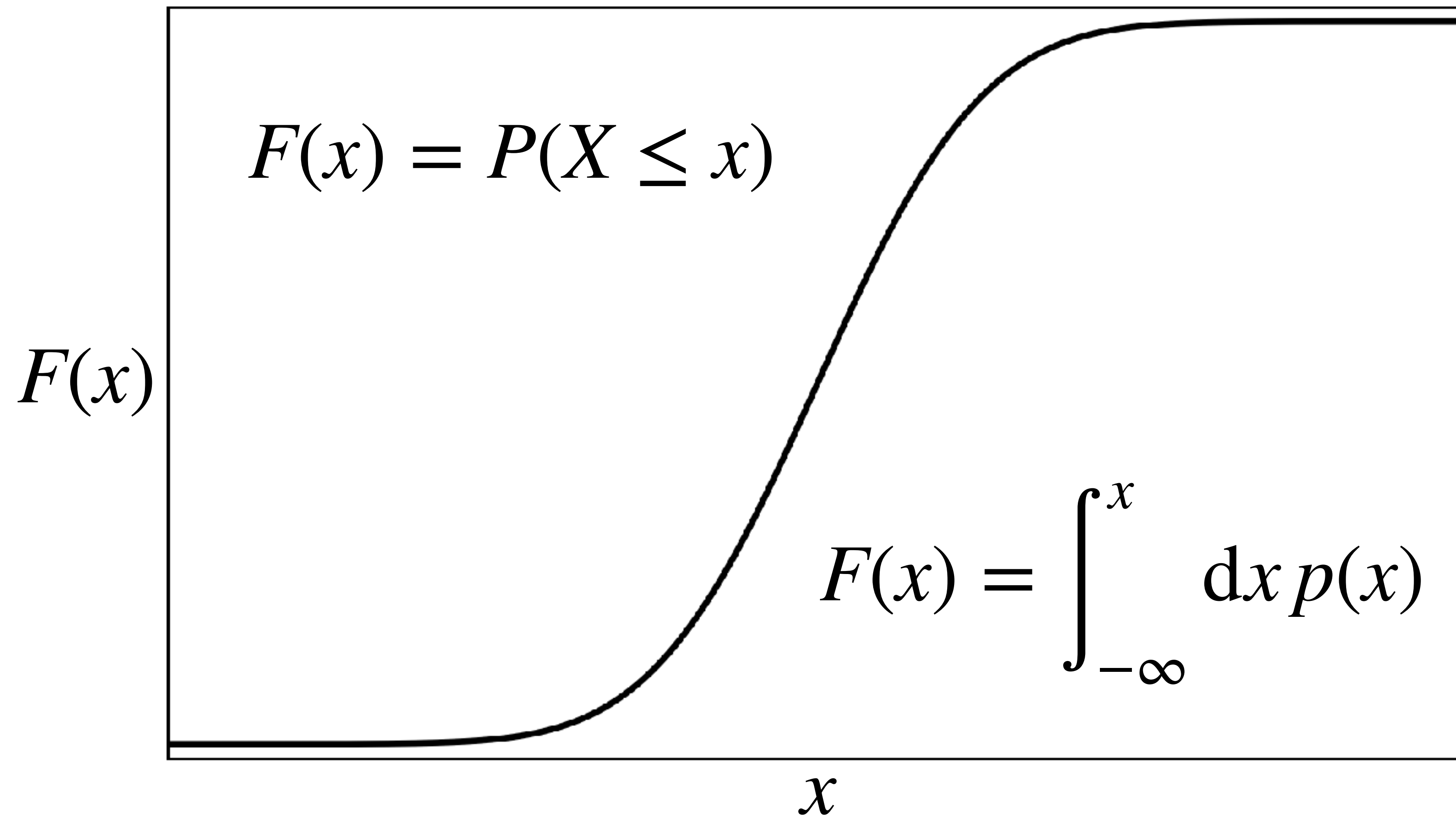
x

x

x

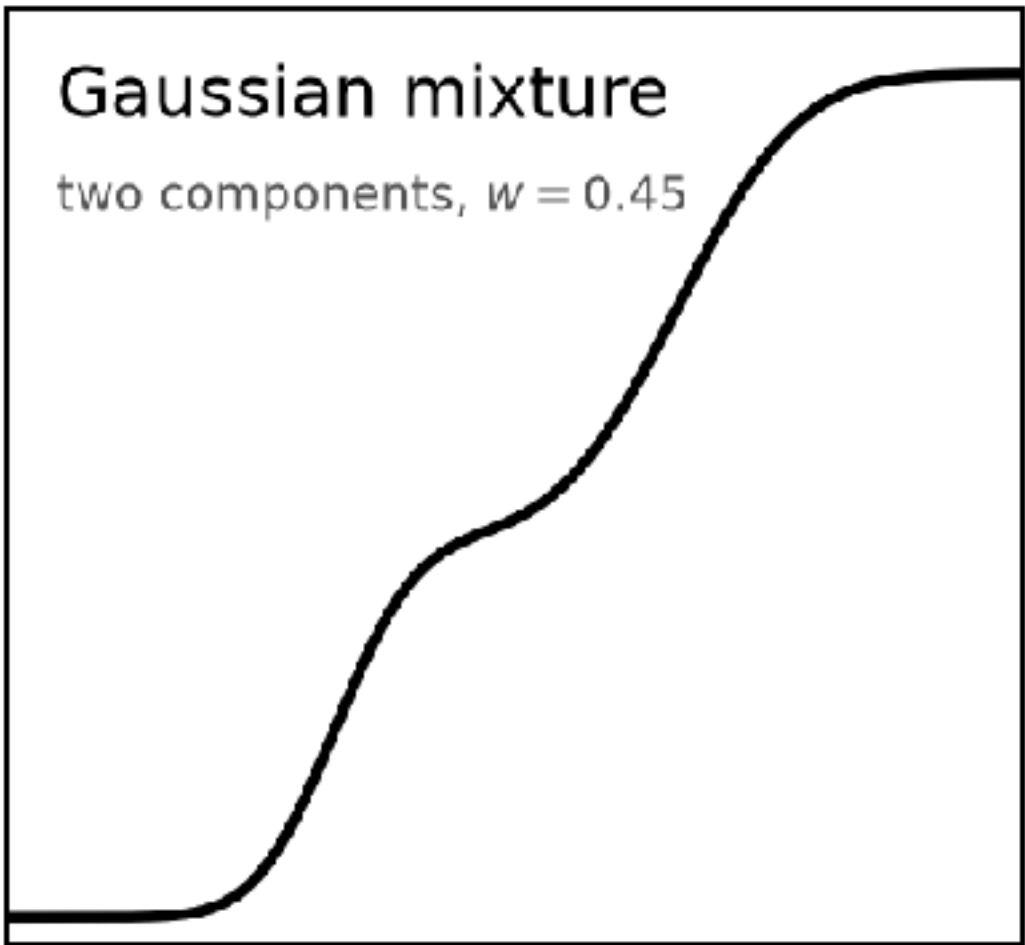
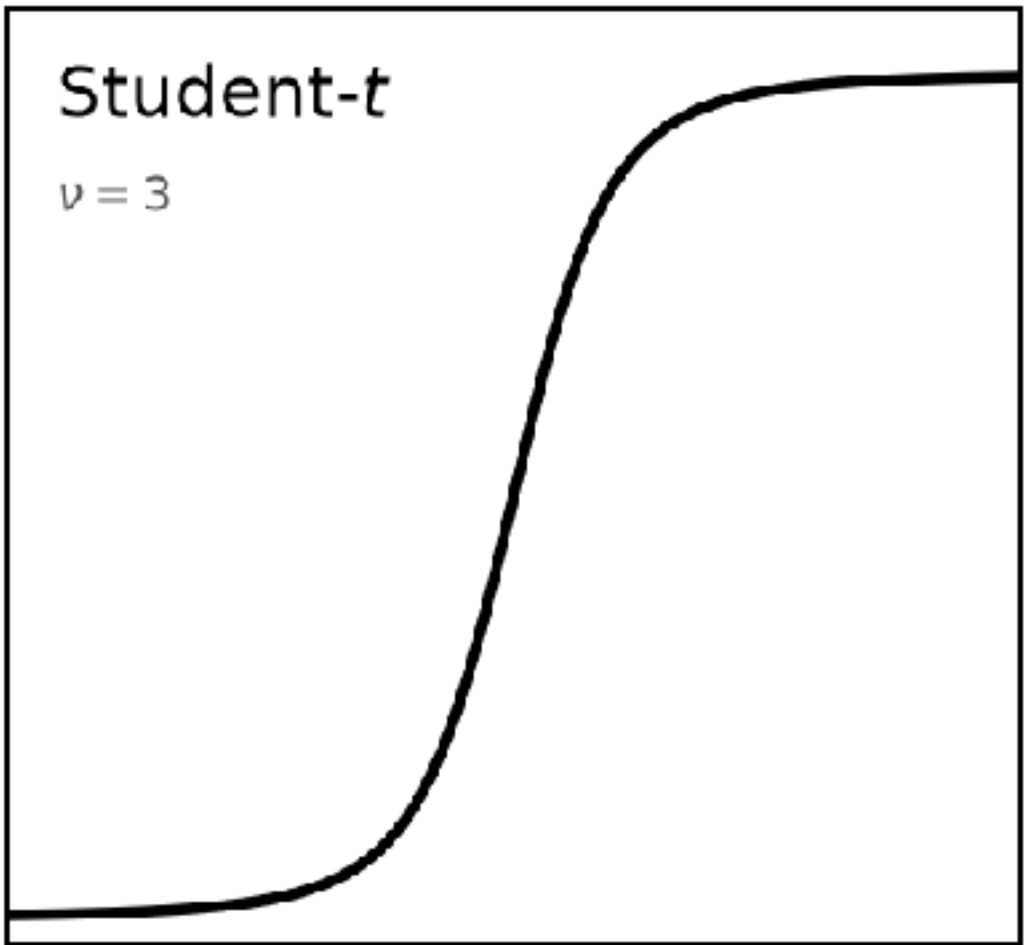
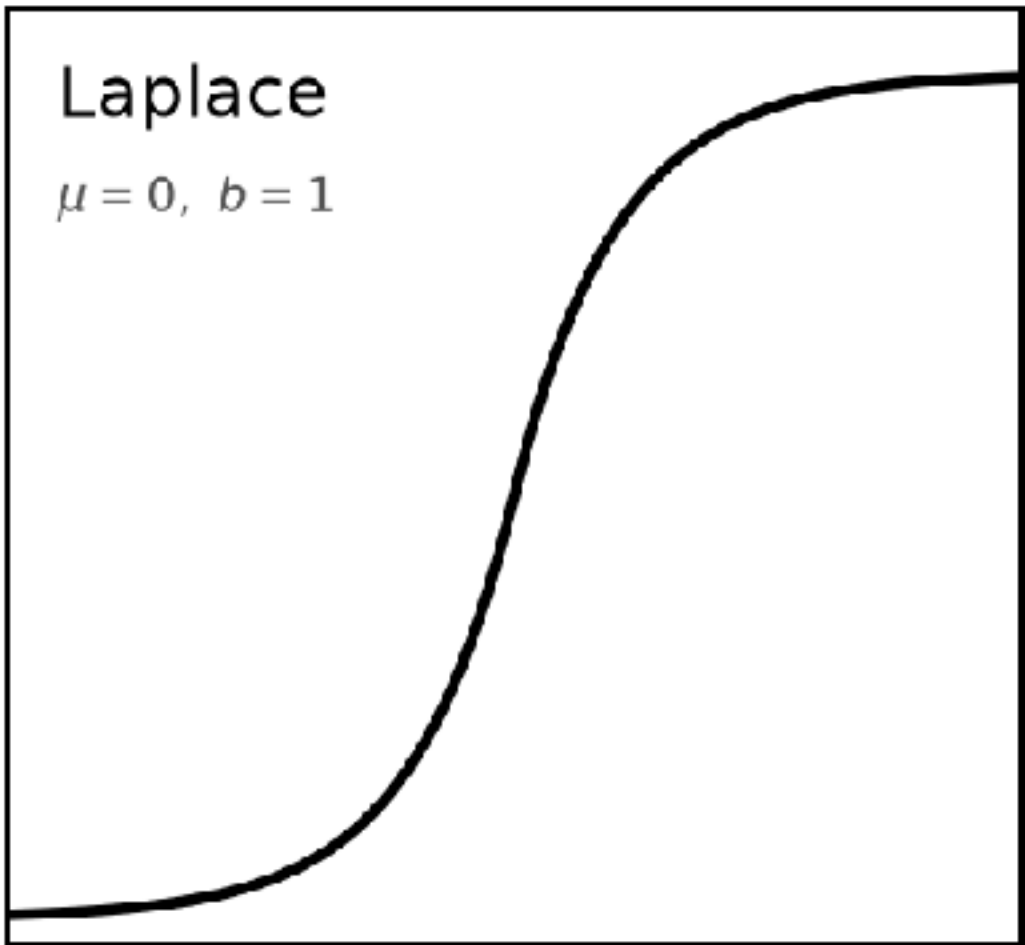
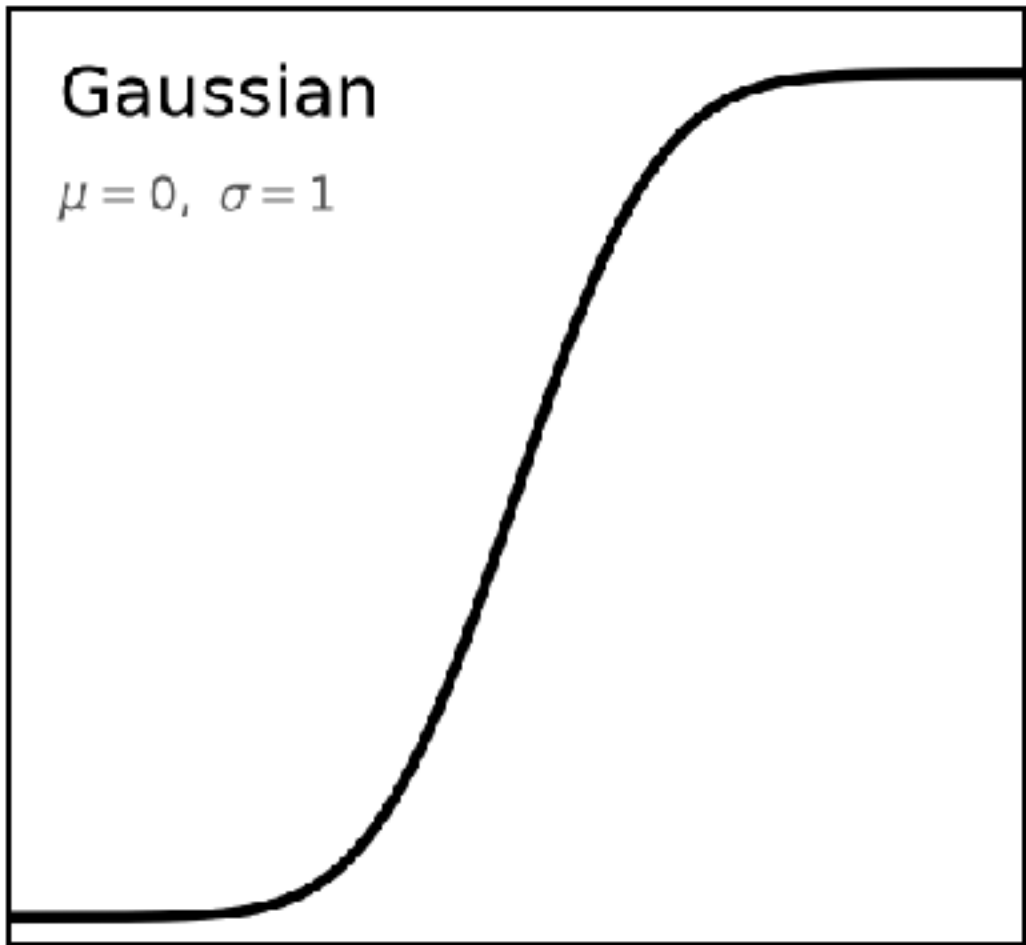
x

Cumulative Density Functions

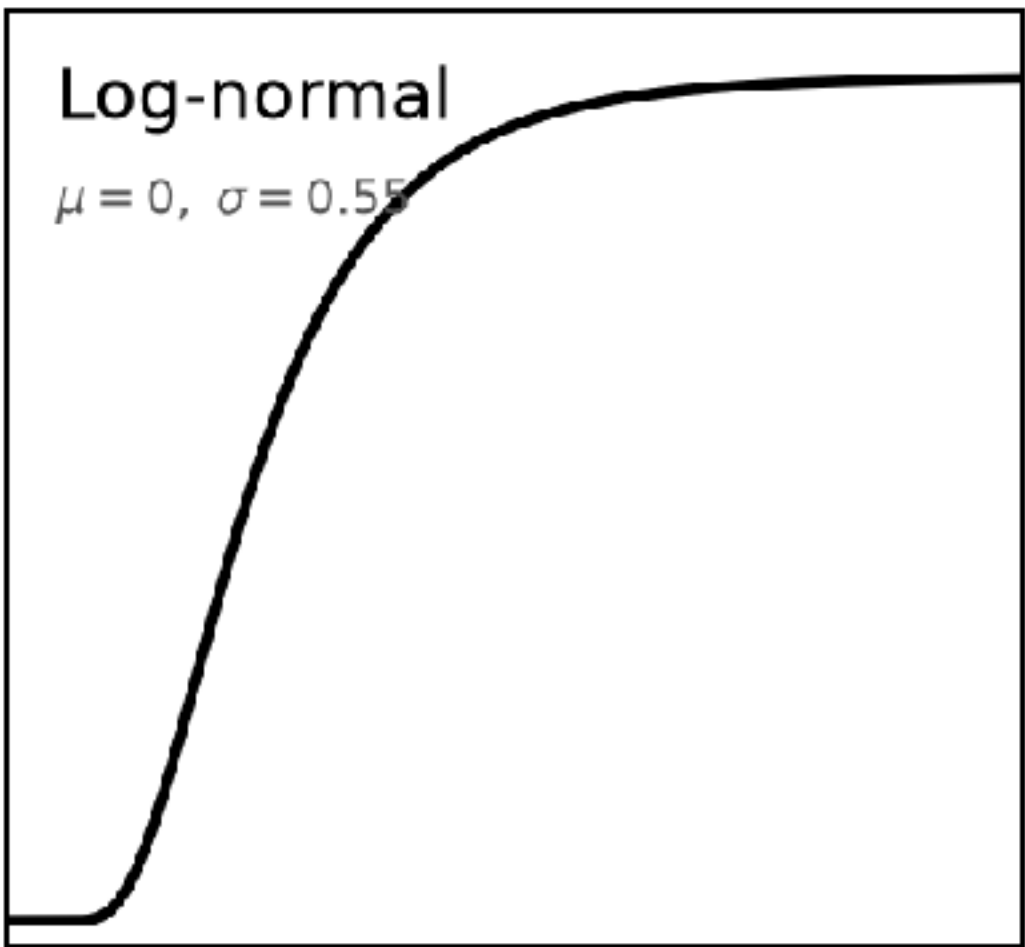
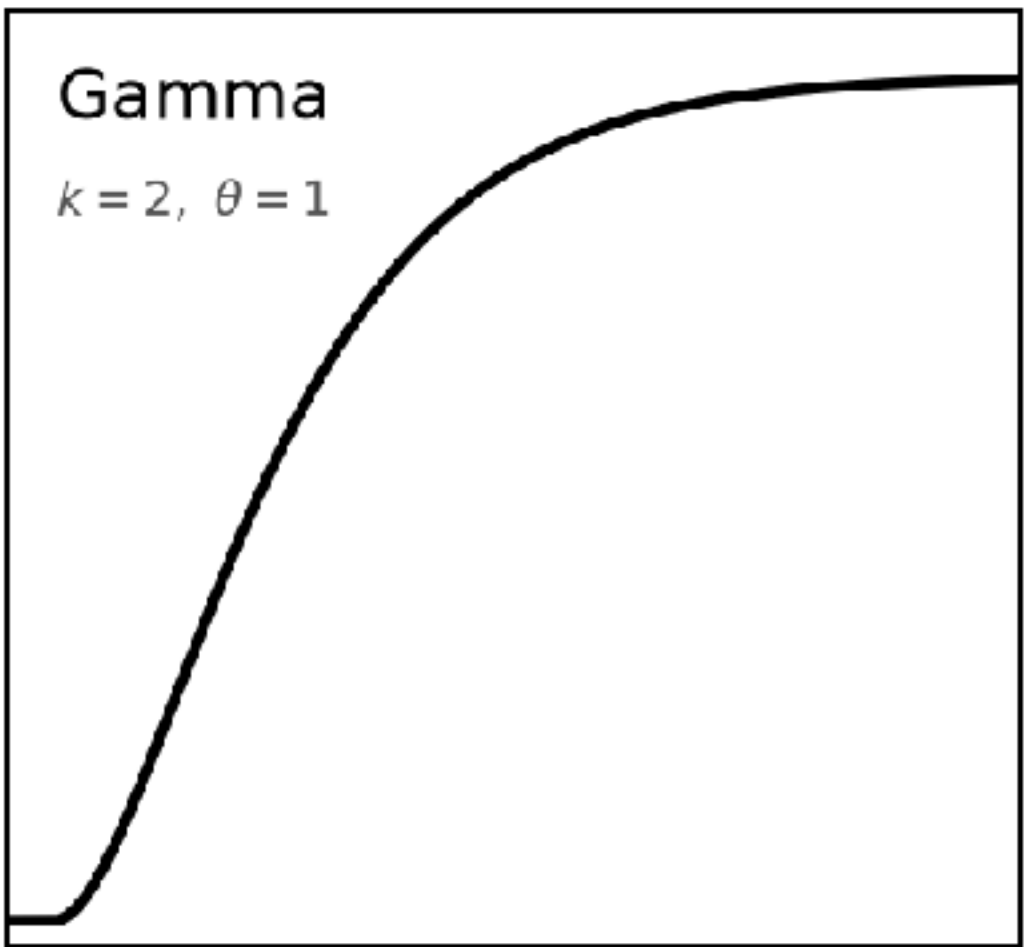
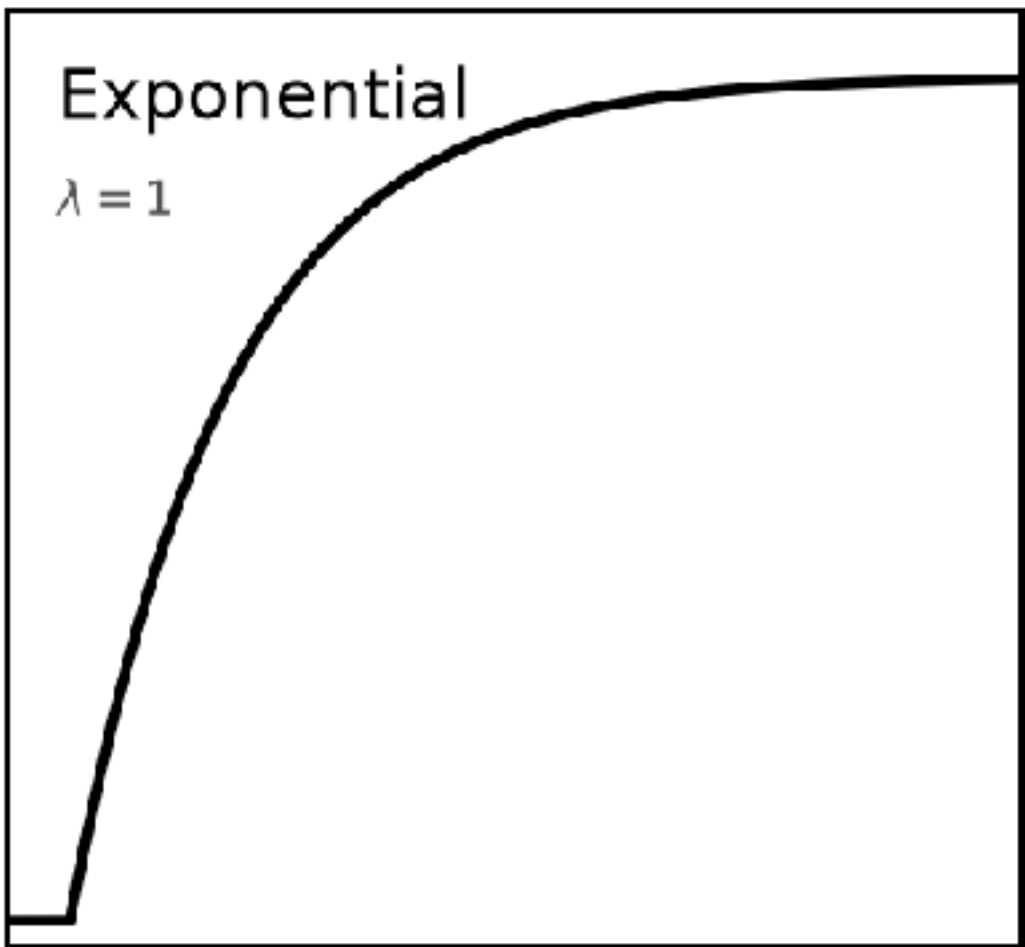
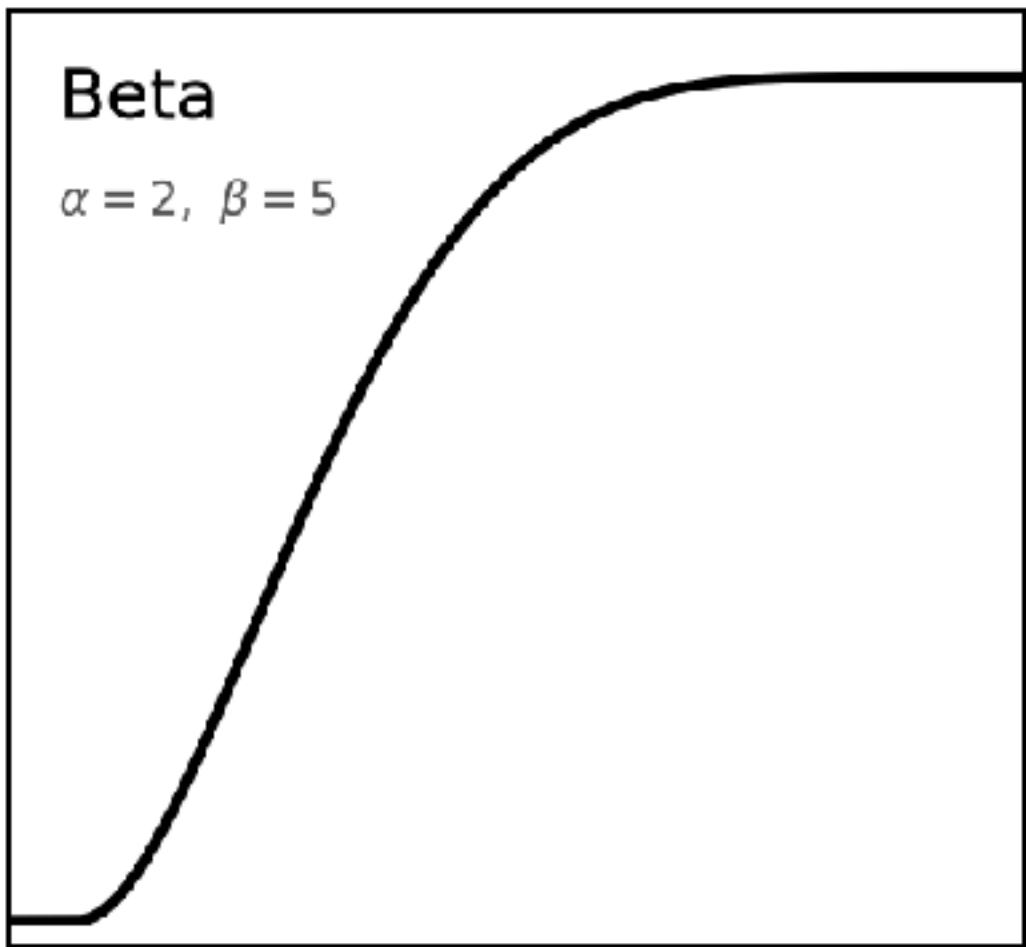


Probability Density Functions

$p(x)$



$p(x)$



x

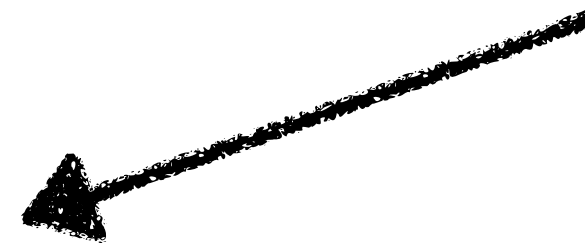
x

x

x

Cumulative Density Functions

$$F(x) = P(X \leq x)$$



Discrete

$$F(x) = P(X = x) - P(X = x^-)$$

Cumulative Density Functions

$$F(x) = P(X \leq x)$$

Discrete

Continuous

$$F(x) = P(X = x) - P(X = x^-)$$

$$F(x) = \int_{-\infty}^x ds p(s)$$

Cumulative Density Functions

$$F(x) = P(X \leq x)$$

Discrete

Continuous

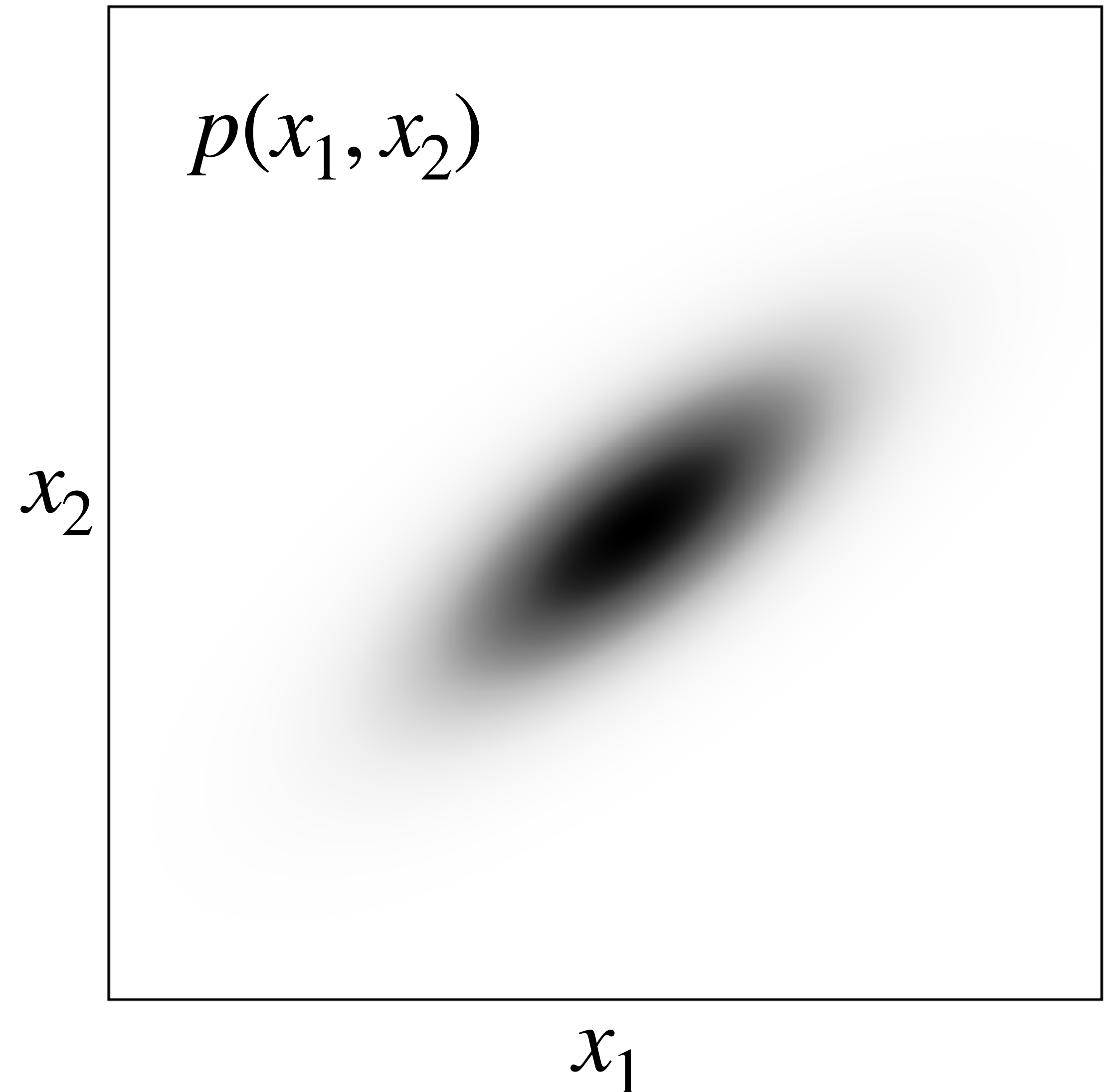
$$F(x) = P(X = x) - P(X = x^-)$$

$$F(x) = \int_{-\infty}^x ds p(s)$$

$$\implies p(x) = \frac{d}{dx} F(x)$$

Joint Distributions

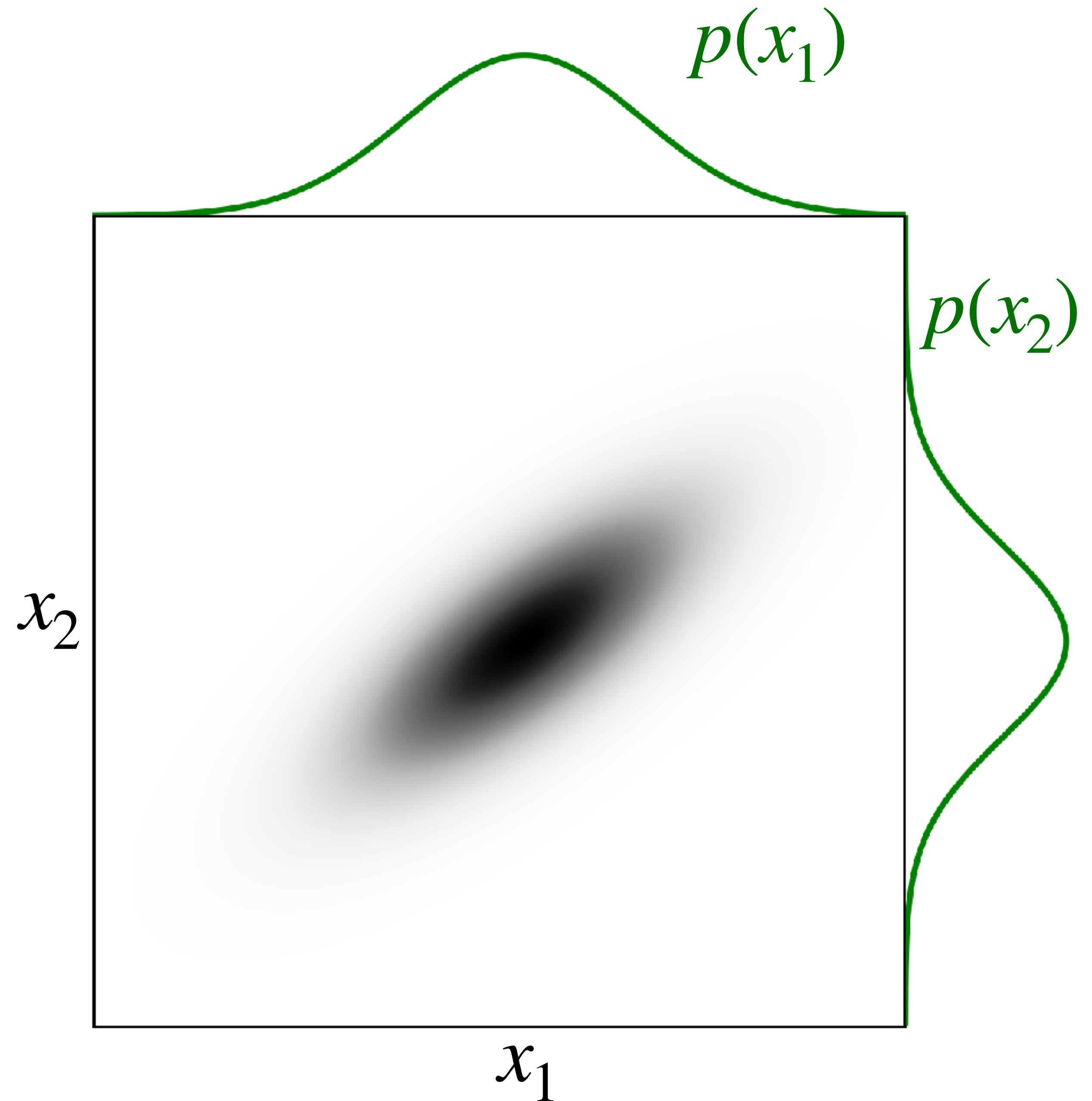
- Probability distribution over multiple random variables $x = (x_1, x_2)$



Marginalisation

- What is the PDF of x_1 when x_2 is not observed?

$$p(x_1) = \int dx_2 p(x_1, x_2)$$

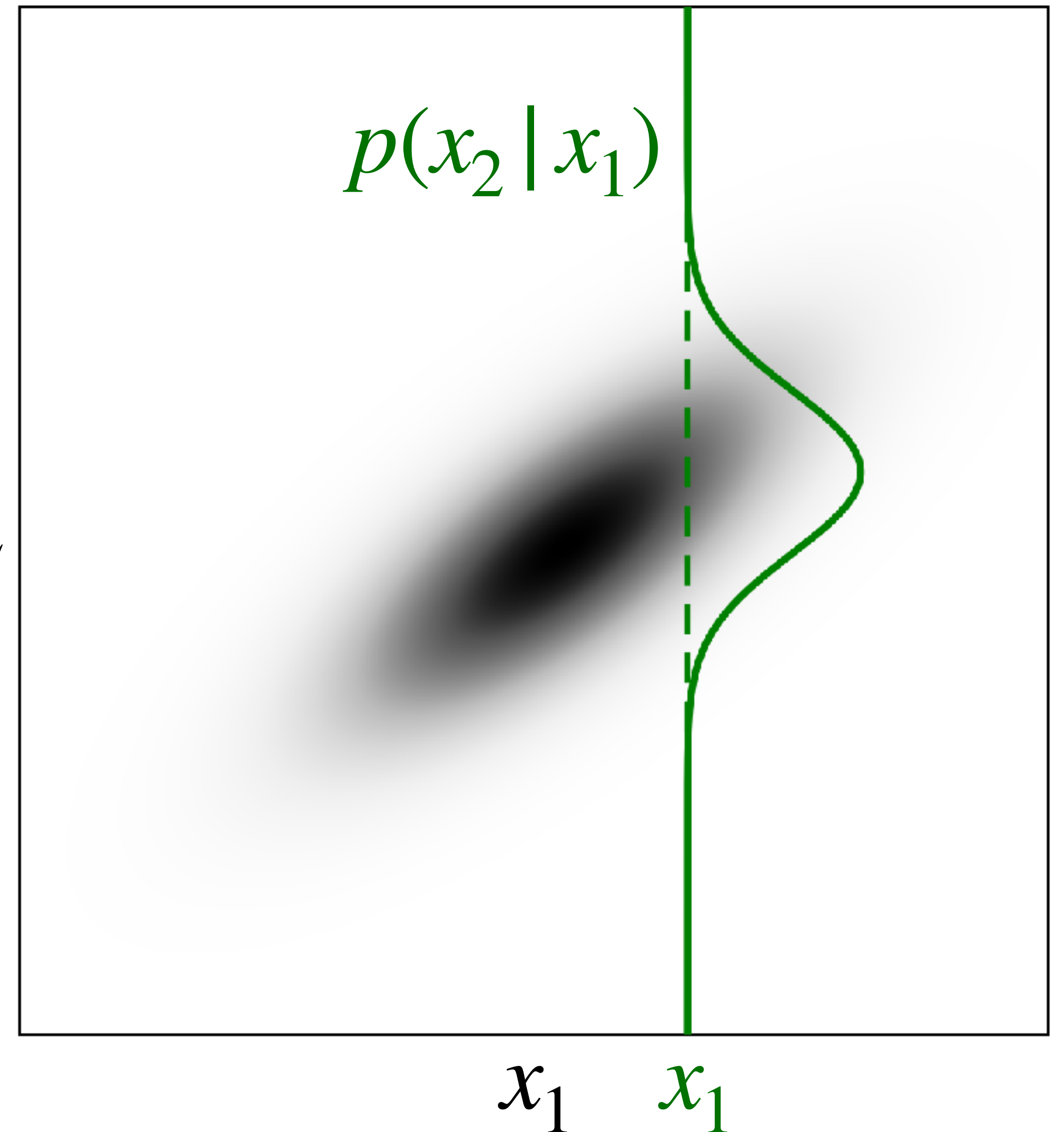


Conditional Probability

- What is the PDF of x_1 when I observe x_2 ?

$$p(x_1, x_2) = p(x_1 | x_2)p(x_2)$$

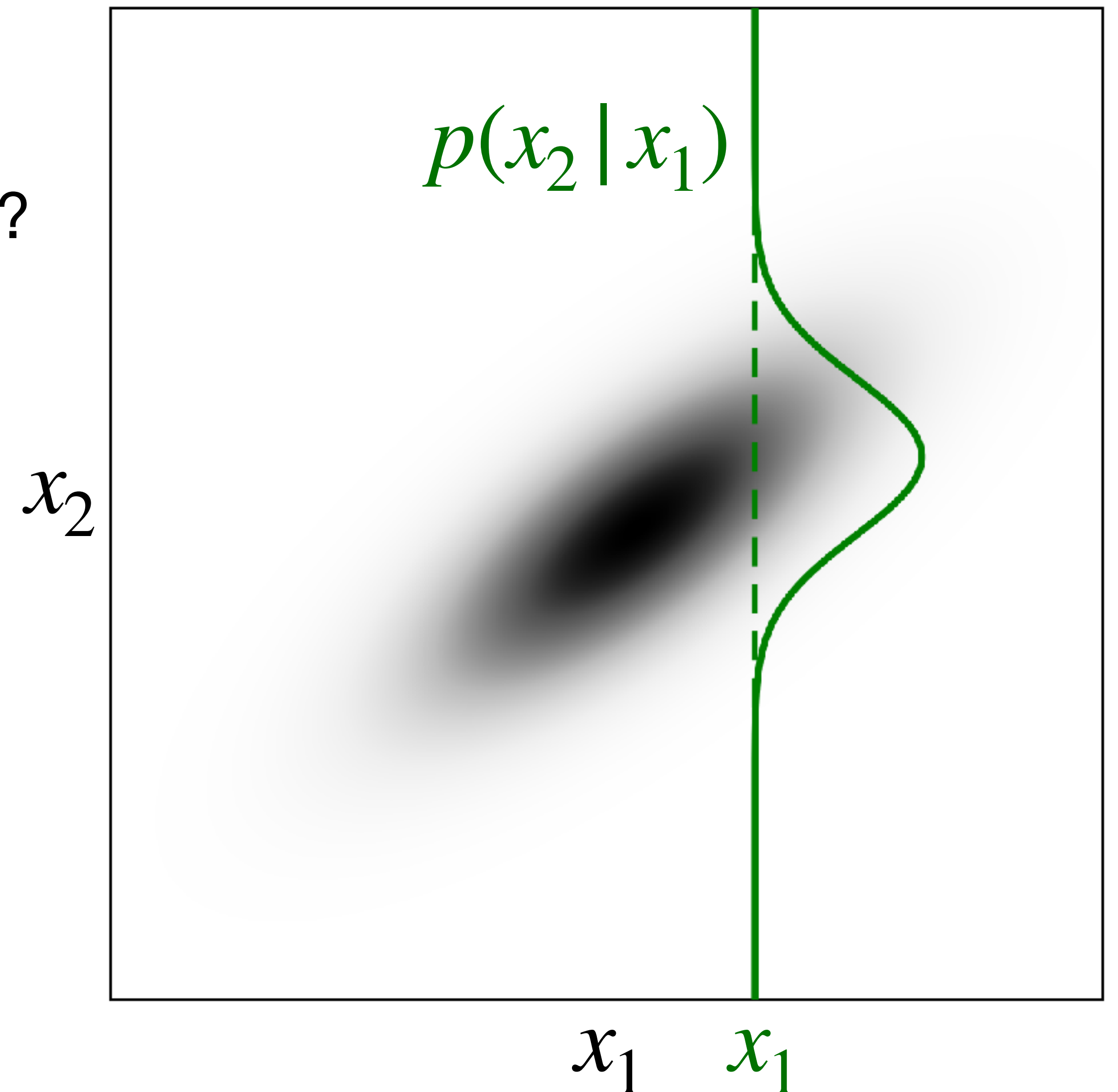
x_2



Conditional Probability

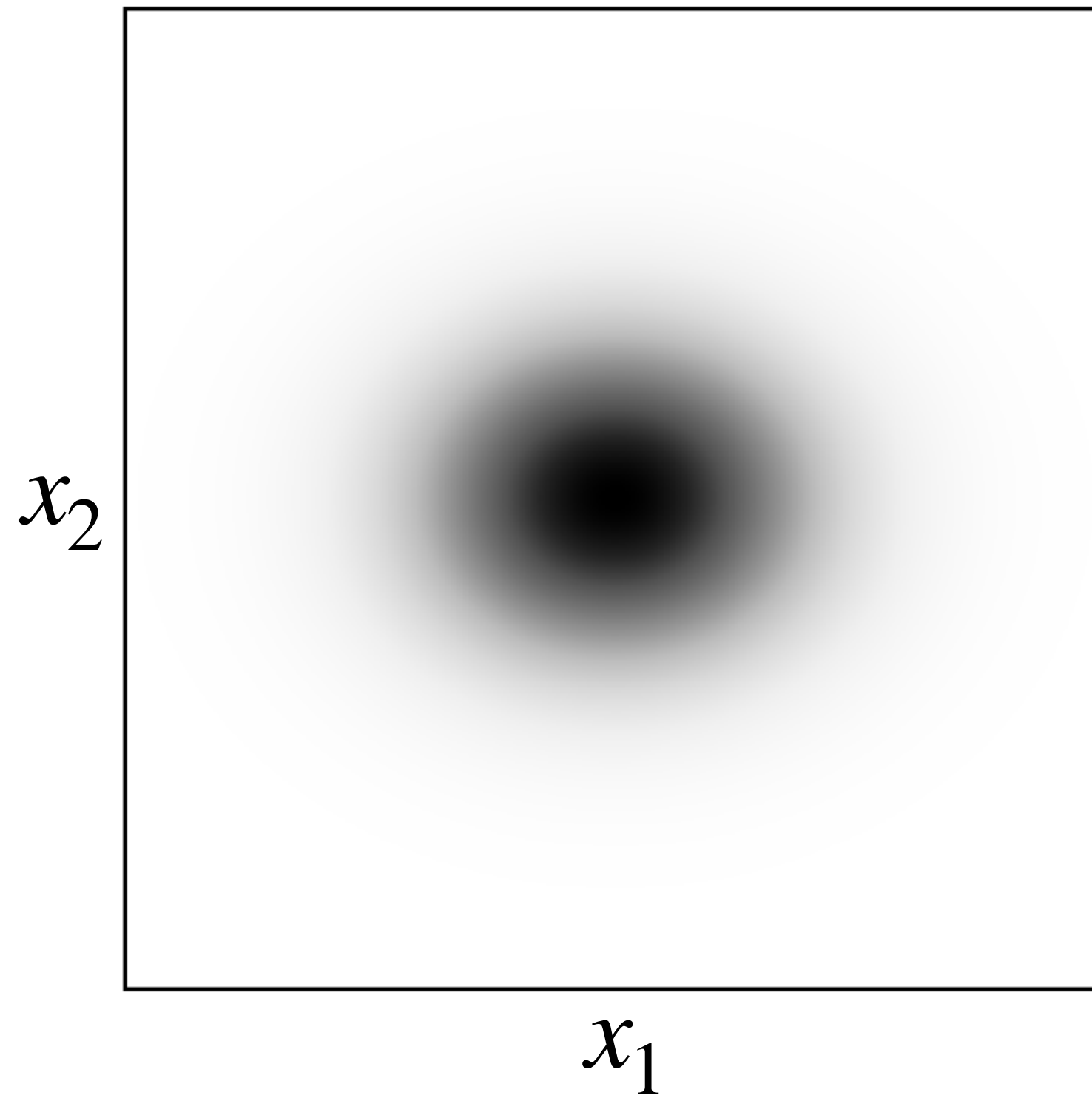
- What is the PDF of x_1 when I observe x_2 ?

$$\begin{aligned} p(x_1, x_2) &= p(x_1 | x_2) p(x_2) \\ &= p(x_2 | x_1) p(x_1) \end{aligned}$$

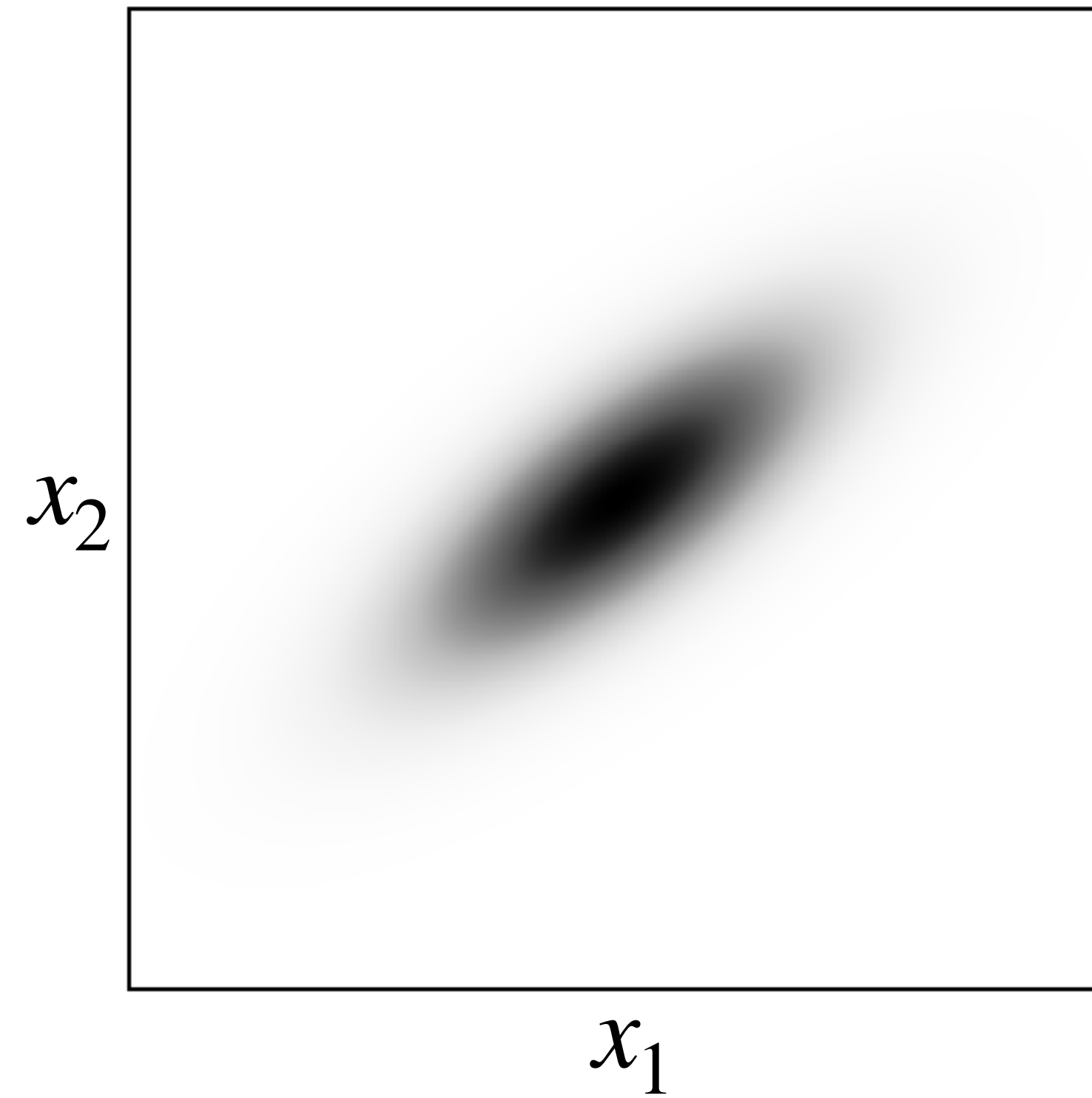


Independence and Correlation

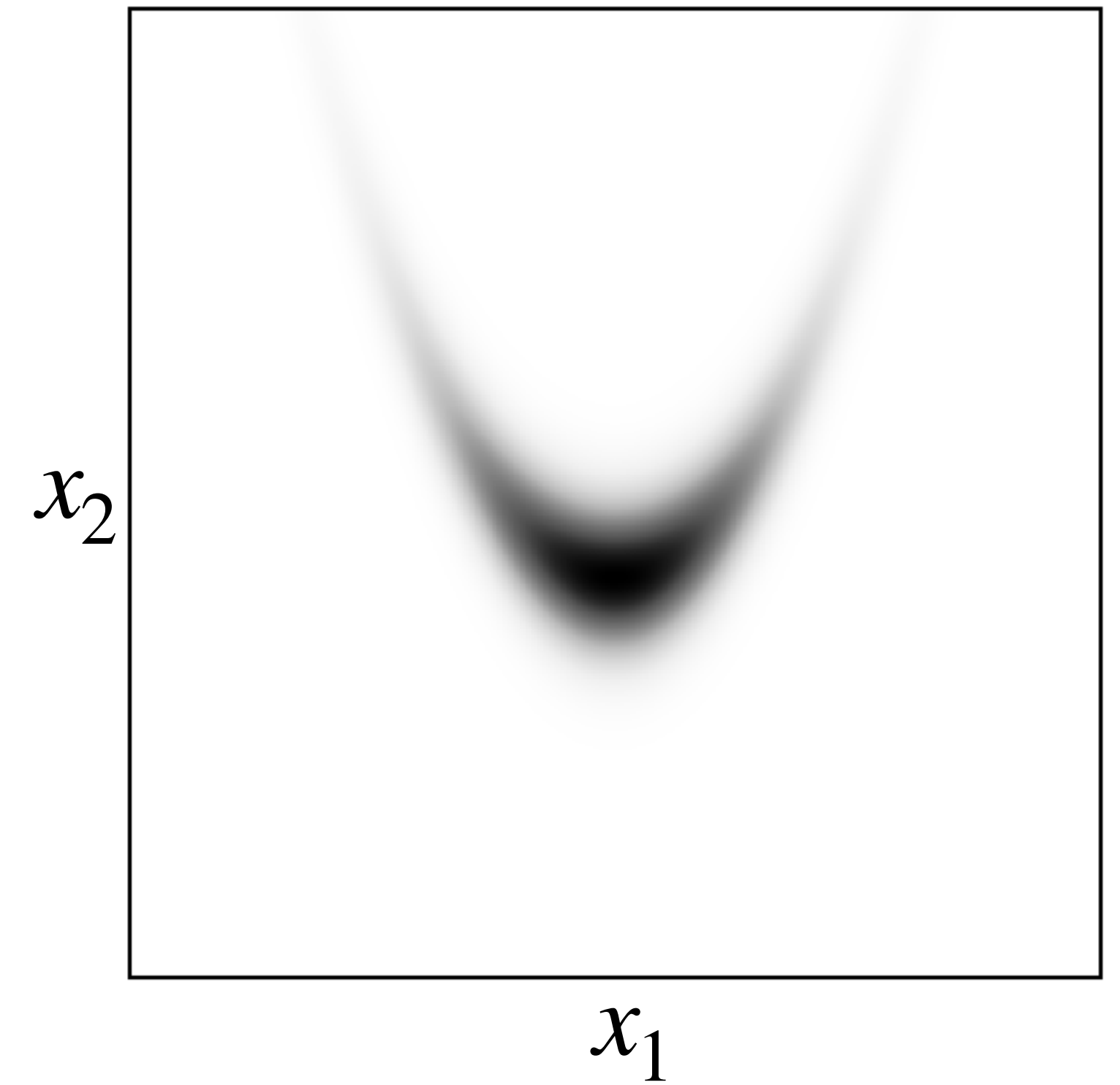
Independent $\rho = 0$



Correlated $\rho = 0.74$

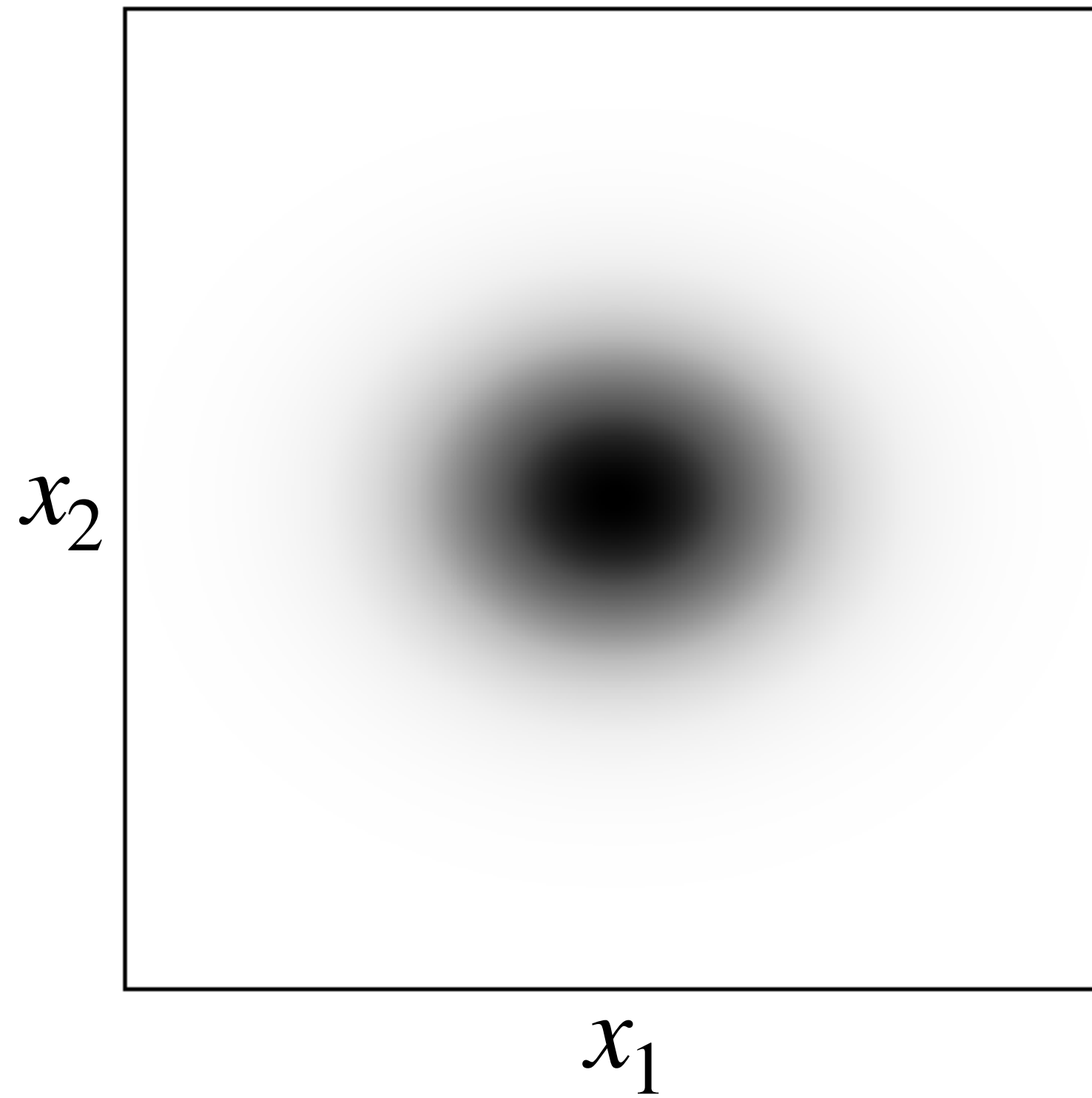


Dependent but uncorrelated $\rho = 0$

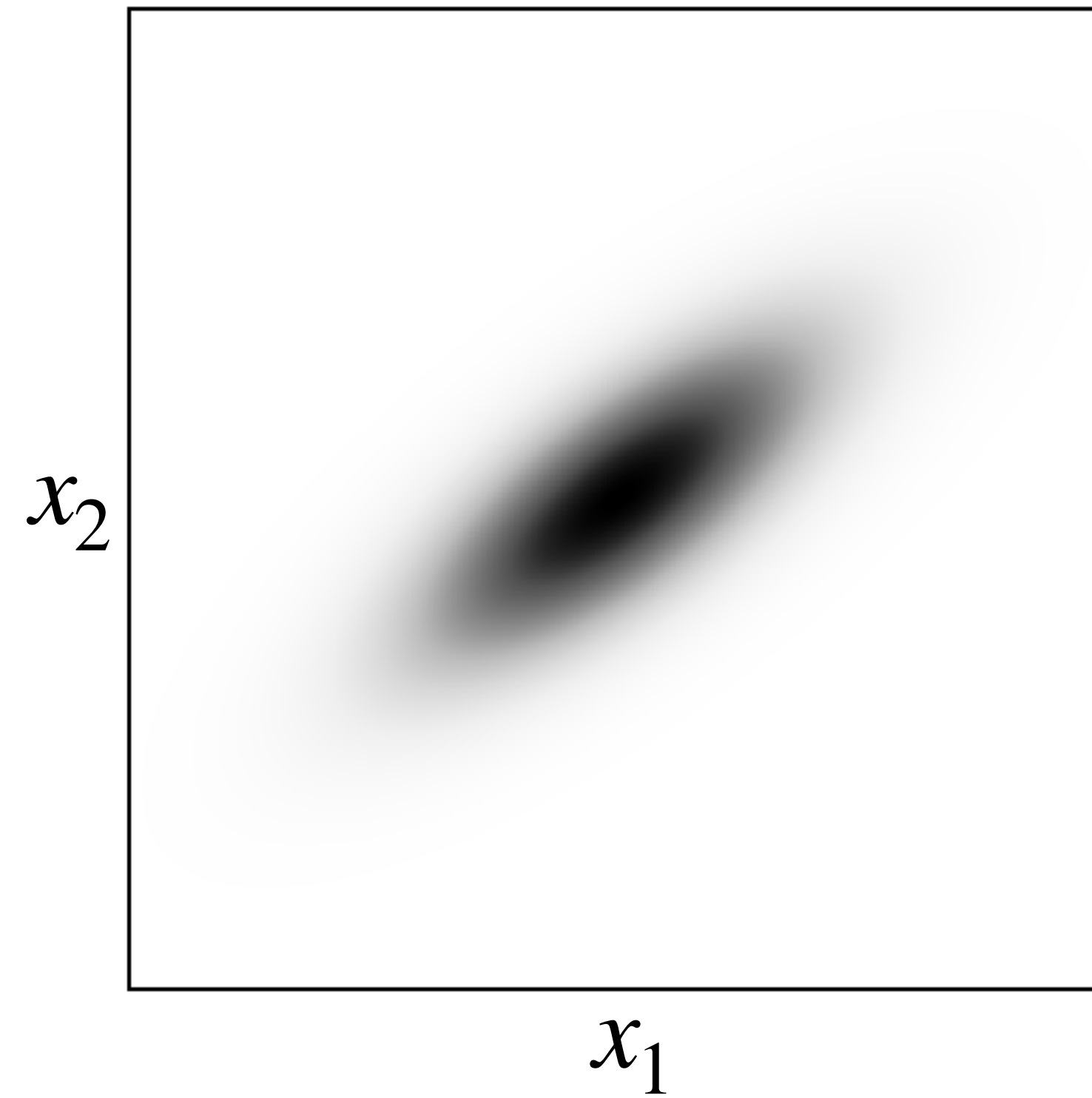


Independence and Correlation

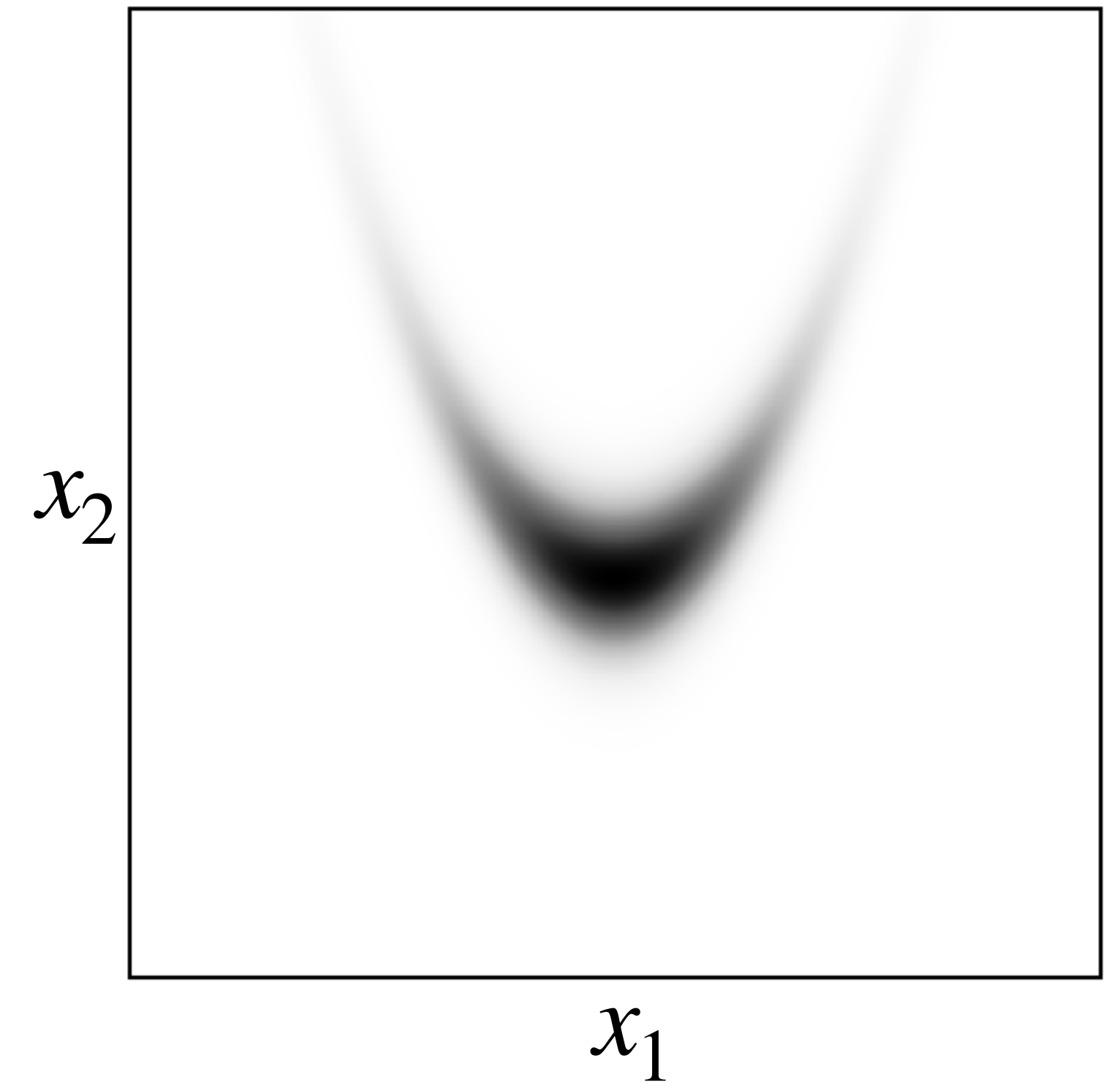
Independent $\rho = 0$



Correlated $\rho = 0.74$



Dependent but uncorrelated $\rho = 0$



→ Independence is a *stronger* assumption than zero correlation, why?

Change-of-variables

- Express PDF $p(x)$ in terms of another PDF $p(z)$

$$p(x) = |J_f(x)| \cdot p(z)$$

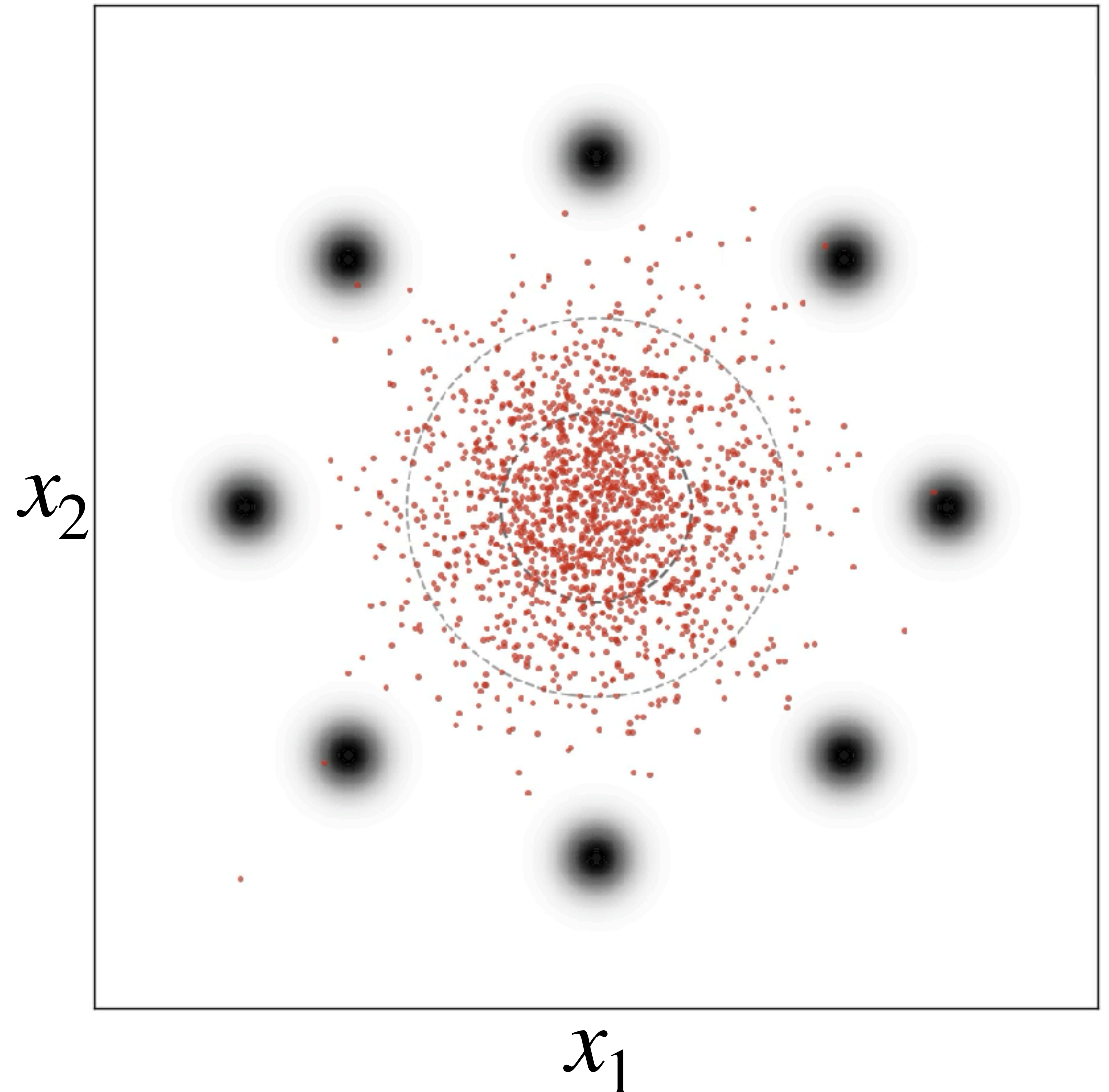
Change-of-variables

- Express PDF $p(x)$ in terms of another PDF $p(z)$

$$p(x) = |J_f(x)| \cdot p(z)$$

- Basis of many ML models, used for analysing data we cannot analytically model.

Change-of-variables from $p(z)$ to $p(x)$

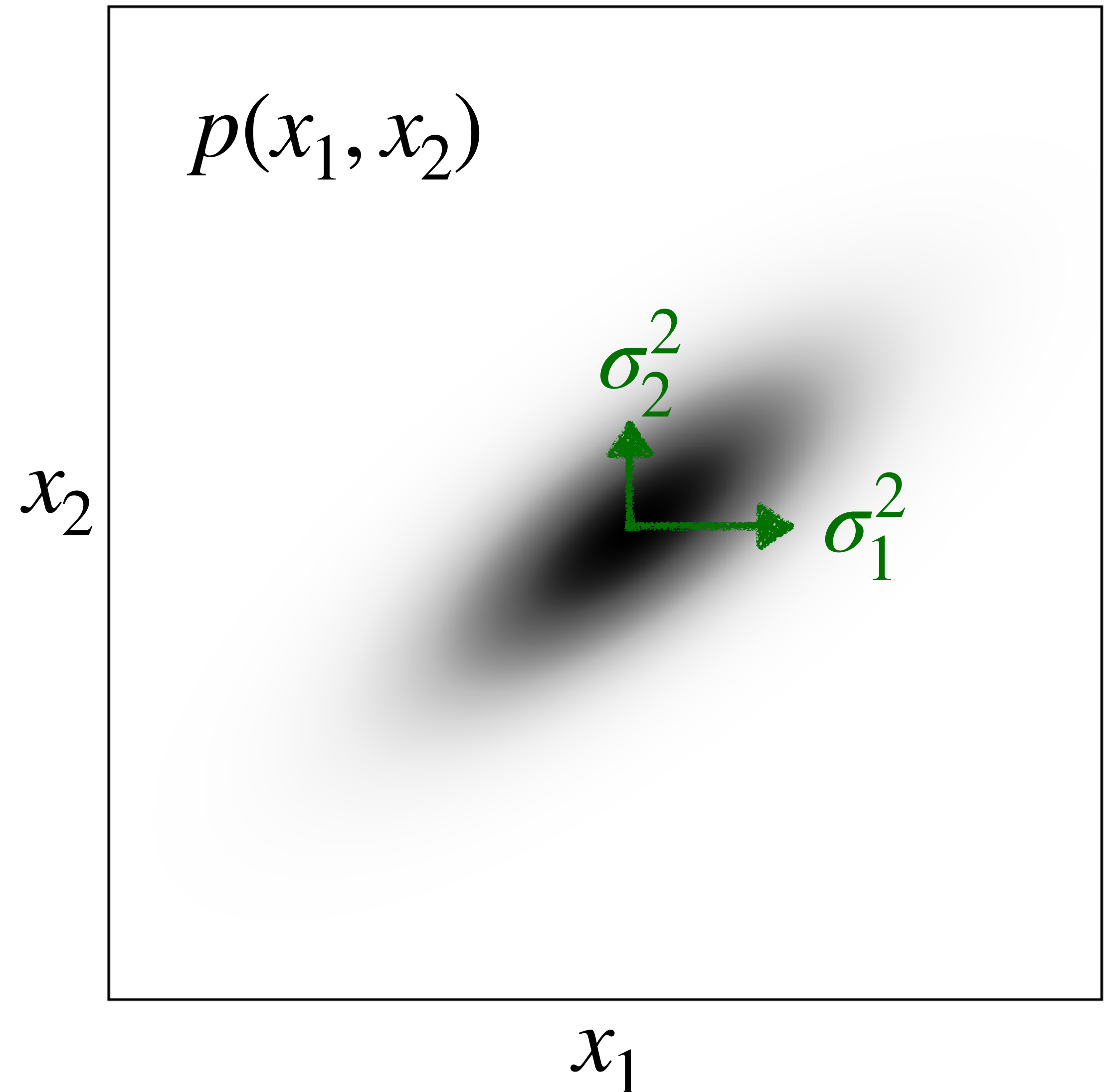


Correlation

- Scale for linear correlation between two variables x_1 and x_2

$$\rho_{12} = \text{Corr}(x_1, x_2)$$

$$= \frac{\text{Cov}(x_1, x_2)}{\sqrt{\text{Var}(x_1)\text{Var}(x_2)}} \in [-1, 1].$$

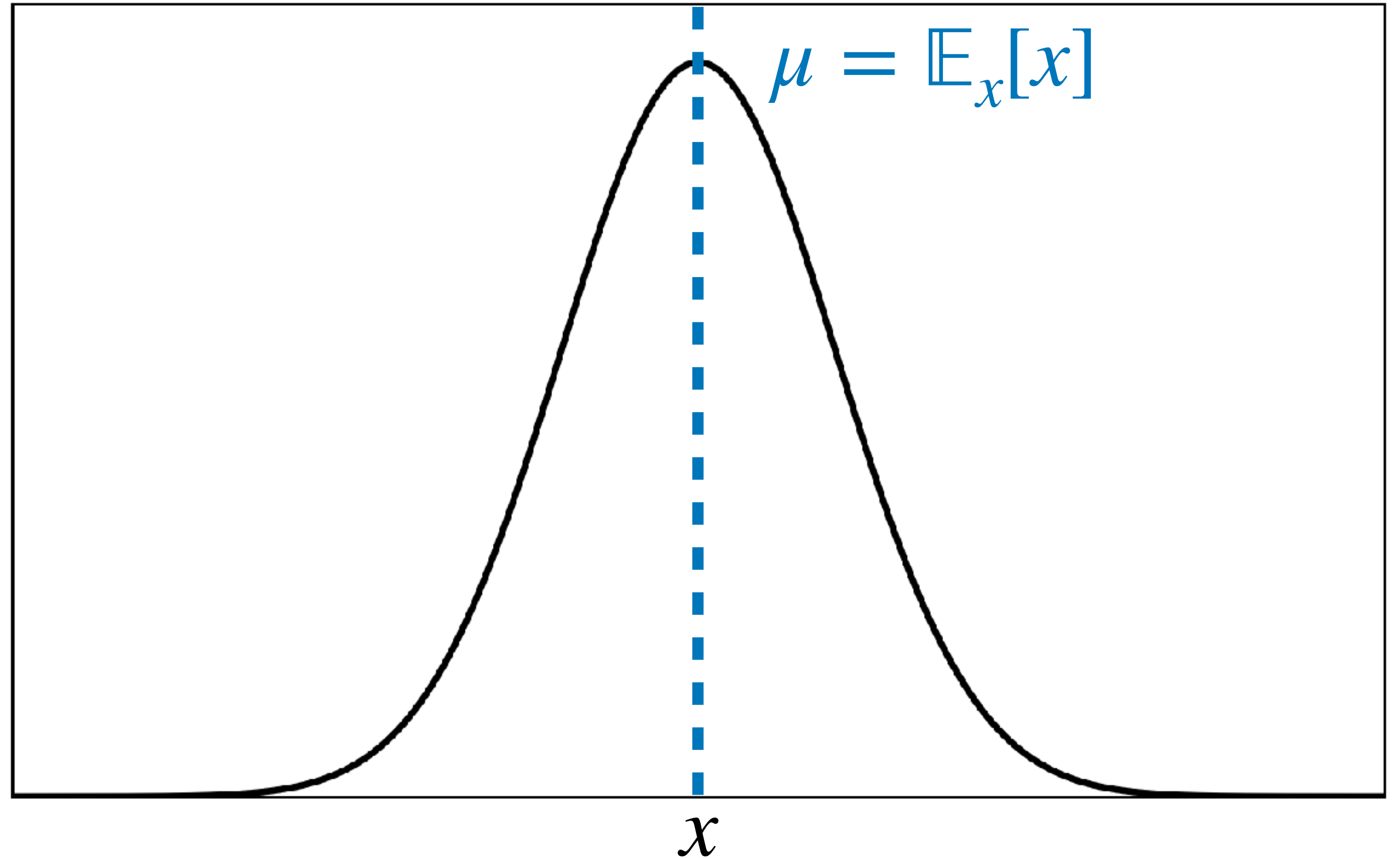


Expectation

- Average value of x under $p(x)$

$$\mathbb{E}_x[x] = \int dx p(x)x$$

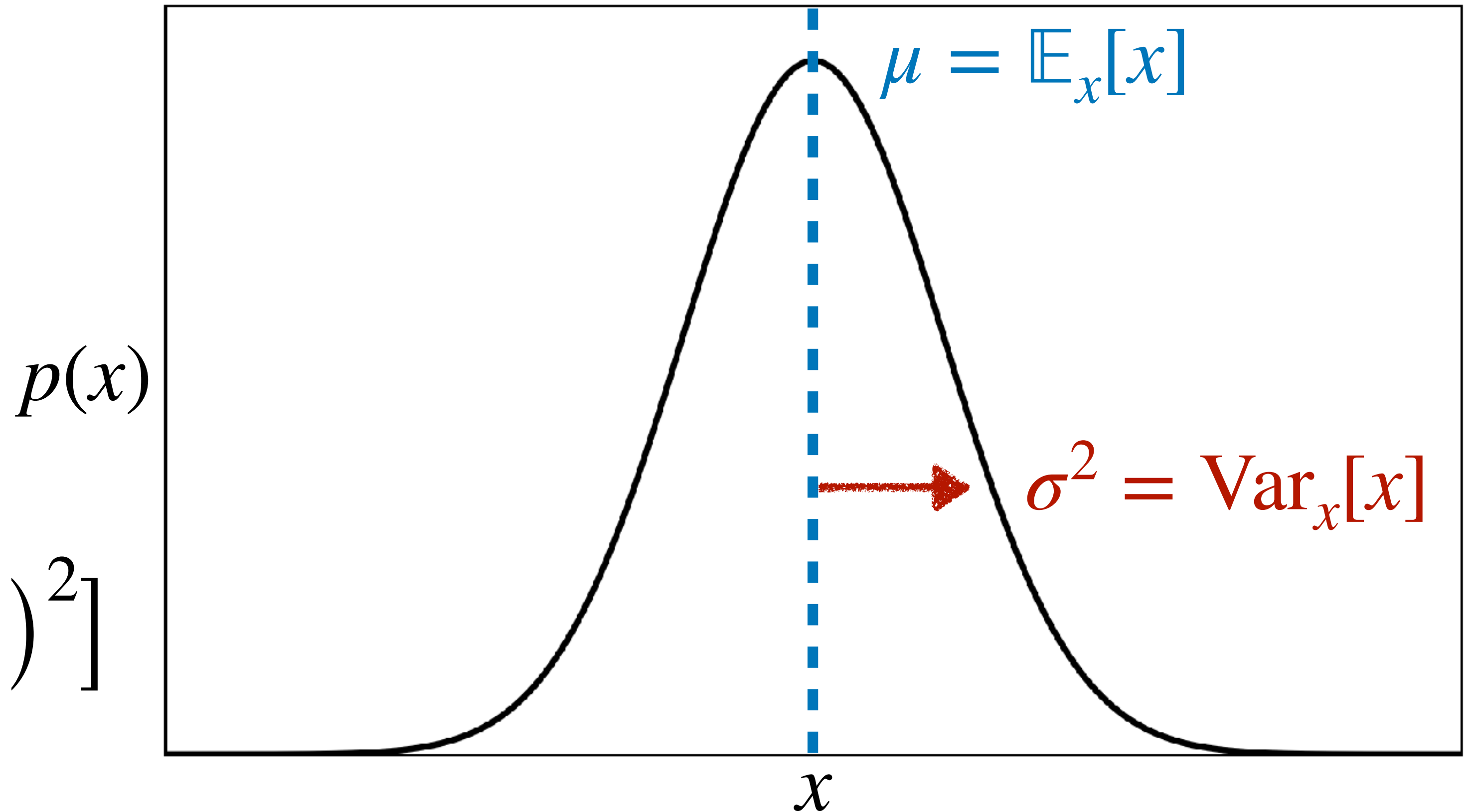
$p(x)$



Variance

- Average scatter of x around $\mathbb{E}_x[x]$ under $p(x)$

$$\text{Var}_x[x] = \mathbb{E}_x[(x - \mathbb{E}_x[x])^2]$$



Moments

- Moments compress a PDF into numbers that can reconstruct its shape.

$$\begin{aligned} m_n &= \mathbb{E}_x \left[(x - \mathbb{E}_x[x])^n \right] \\ &= \int dx p(x) (x - \mathbb{E}[x])^n \end{aligned}$$

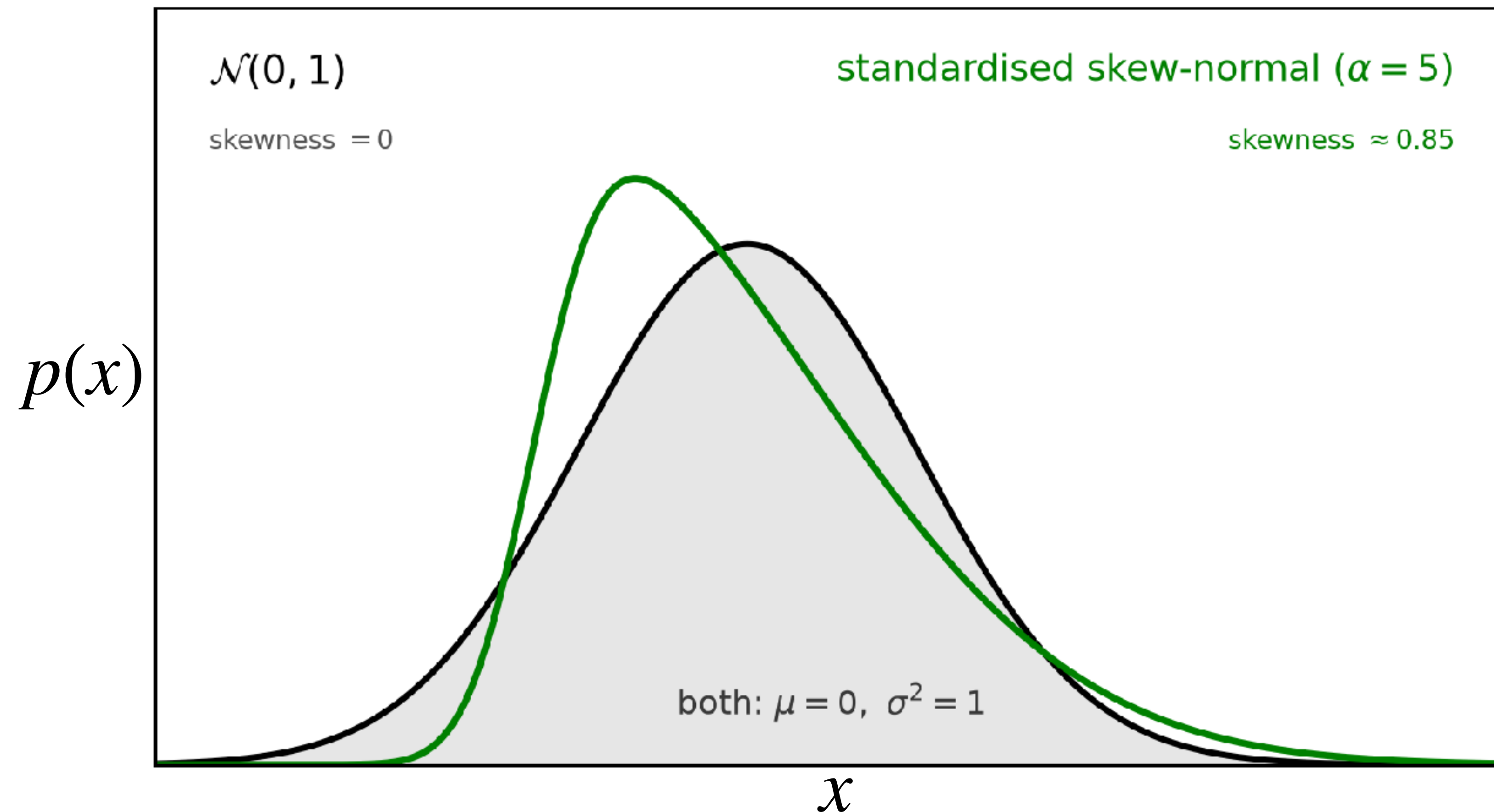
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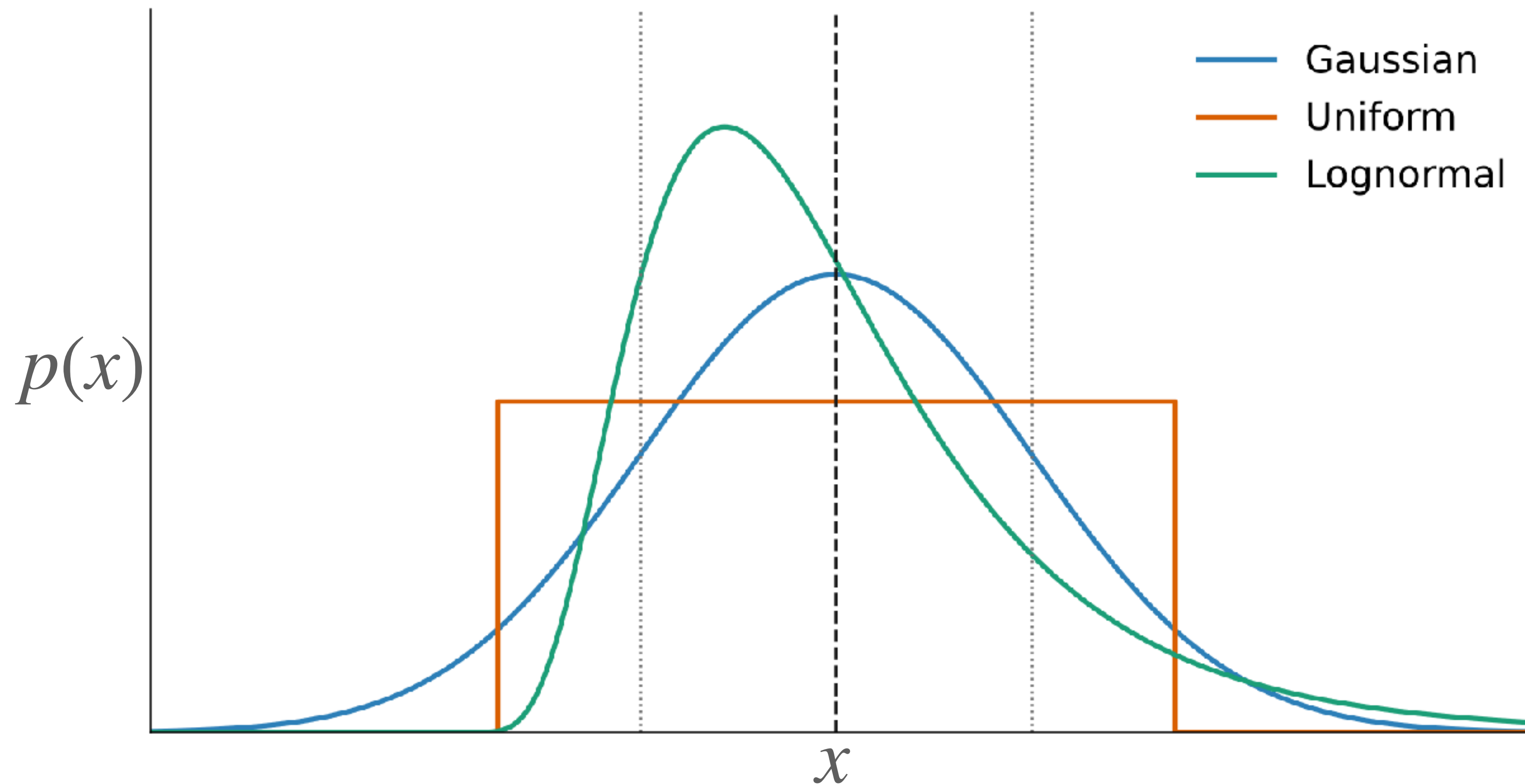
Moments

- Moments compress a PDF into numbers that can reconstruct its shape.



Moments

- Moments compress a PDF into numbers that can reconstruct its shape.



Covariance

- How do two random variables scatter together?

$$\begin{aligned}\text{Cov}(x_1, x_2) &= \mathbb{E}[(x_1 - \mathbb{E}[x_1])(x_2 - \mathbb{E}[x_2])], \\ &= \mathbb{E}[x_1 x_2] - \mathbb{E}[x_1]\mathbb{E}[x_2].\end{aligned}$$

Covariance

- If x is multi-dimensional, we have the covariance matrix

$$\text{Cov}(x) = \mathbb{E} \left[(x - \mathbb{E}[x])(x - \mathbb{E}[x])^T \right],$$

Covariance

- If x is multi-dimensional, we have the covariance matrix

$$\text{Cov}(x) = \mathbb{E} \left[(x - \mathbb{E}[x])(x - \mathbb{E}[x])^T \right],$$

$$= \begin{pmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) \end{pmatrix}.$$

Covariance

- Why is the covariance so important?

$$\text{Cov}(x) = \mathbb{E} \left[(x - \mathbb{E}[x])(x - \mathbb{E}[x])^T \right],$$

- It is high-dimensional, and every moment $n > 2$ is a non-linear function of it

Central Limit Theorem (CLT)

- Dataset $X = \{x_i, \dots\}_{i=1}^n$ drawn i.i.d. from any PDF (mean μ , finite variance σ^2)

$$Z_n = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}$$

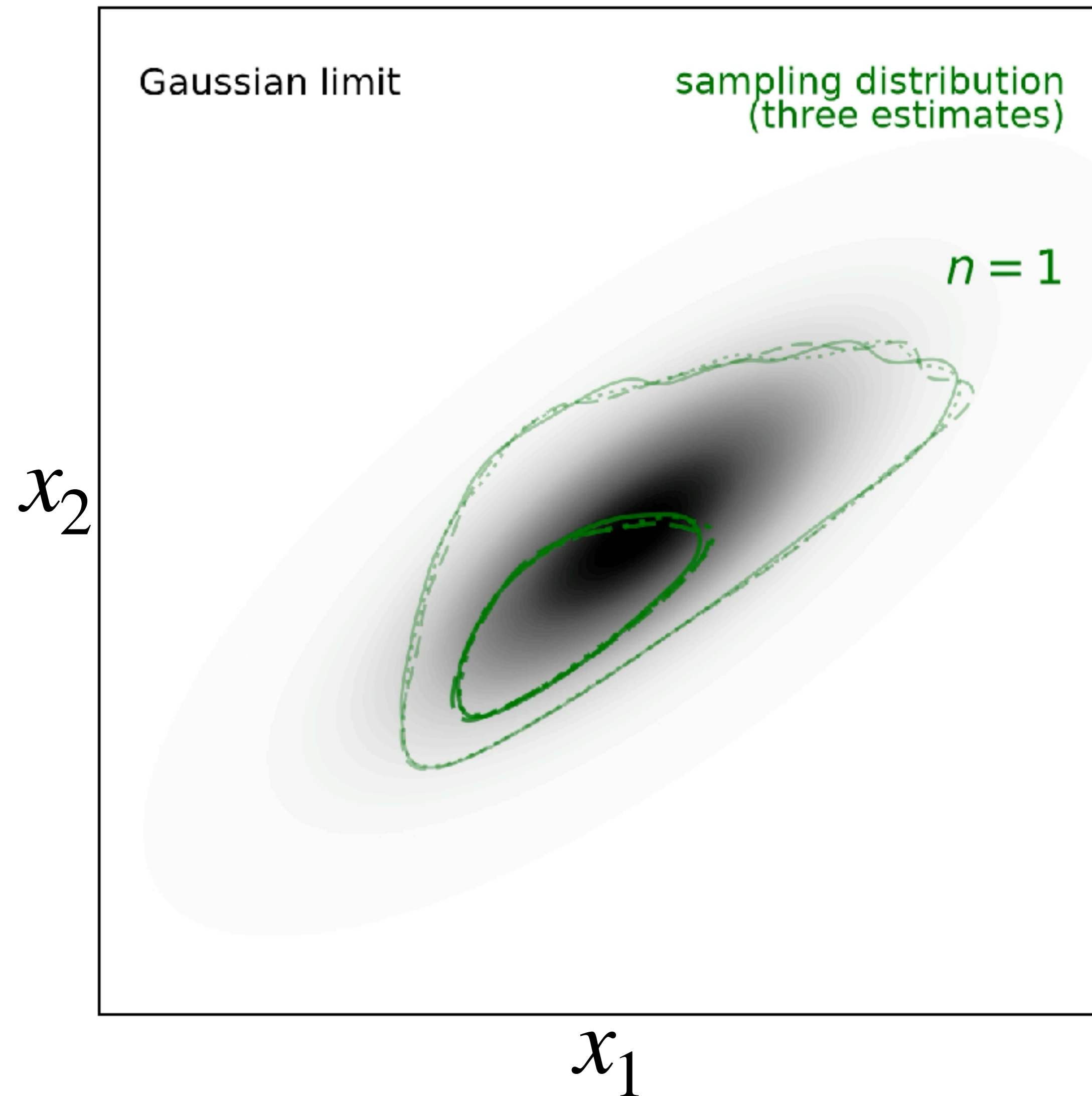
- What shape does the sampling distribution of $\bar{X}_n$ have?

Central Limit Theorem (CLT)

- Dataset $X = \{x_i, \dots\}_{i=1}^n$ drawn i.i.d. from any PDF (finite variance)

$$\begin{aligned} Z_n &= \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \\ &= \frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}} \sim \mathcal{N}(0,1). \end{aligned}$$

Central Limit Theorem (CLT)



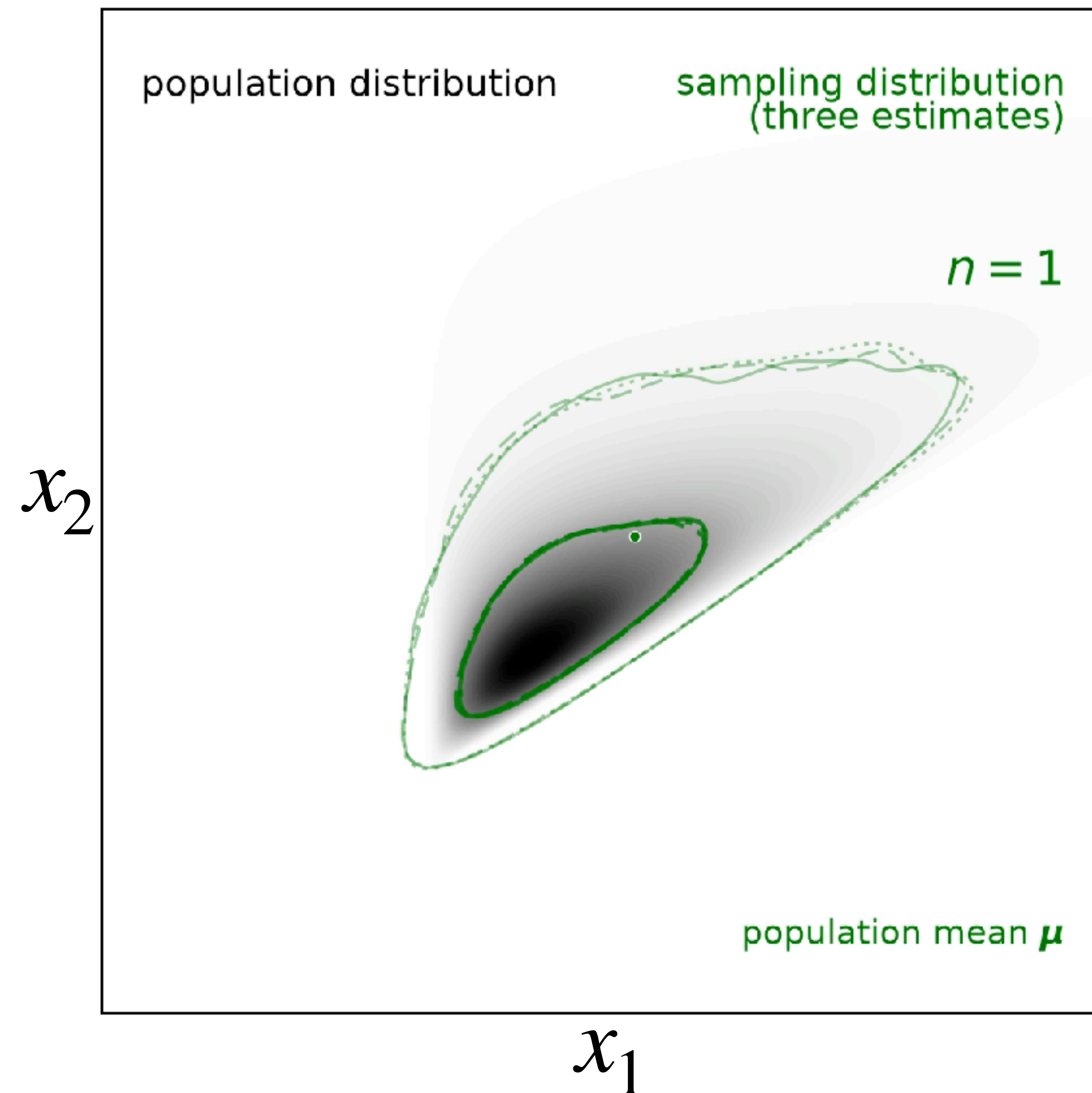
Law of Large Numbers (LLN)

- How does the scatter in $\bar{X}_n$ change with n ?

$$P(|\bar{X}_n - \mu| \geq \varepsilon) \longrightarrow 0.$$

- Larger $n \implies \bar{X}_n$ tends to scatter in smaller region around mean μ

Law of Large Numbers (LLN)



Monte-Carlo estimation

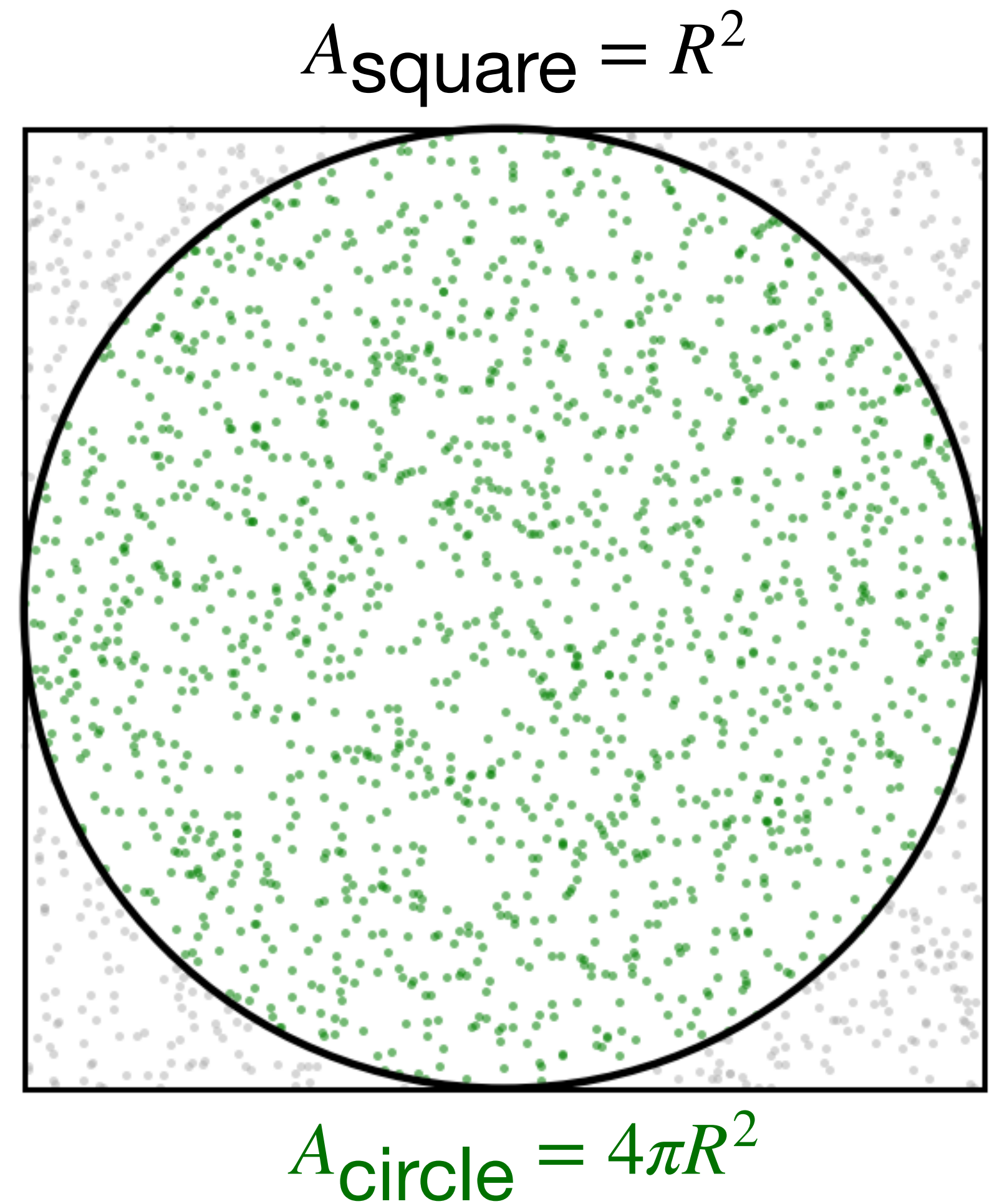
- Why are the LLN and CLT important?
- MC estimation: turn an integral into an expectation

$$I = \mathbb{E}_x[f(x)] = \int dx p(x) f(x)$$

- LLN and CLT dictate narrowing scatter of estimate around truth

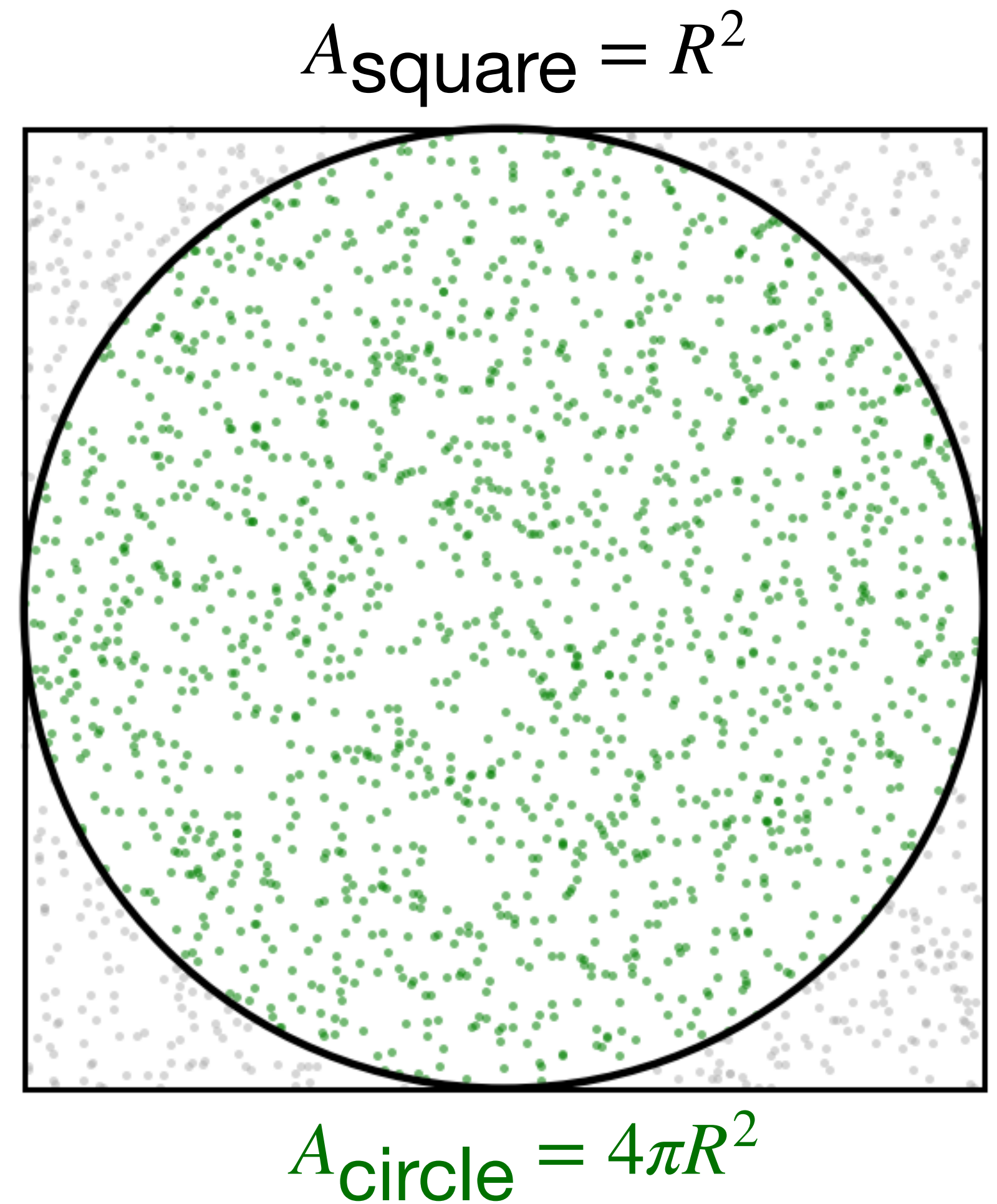
Monte-Carlo estimation of π

- Sample points in a box $(x, y) \sim \mathcal{U}[0,1]$



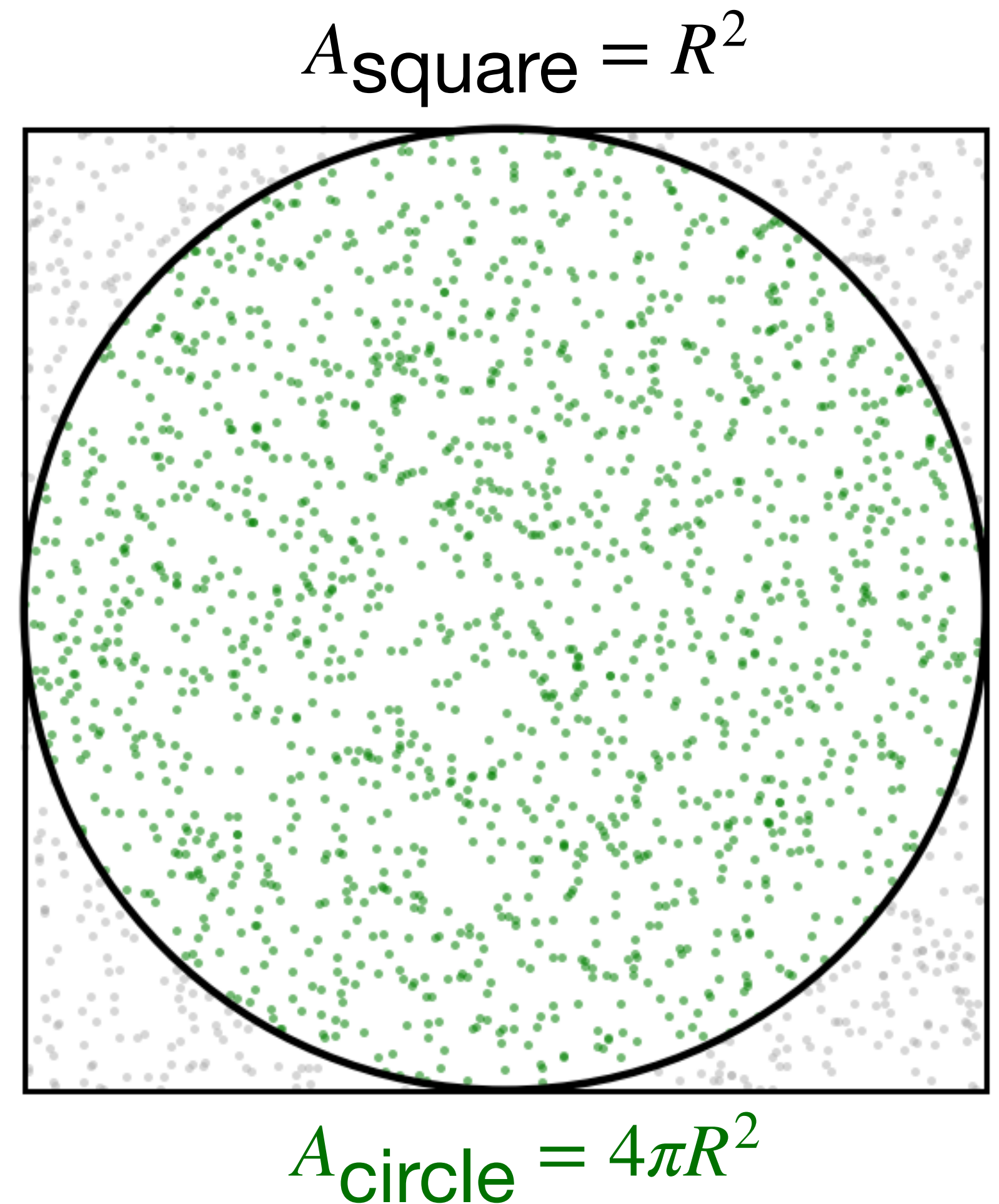
Monte-Carlo estimation of π

- Sample points in a box $(x, y) \sim \mathcal{U}[0,1]$
- Inside circle? $A(x, y) = \mathbb{I}\{x^2 + y^2 \leq 1\}$



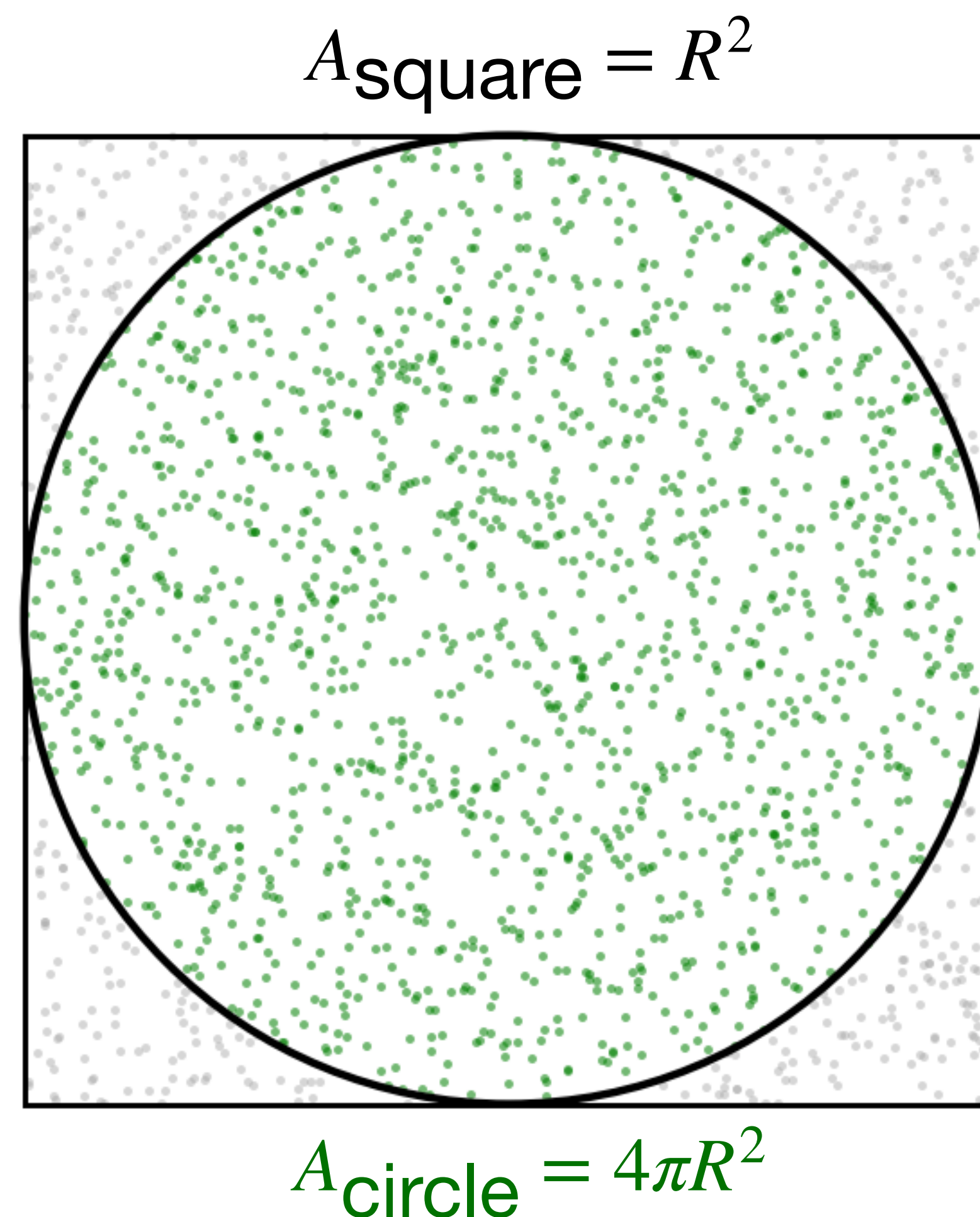
Monte-Carlo estimation of π

- Sample points in a box $(x, y) \sim \mathcal{U}[0,1]$
- Inside circle? $A(x, y) = \mathbb{I}\{x^2 + y^2 \leq 1\}$
- Average "inside circle?": $\mathbb{E}_{x,y}[A(x, y)] = \frac{\pi}{4}$



Monte-Carlo estimation of π

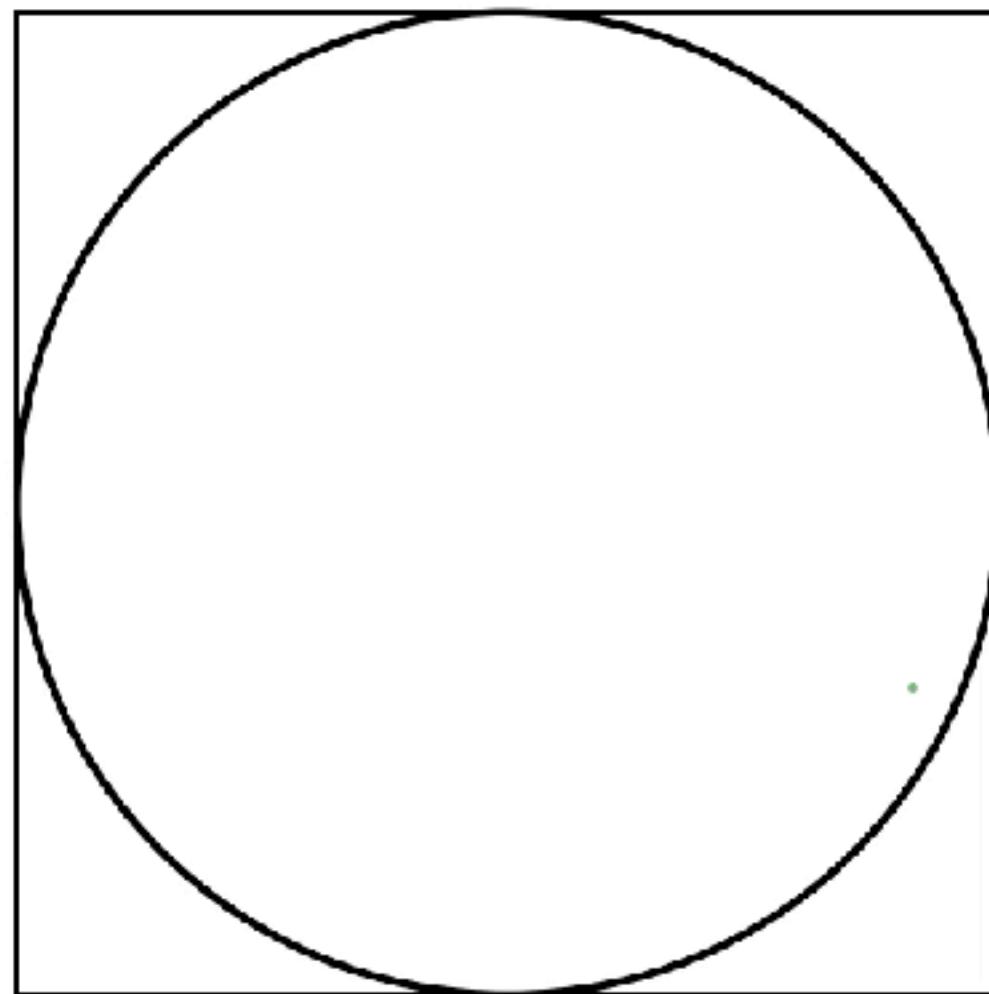
- Sample points in a box $(x, y) \sim \mathcal{U}[0,1]$
- Inside circle? $A(x, y) = \mathbb{I}\{x^2 + y^2 \leq 1\}$
- Average "inside circle?": $\mathbb{E}_{x,y}[A(x, y)] = \frac{\pi}{4}$
- Sample average: $\hat{\pi}_N = 4\bar{A}_N \approx 4\mathbb{E}_{x,y}[A(x, y)]$



Monte-Carlo estimation of π

- Why are the LLN and CLT important?

One experiment for π

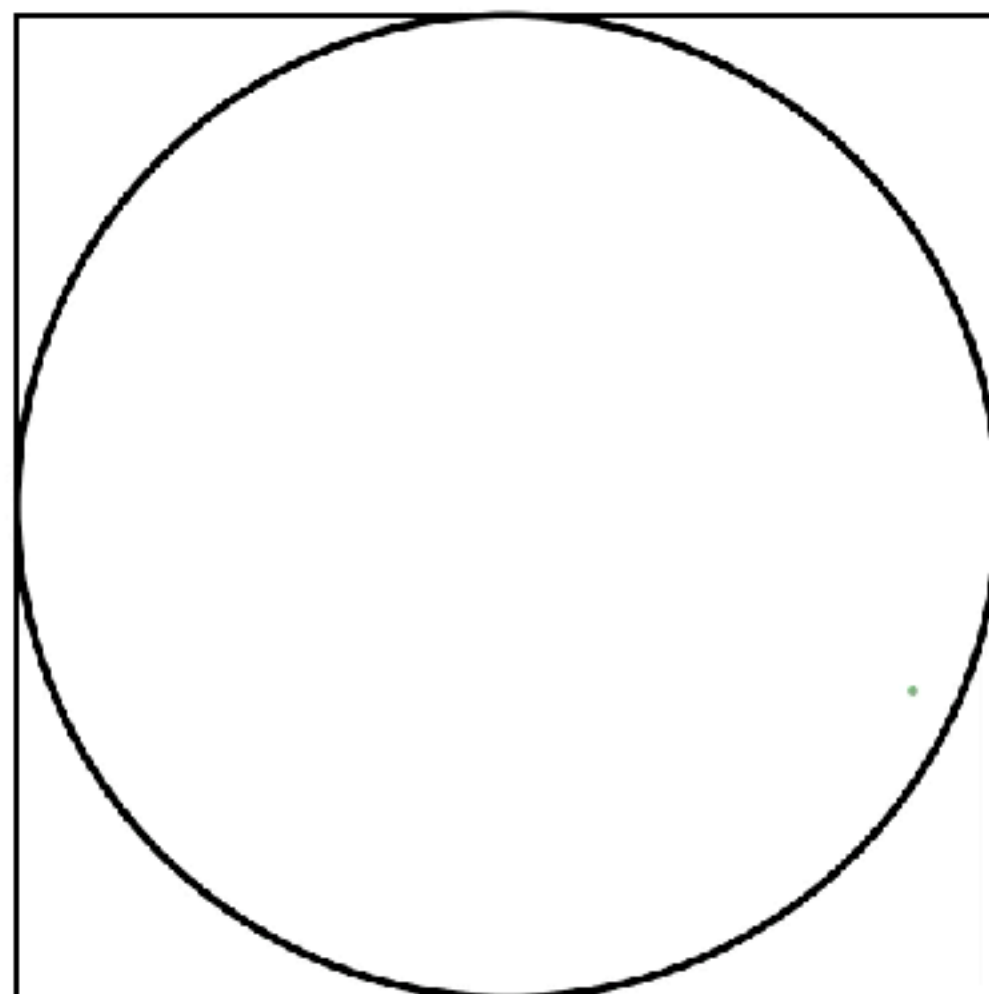


$n = 1, N_{\text{in}} = 1, \hat{\pi}_n = 4.000$

Monte-Carlo estimation of π

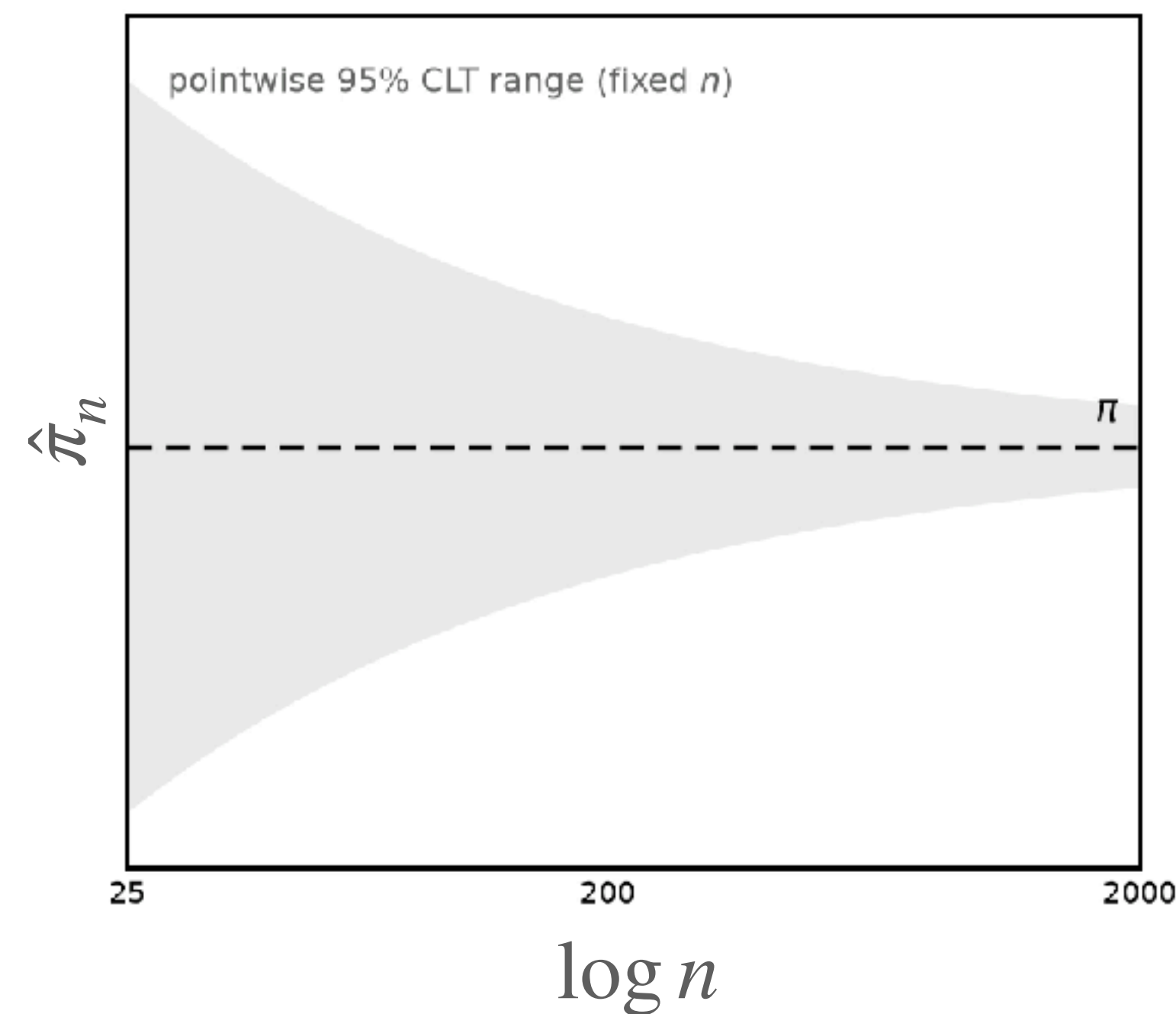
- Why are the LLN and CLT important?

One experiment for π



$n = 1, N_{in} = 1, \hat{\pi}_n = 4.000$

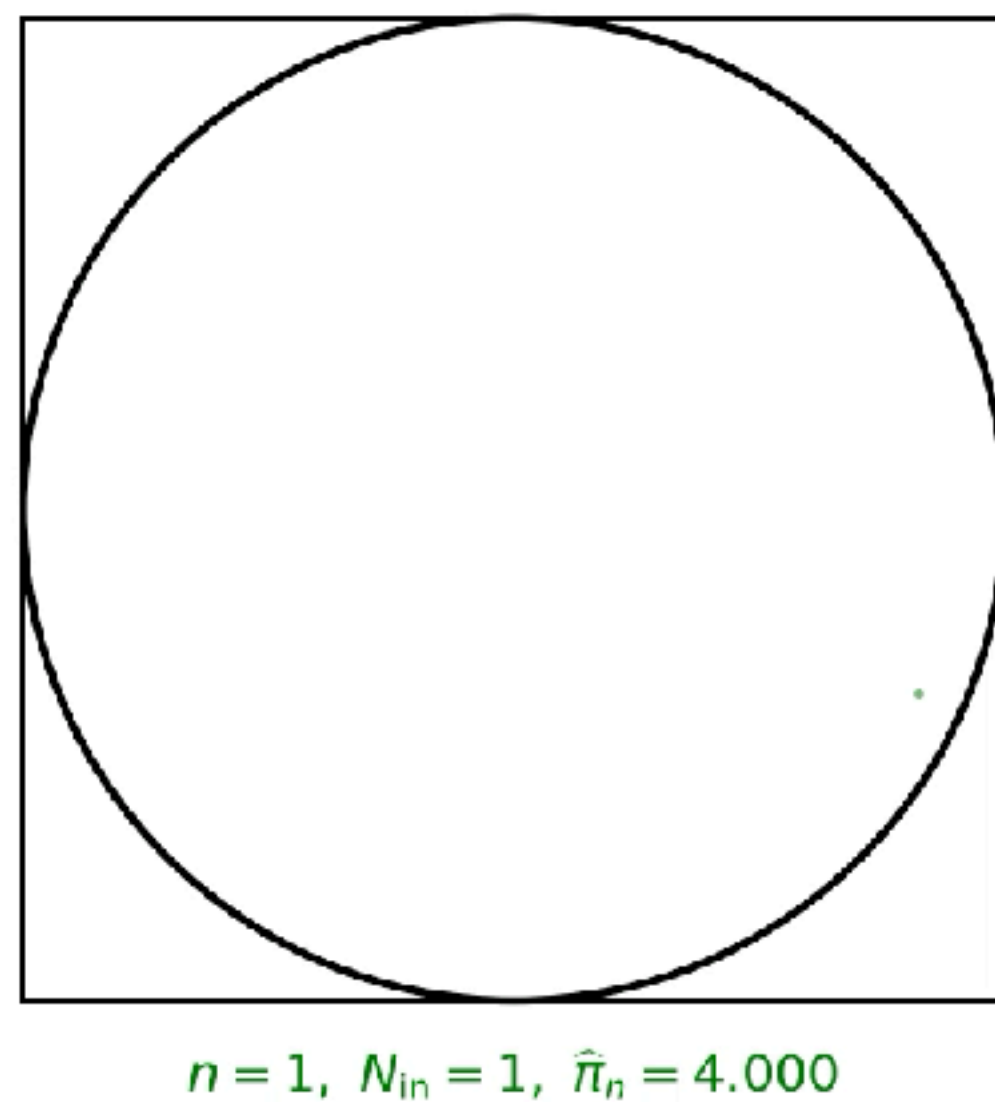
Same experiment, running estimate



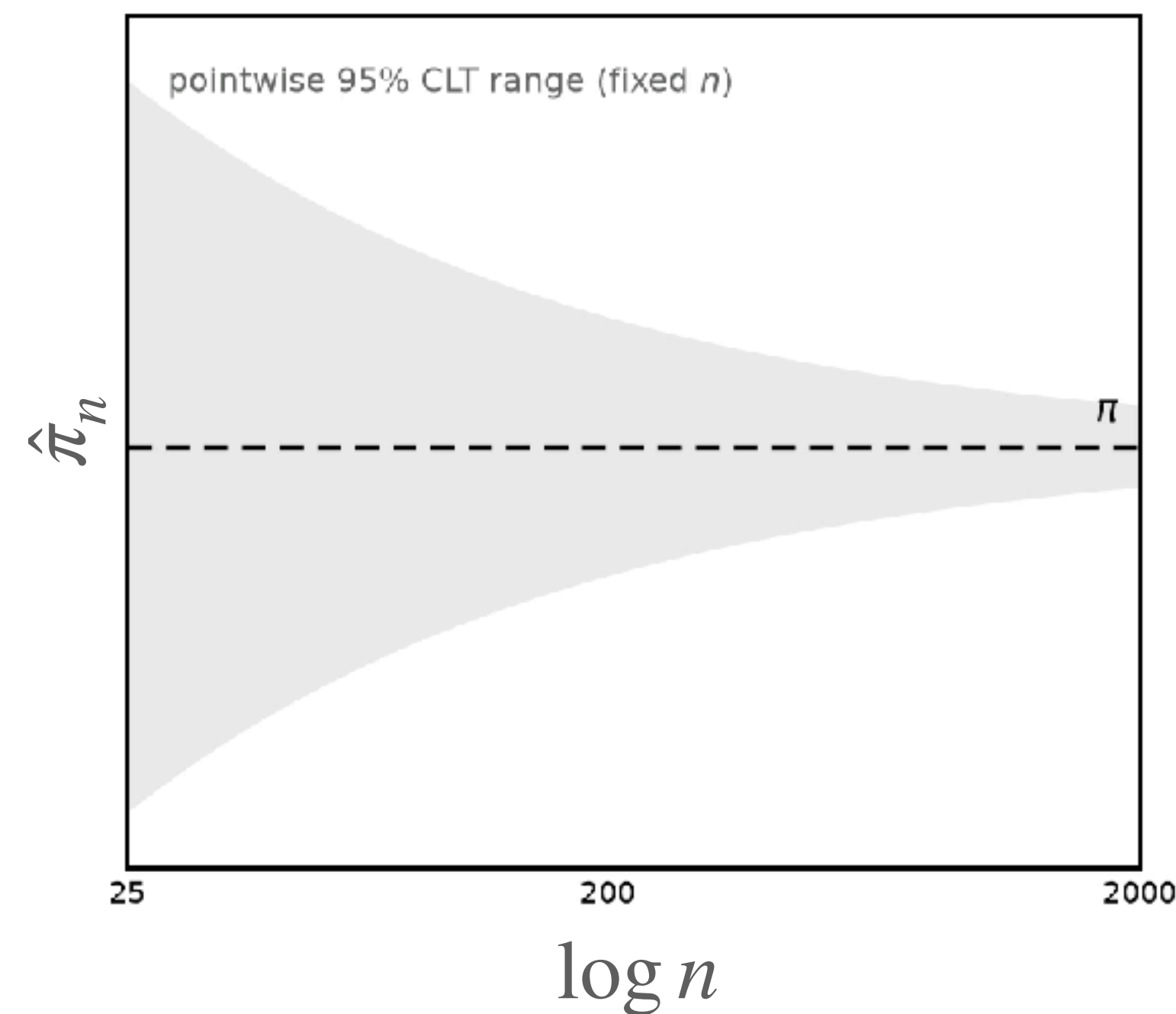
Monte-Carlo estimation of π

- Why are the LLN and CLT important?

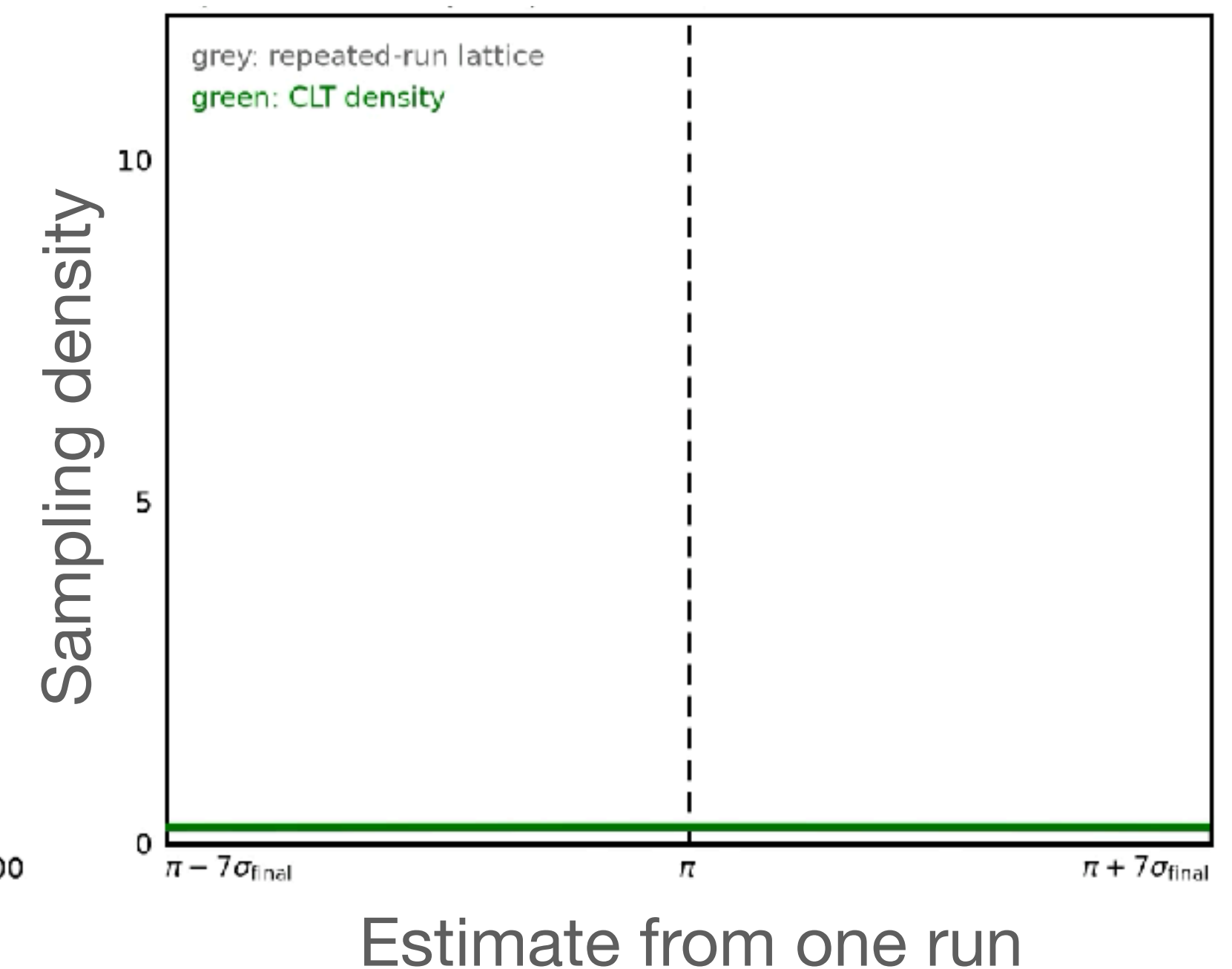
One experiment for π



Same experiment, running estimate



Repeated identical experiments



Monte-Carlo estimation of π

- How does the estimator $\hat{\pi}$ behave as a function of n ?

What we saw today

- So far we have talked about
 - randomness and random variables
 - axioms for probability calculus define probability distributions
 - probability, distributions of probability over values of a random variable
 - multivariate random variables
 - using probability rules to calculate distributions of other variables
 - independence and correlation

Write down something you learned today and
something you need to hear more about next lecture

 jed.homer@physik.lmu.de

Example: Heavens galaxy images

Independence vs. Correlation

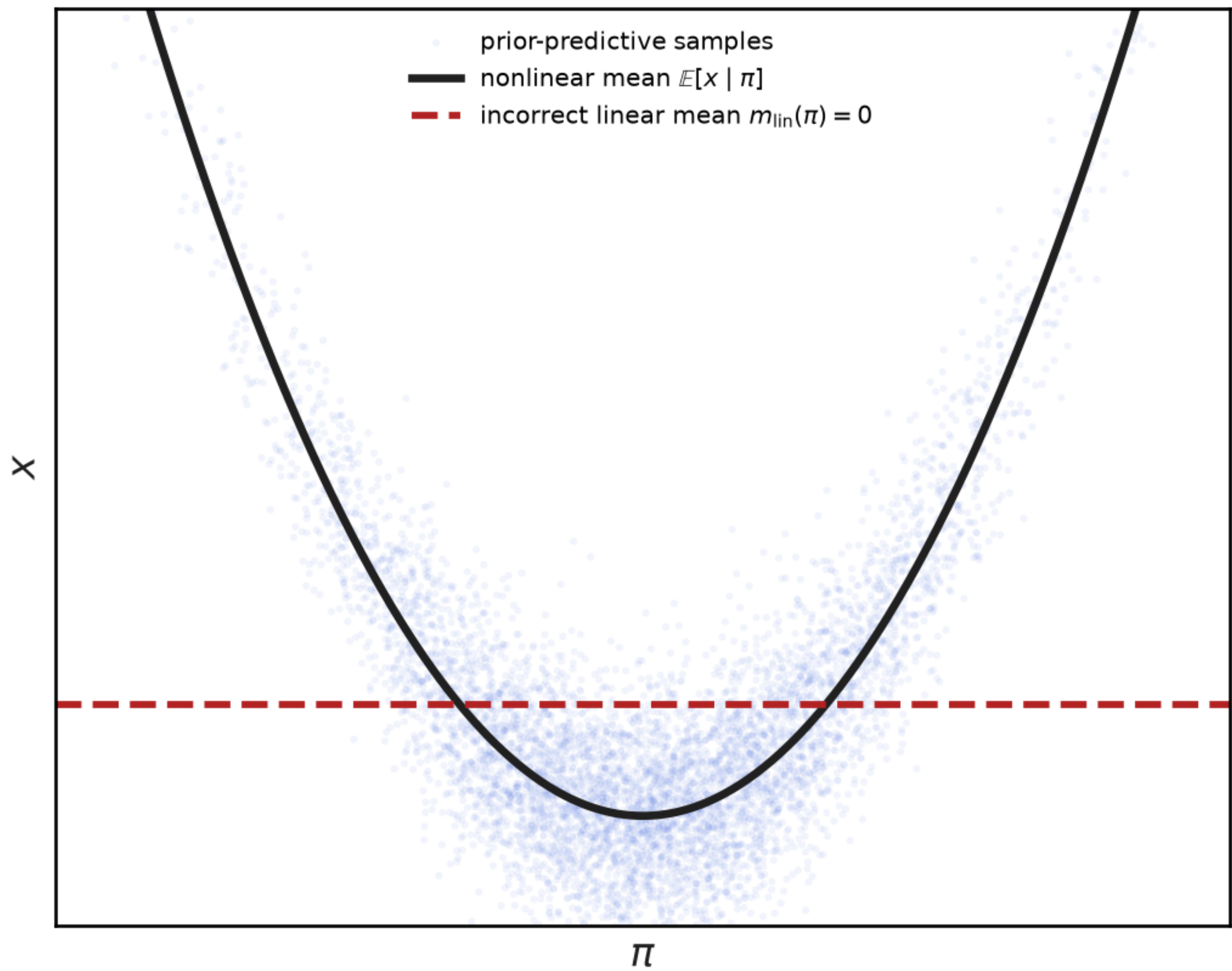
- What if a dataset tells us $\text{Corr}(\pi, x) = 0$ but π and x are **not** independent?
- Assume $\pi \sim \mathcal{G}[\pi | 0, 1]$ and $x | \pi \sim \mathcal{G}\left[x \middle| \frac{\pi^2 - 1}{\sqrt{2}}, \sigma^2\right]$.

Independence vs. Correlation

$$x \mid \pi \sim \mathcal{G} \left[x \mid \frac{\pi^2 - 1}{\sqrt{2}}, \sigma^2 \right] \implies \mathbb{E}[x \mid \pi] = \frac{\pi^2 - 1}{\sqrt{2}}$$

- Data x scatter around a non-linear function $\mathbb{E}[x \mid \pi]$ of π but...
 - $\mathbb{E}[x \mid \pi]$ quantifies the information content of x on π ,
 - the function cannot identify the sign of x ,
 - the

the linear mean misses the nonlinear dependence



the erroneous analysis loses the information in the data

